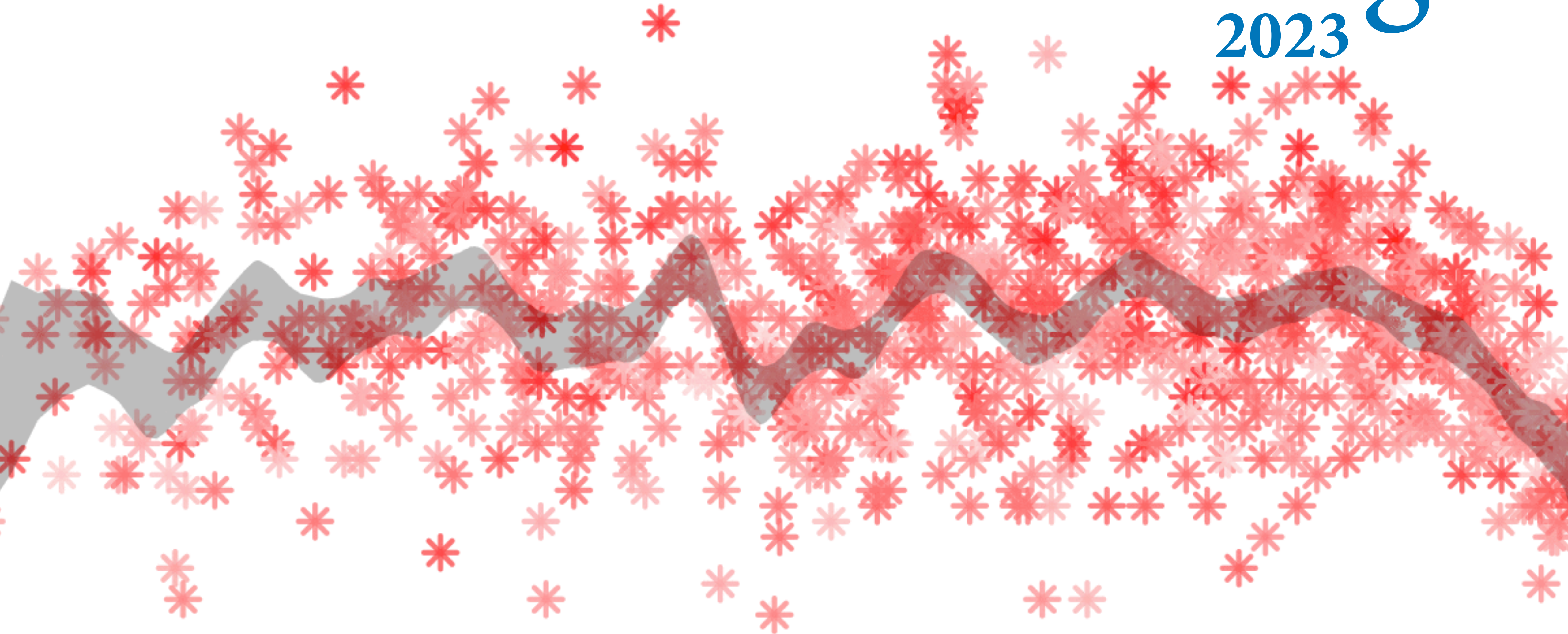


Statistical Rethinking

2023

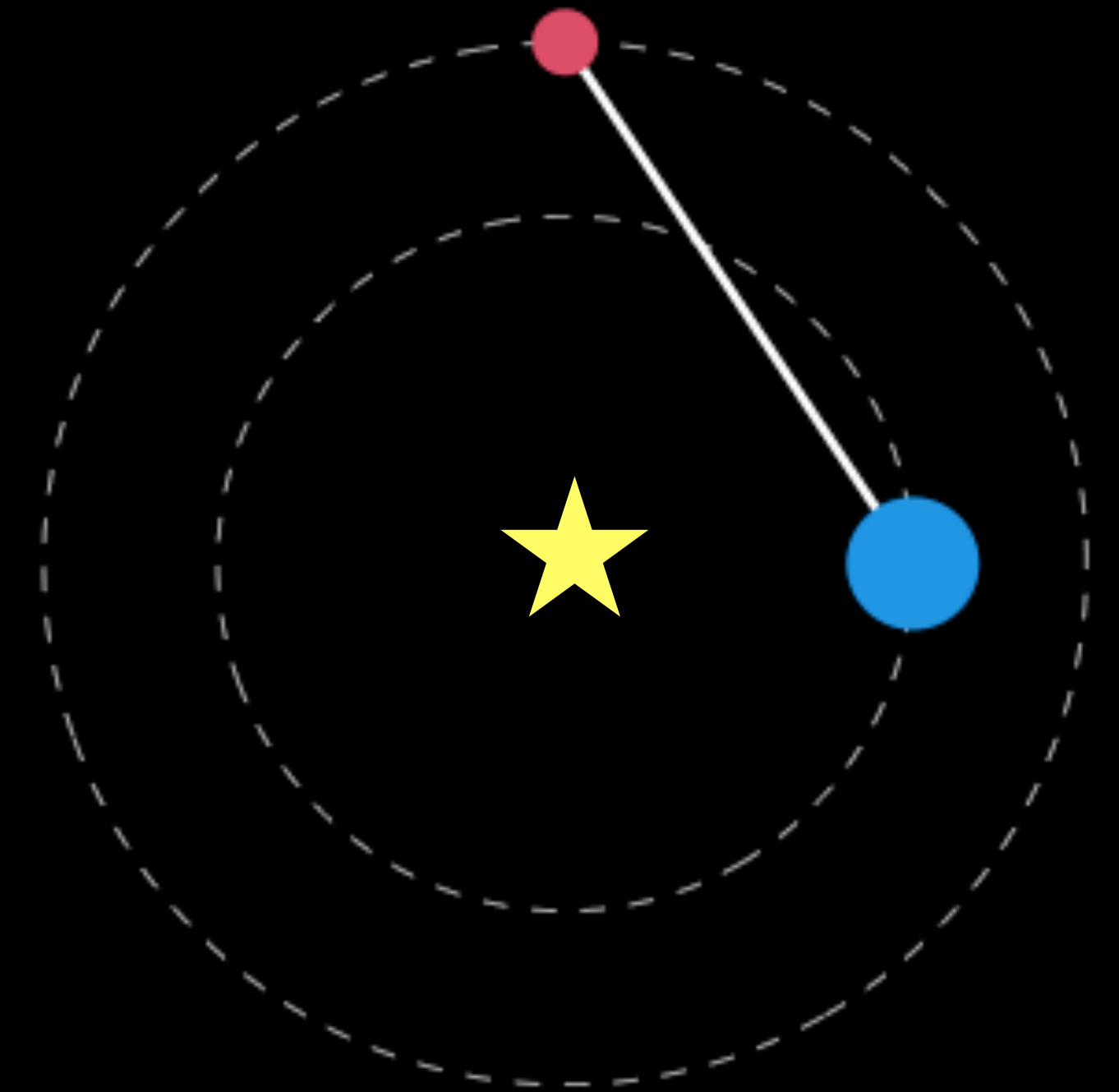


7. Fitting Over & Under

Wrong models can still predict very well



Mikołaj Kopernik (1473–1543)



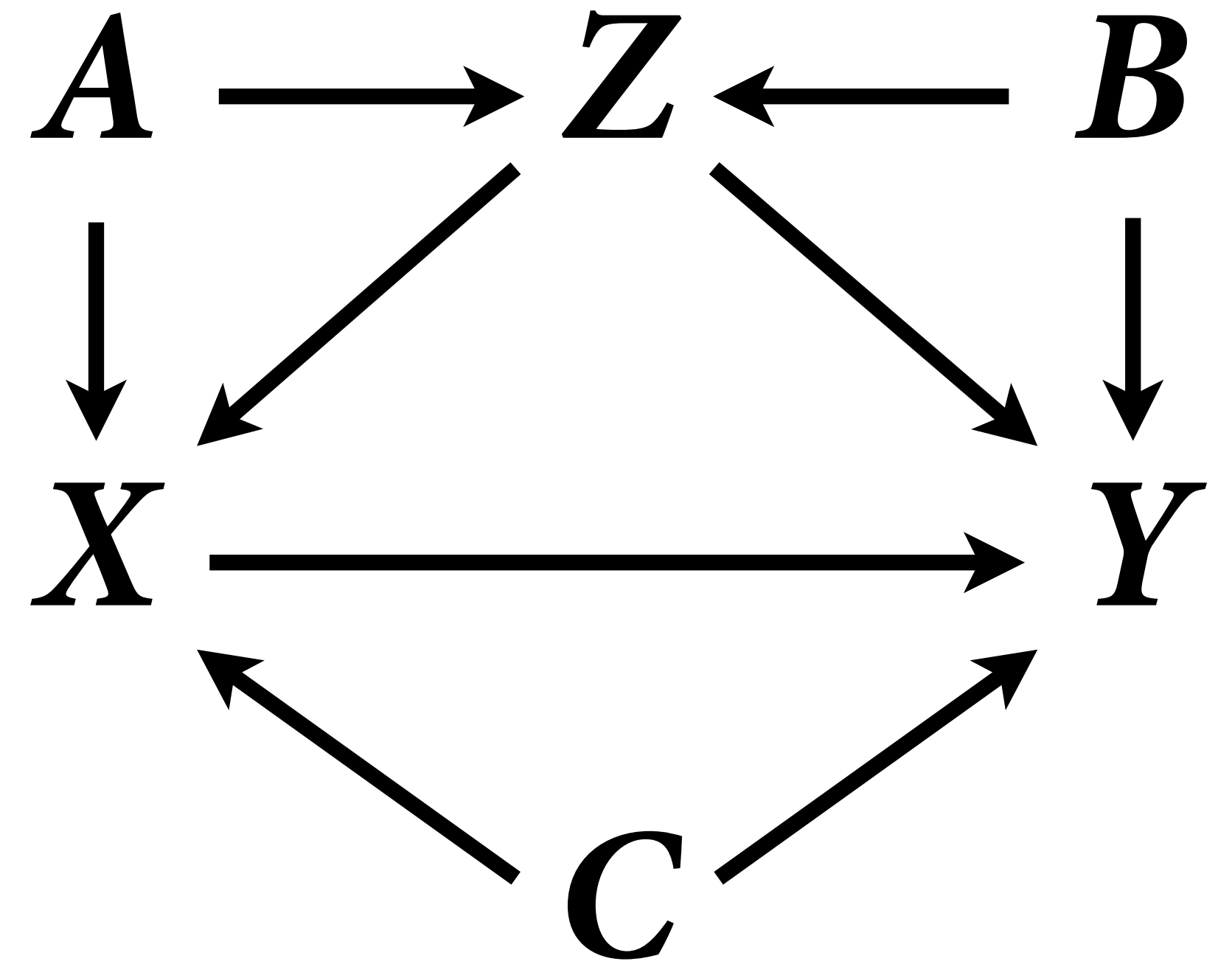
Predicting and testing causation are separate goals with separate tools

Infinite causes, finite data

Estimator might **exist**, but not be **useful**

Struggle against causation: How to use causal assumptions to design estimators, contrast alternative models

Struggle against data: How to make the estimators work



Problems of Prediction

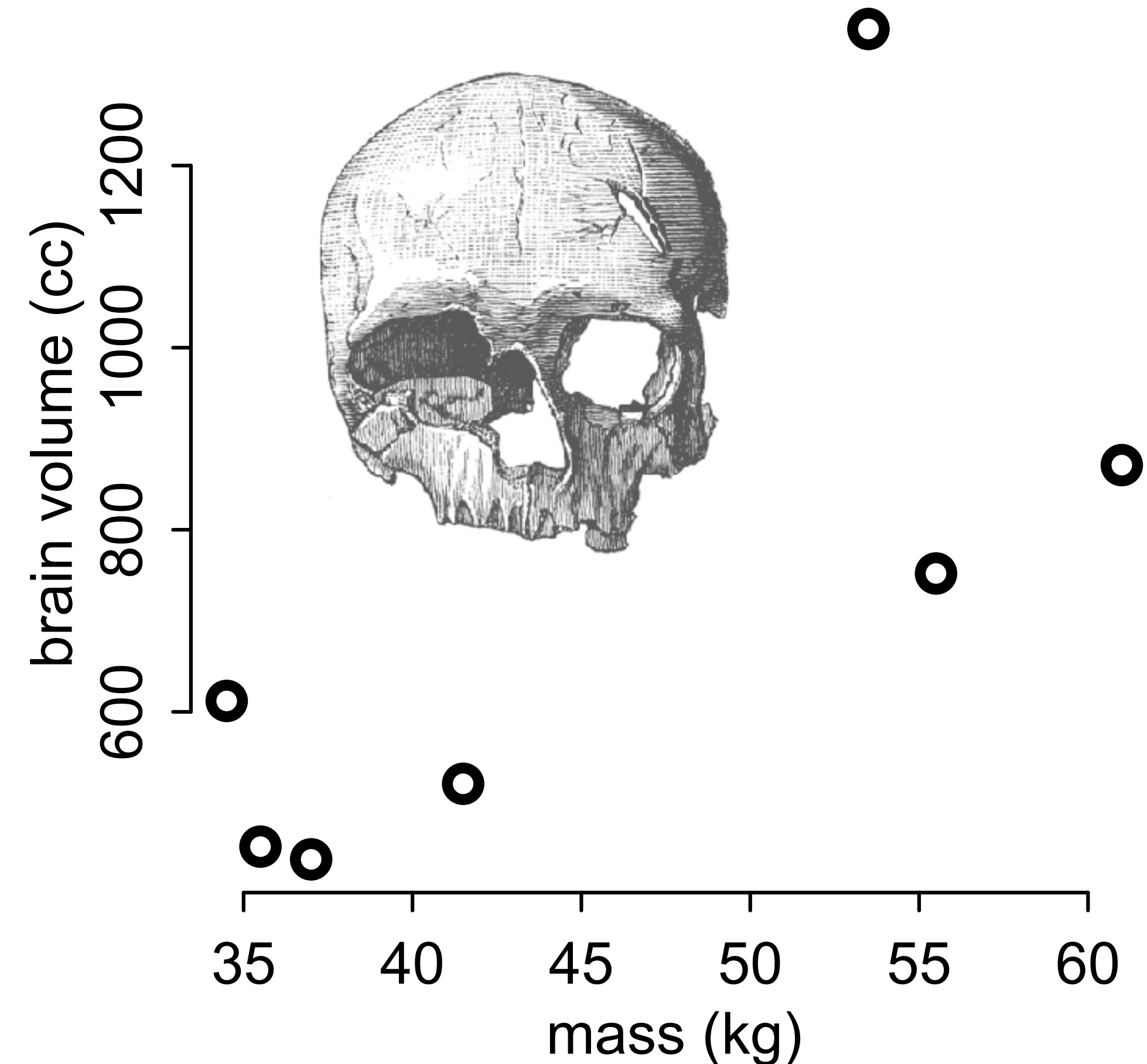
-> Questions/Aims

What function describes these points?
(fitting, compression)

What function explains these points?
(causal inference)

What would happen if we changed a point's mass? (intervention)

What is the next observation from the same process? (prediction)



Leave-one-out cross-validation

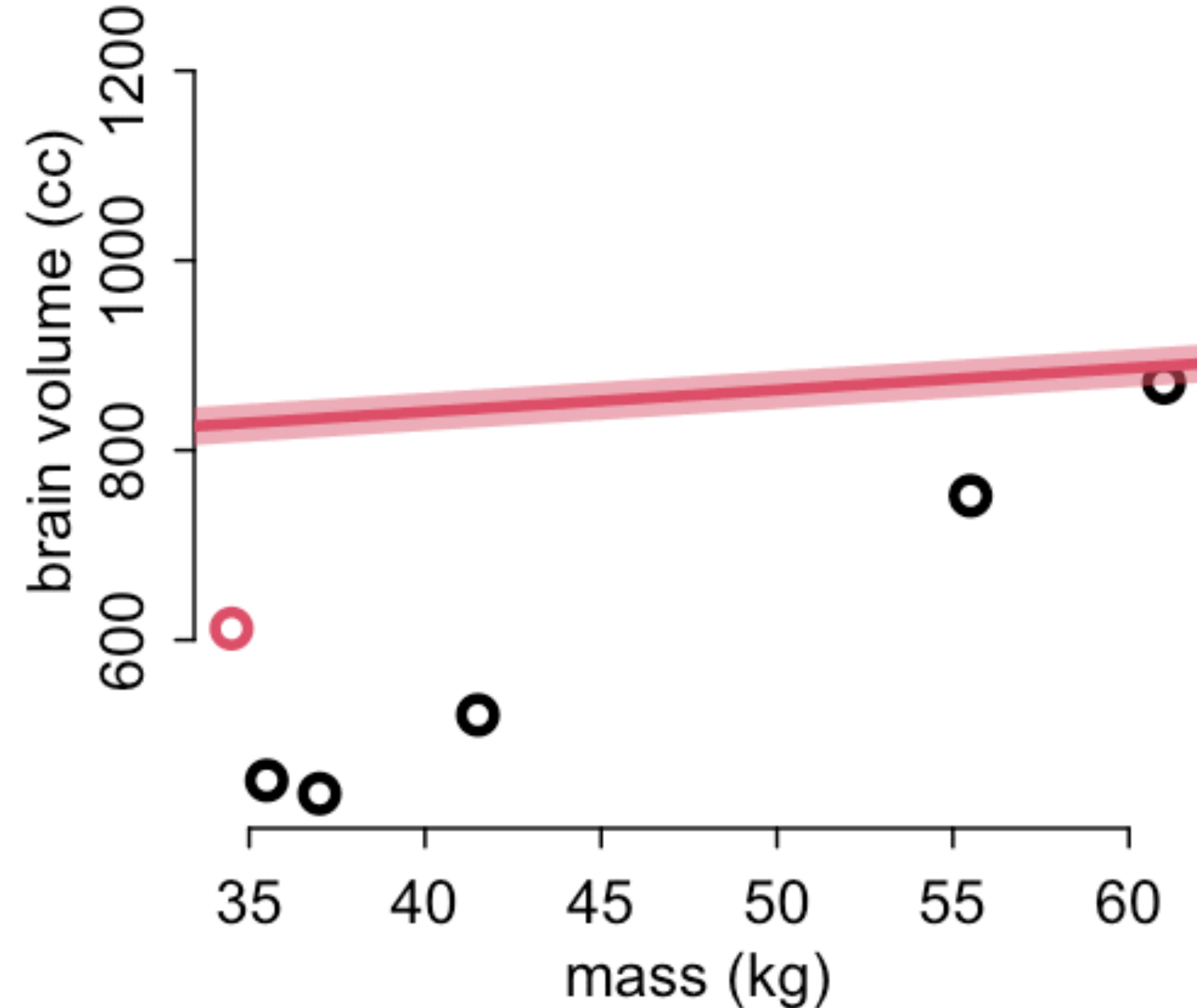
(1) Drop one point

(2) Fit line to remaining

(3) Predict dropped point

(4) Repeat (1) with next point

(5) Score is error on dropped



Leave-one-out cross-validation

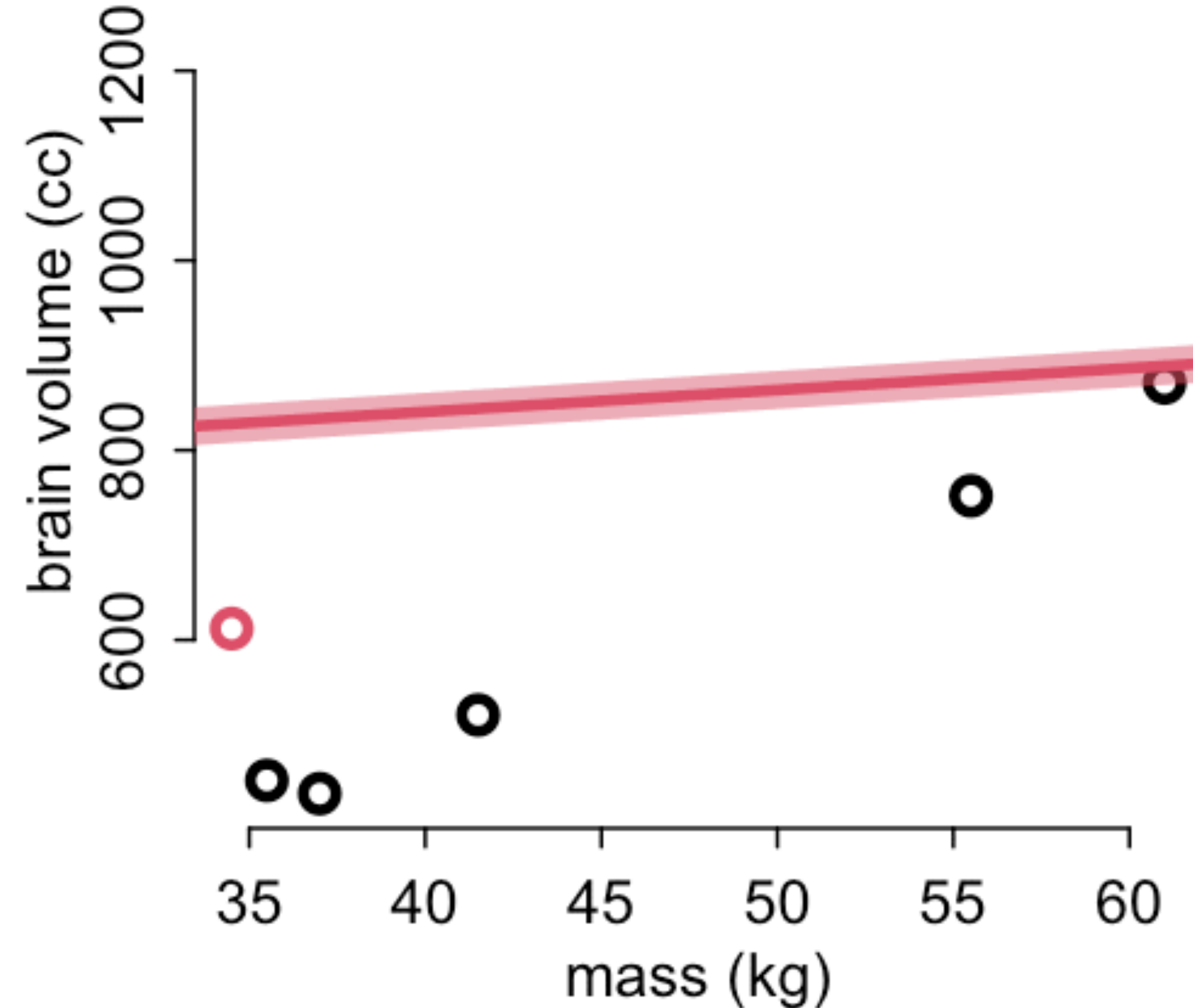
(1) Drop one point

(2) Fit line to remaining

(3) Predict dropped point

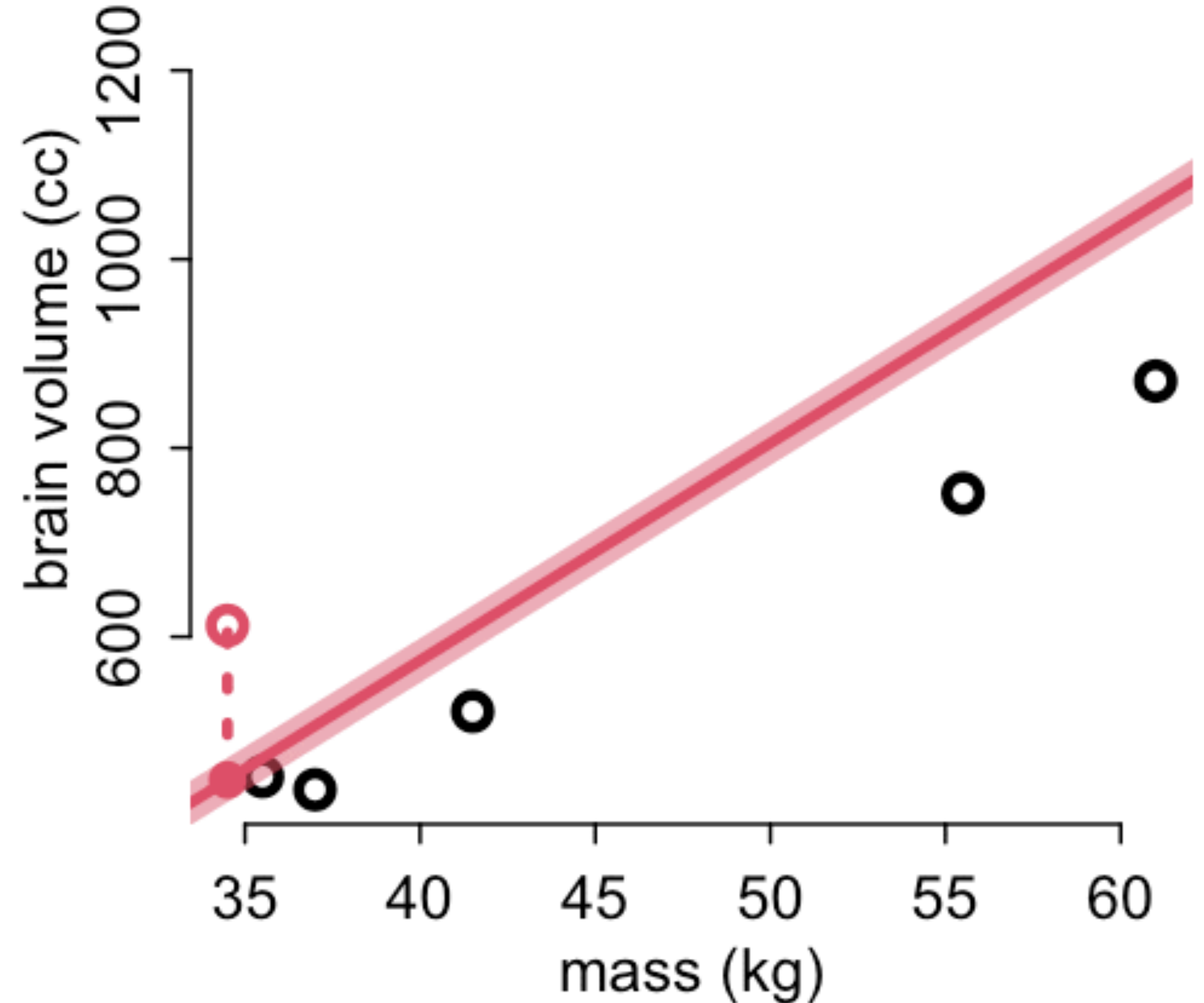
(4) Repeat (1) with next point

(5) Score is error on dropped



Leave-one-out cross-validation

- (1) Drop one point
- (2) Fit line to remaining
- (3) Predict dropped point
- (4) Repeat (1) with next point
- (5) Score is error on dropped



Leave-one-out cross-validation

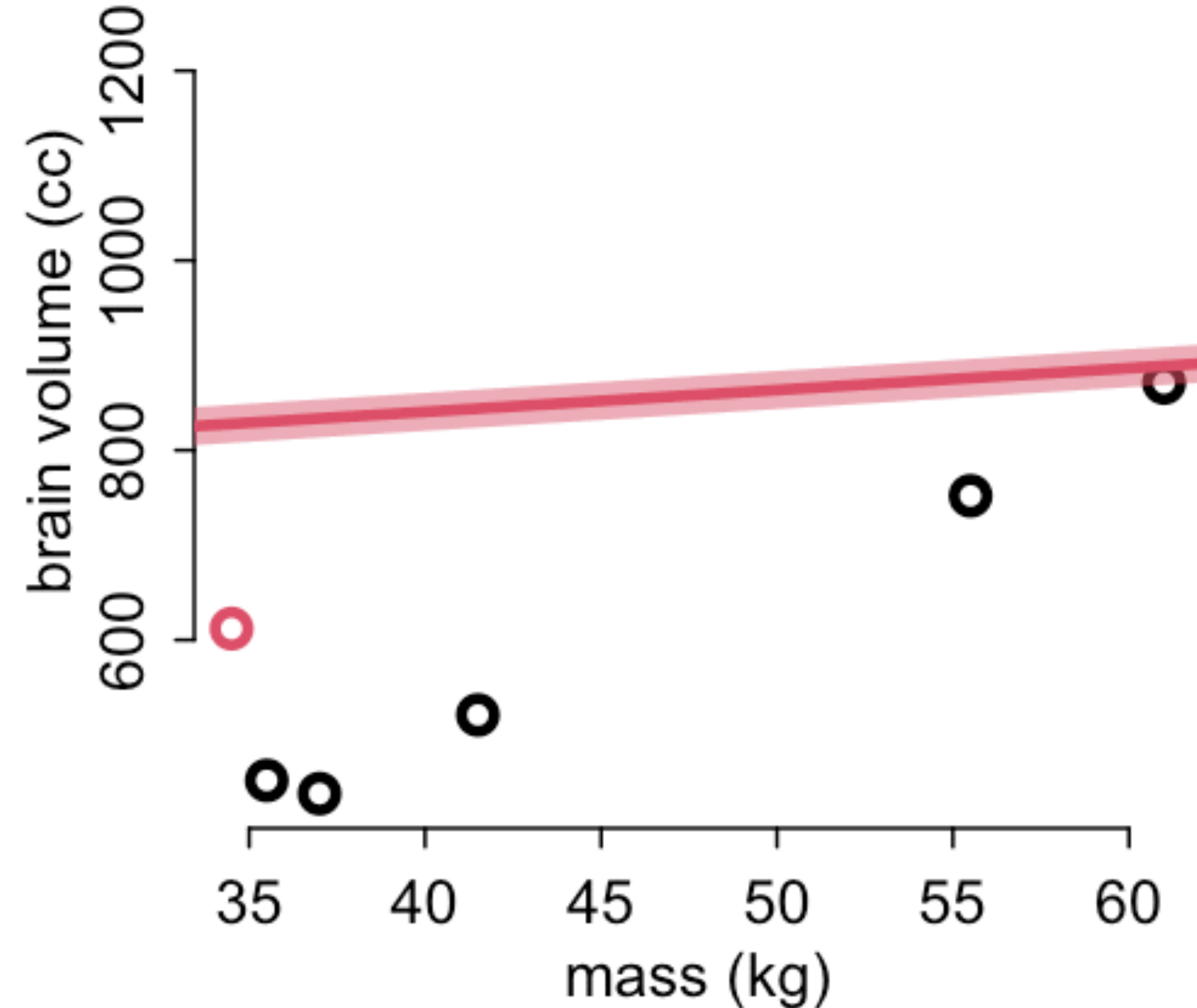
(1) Drop one point

(2) Fit line to remaining

(3) Predict dropped point

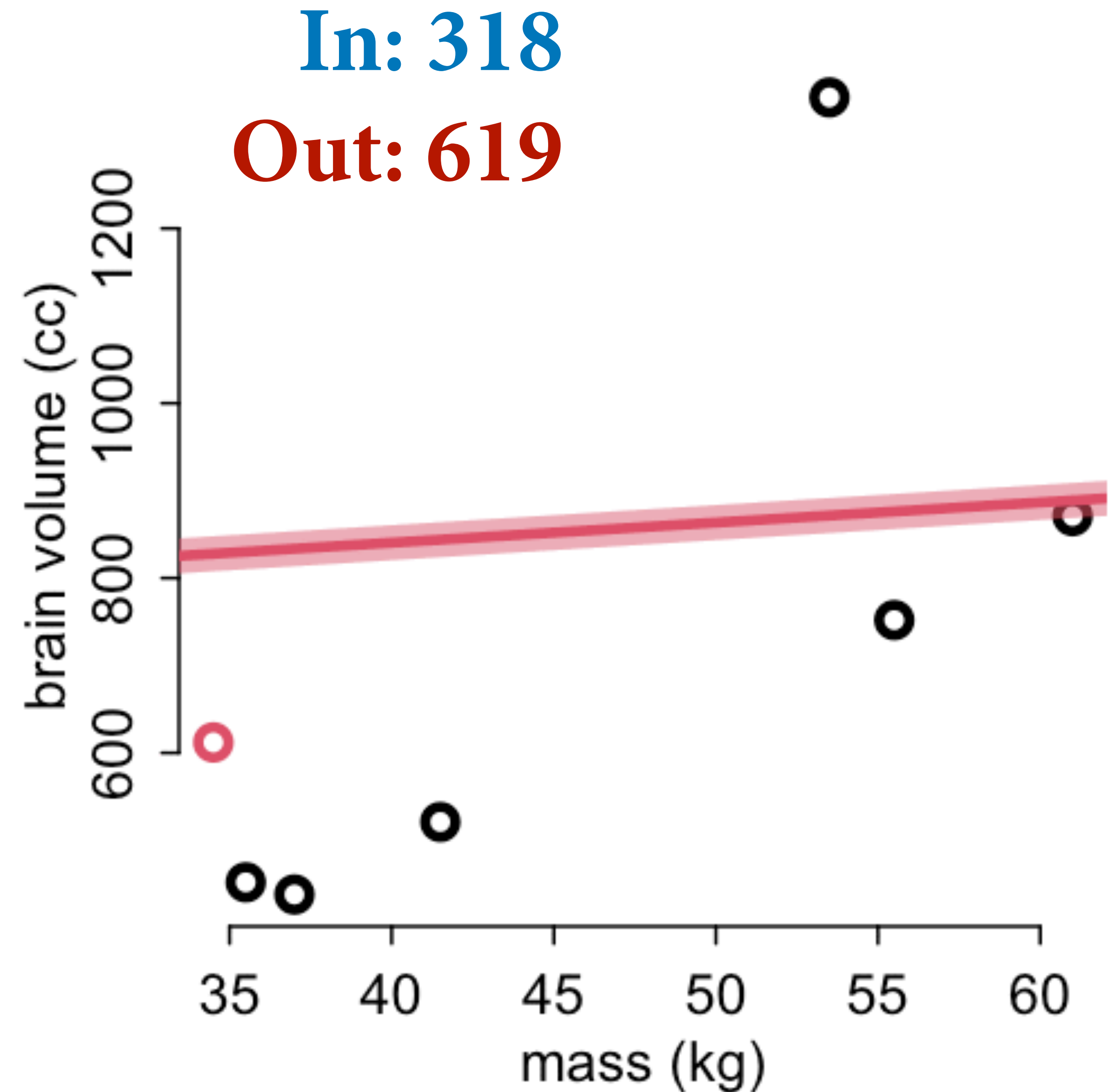
(4) Repeat (1) with next point

(5) Score is error on dropped



Leave-one-out cross-validation

- (1) Drop one point
- (2) Fit line to remaining
- (3) Predict dropped point
- (4) Repeat (1) with next point
- (5) Score is error on dropped

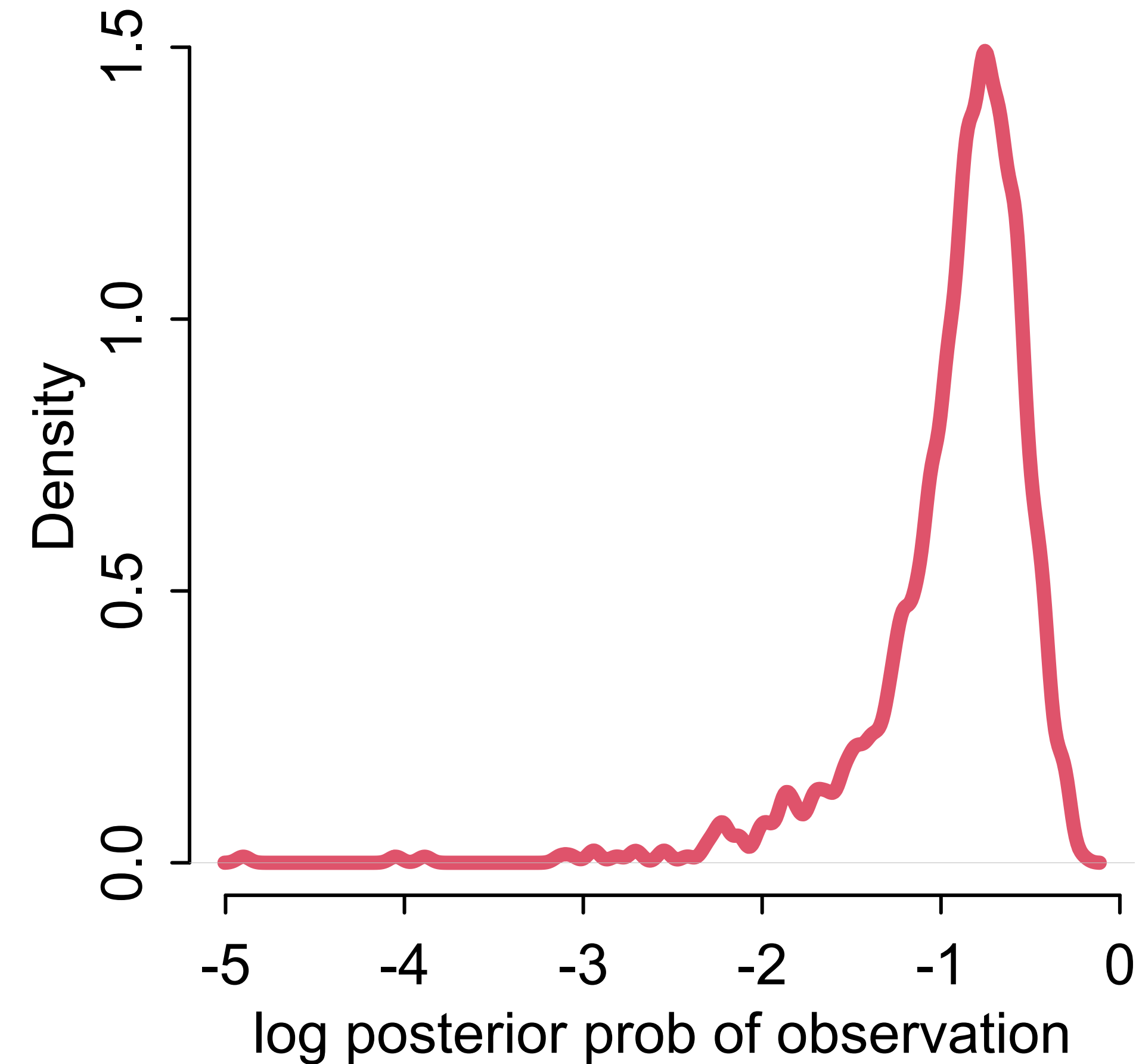


Bayesian Cross-Validation

We use the entire posterior, not just a point prediction

Cross-validation score is:

$$\text{lppd}_{\text{CV}} = \sum_{i=1}^N \frac{1}{S} \sum_{s=1}^S \log \Pr(y_i | \theta_{-i,s})$$



Bayesian Cross-Validation

log pointwise predictive density

N data points

S samples from posterior

$$\text{lppd}_{\text{CV}} = \sum_{i=1}^N \frac{1}{S} \sum_{s=1}^S \log \text{Pr}(y_i | \theta_{-i,s})$$

average log probability for point i

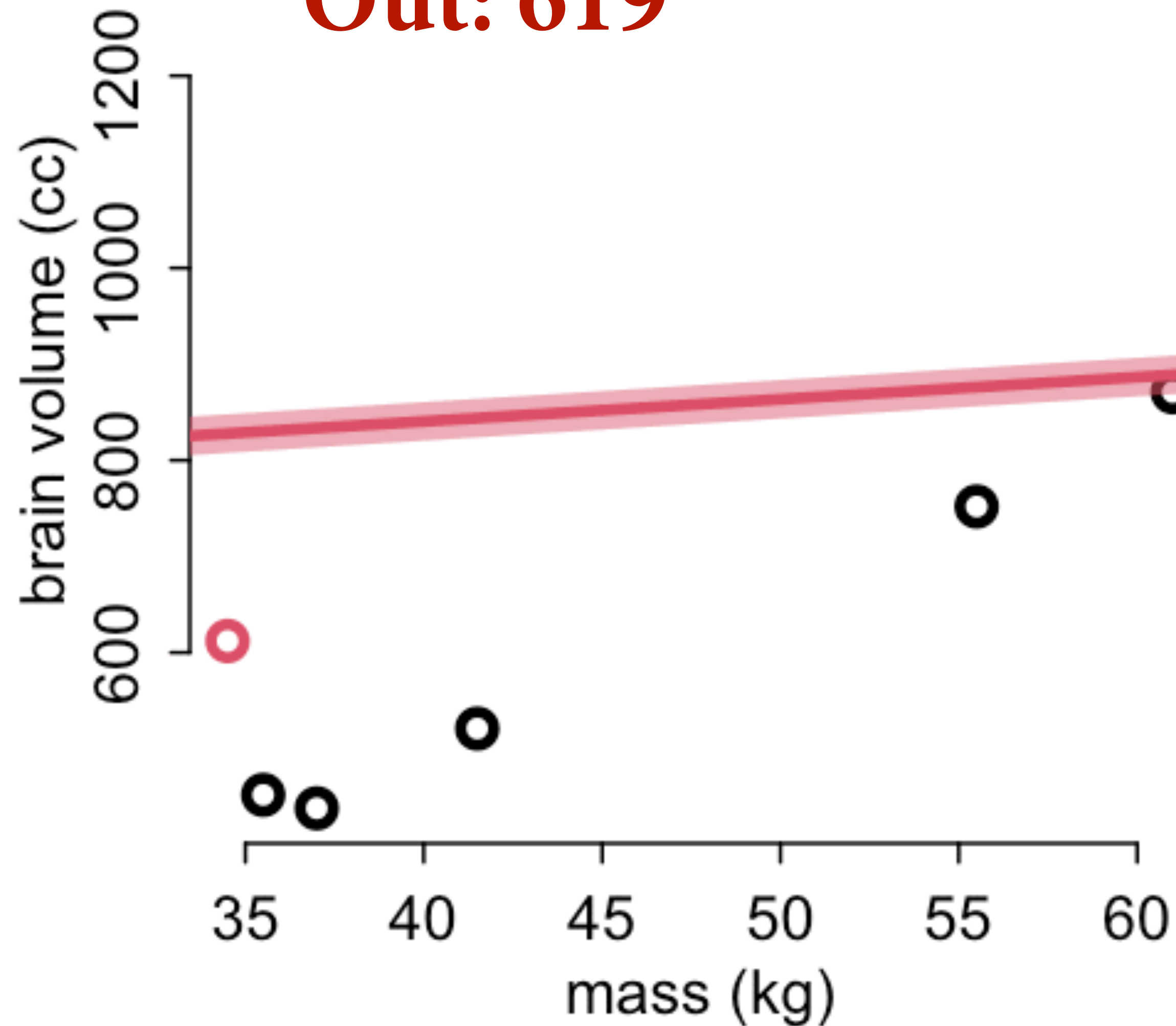
log probability of each point i, computed with posterior that omits point i

The diagram illustrates the formula for log pointwise predictive density (lppd_{CV}). The formula is
$$\text{lppd}_{\text{CV}} = \sum_{i=1}^N \frac{1}{S} \sum_{s=1}^S \log \text{Pr}(y_i | \theta_{-i,s})$$
. Annotations include: 'log pointwise predictive density' pointing to the left side of the equation; 'N data points' pointing to the summation index $i=1$ to N ; 'S samples from posterior' pointing to the summation index $s=1$ to S ; 'average log probability for point i' pointing to the $\frac{1}{S}$ term; and 'log probability of each point i, computed with posterior that omits point i' pointing to the $\log \text{Pr}(y_i | \theta_{-i,s})$ term.

Adding polynomial terms to the regression

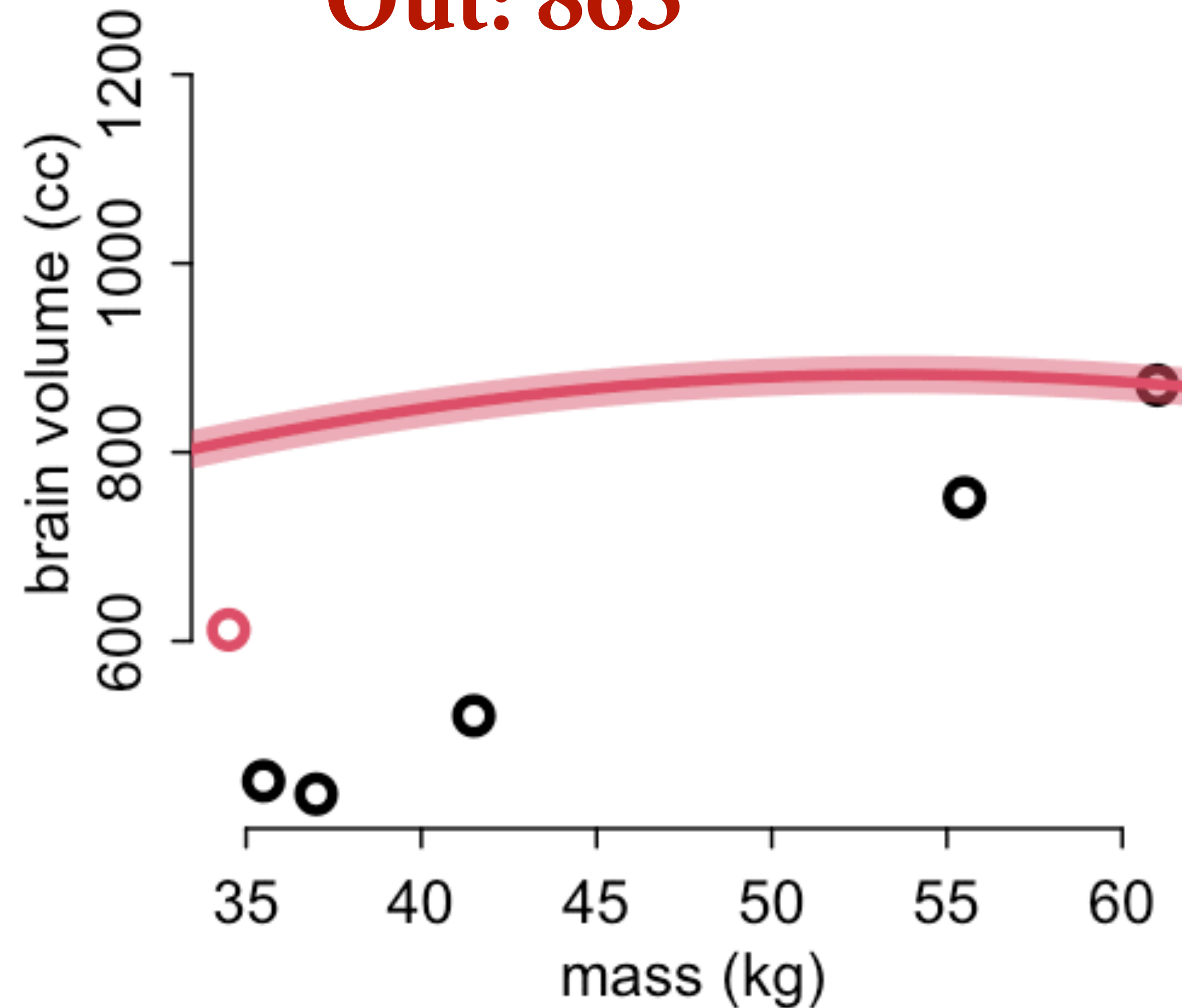
[1] In: 318

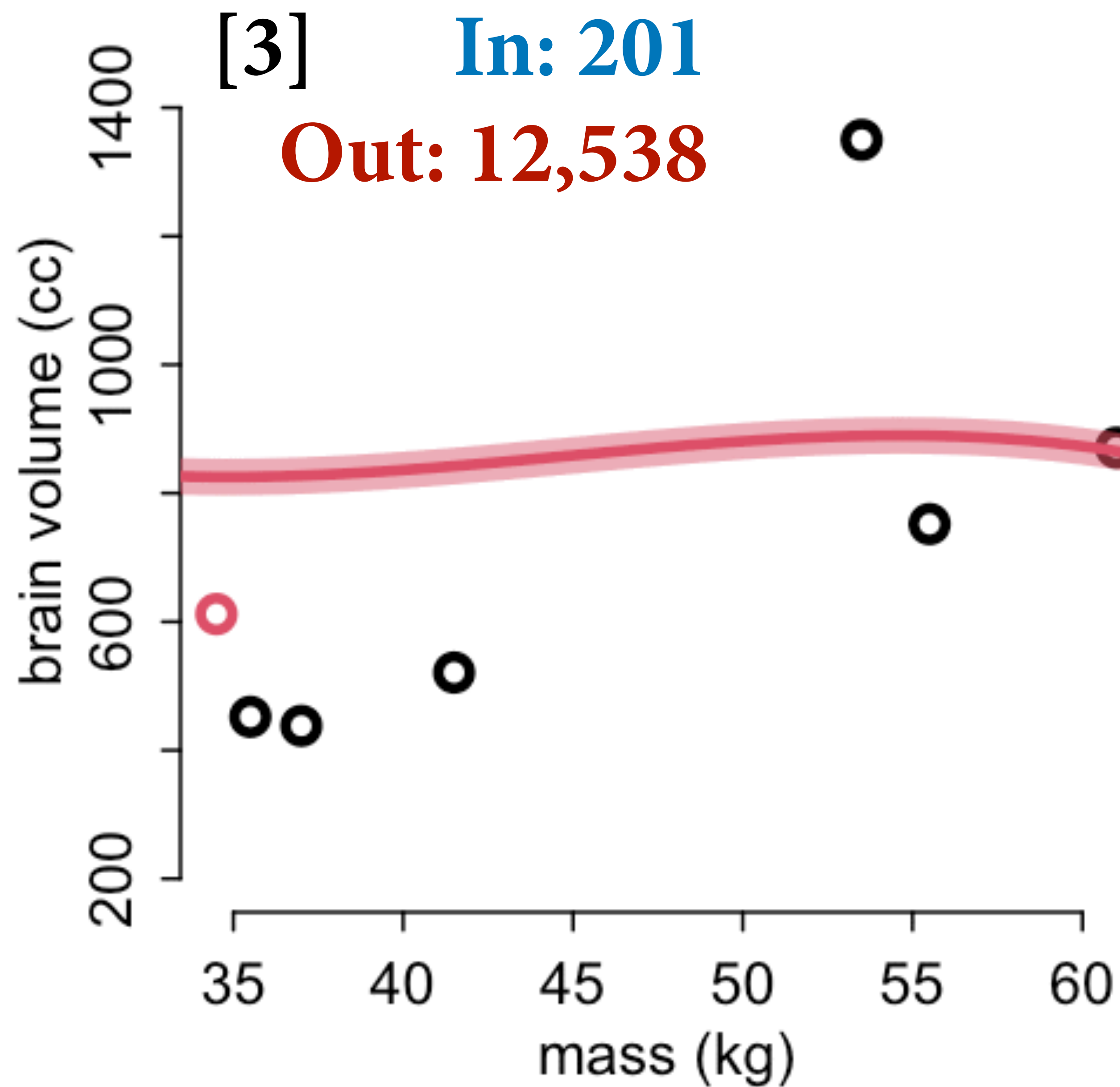
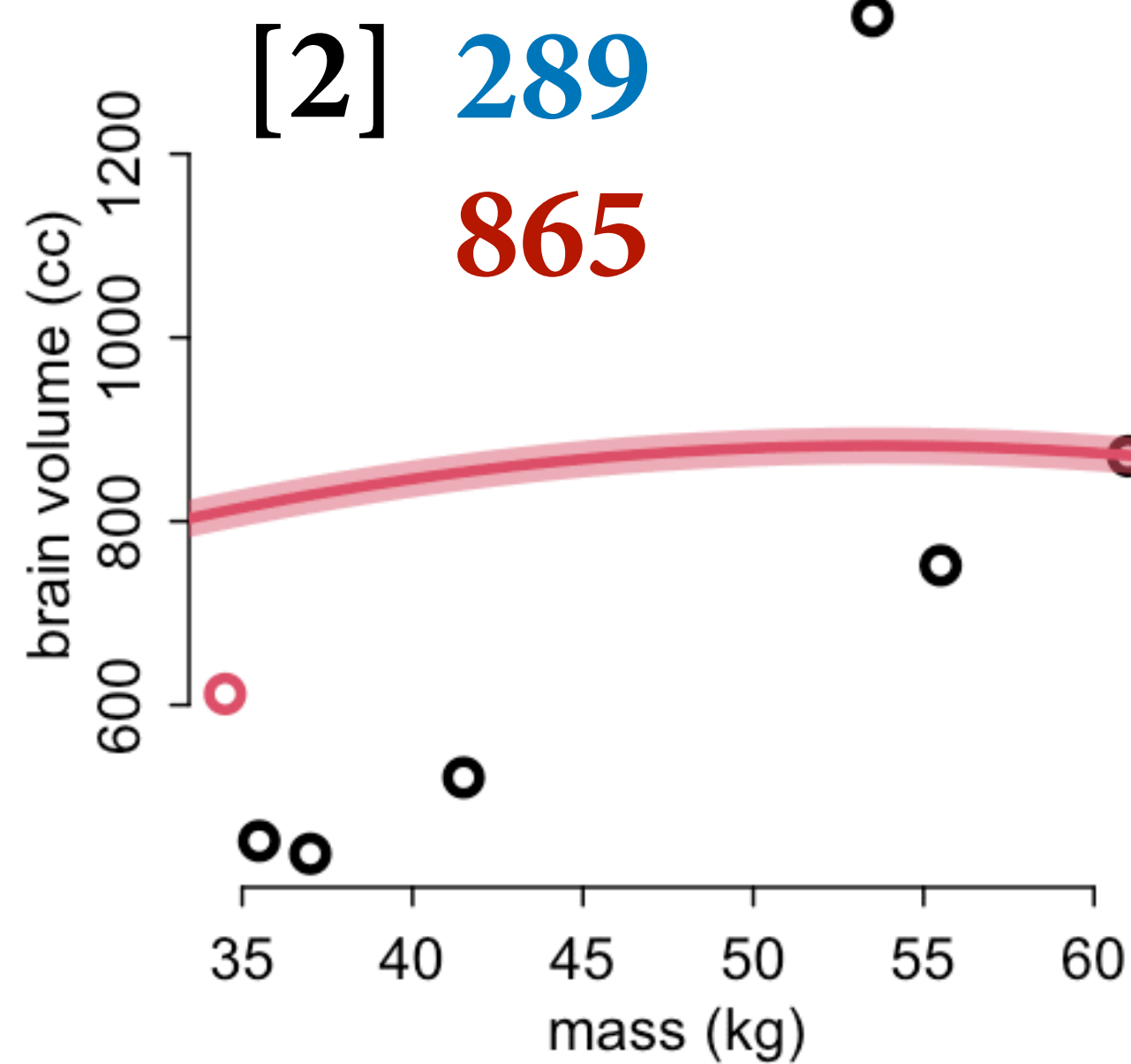
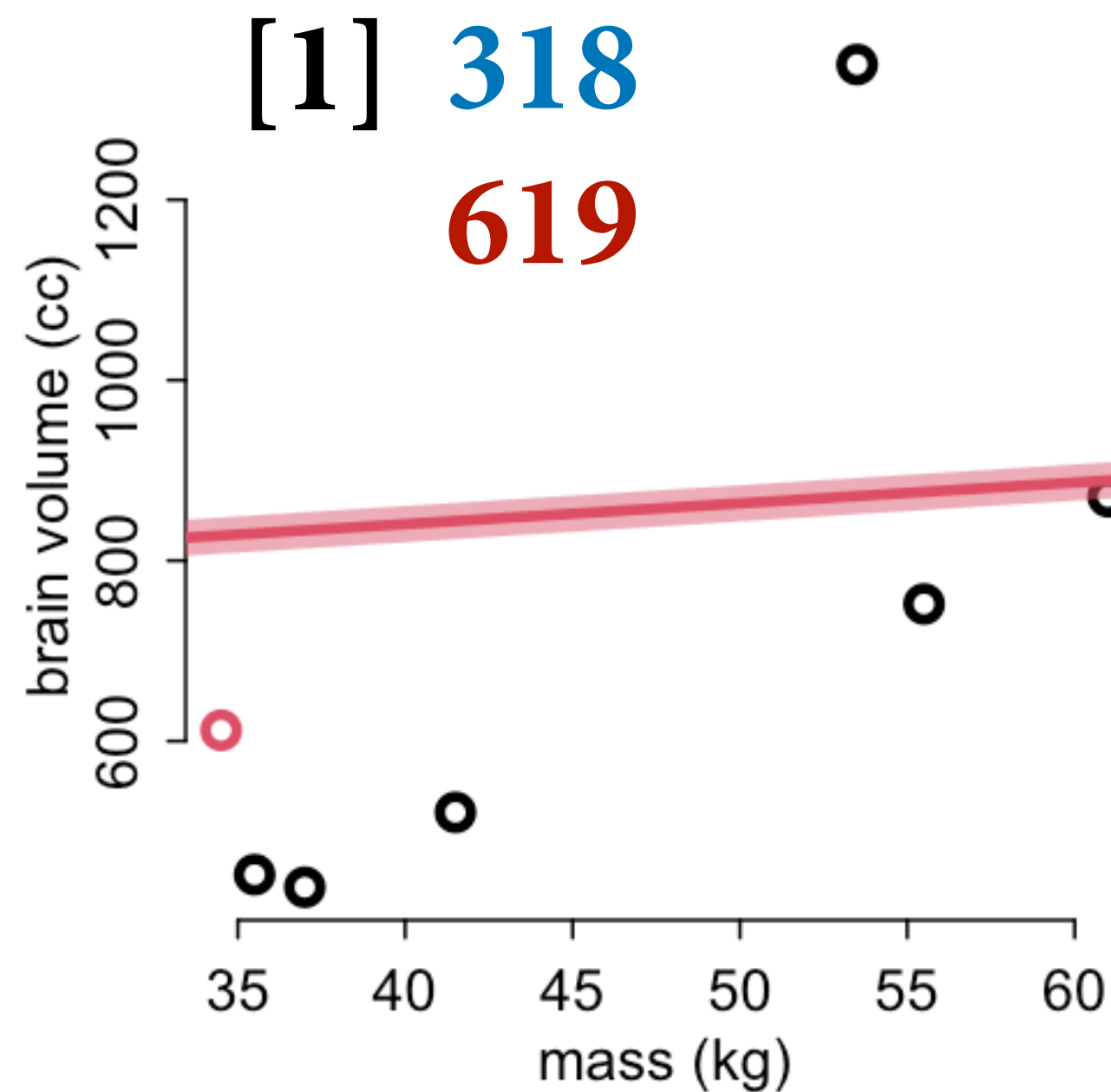
Out: 619

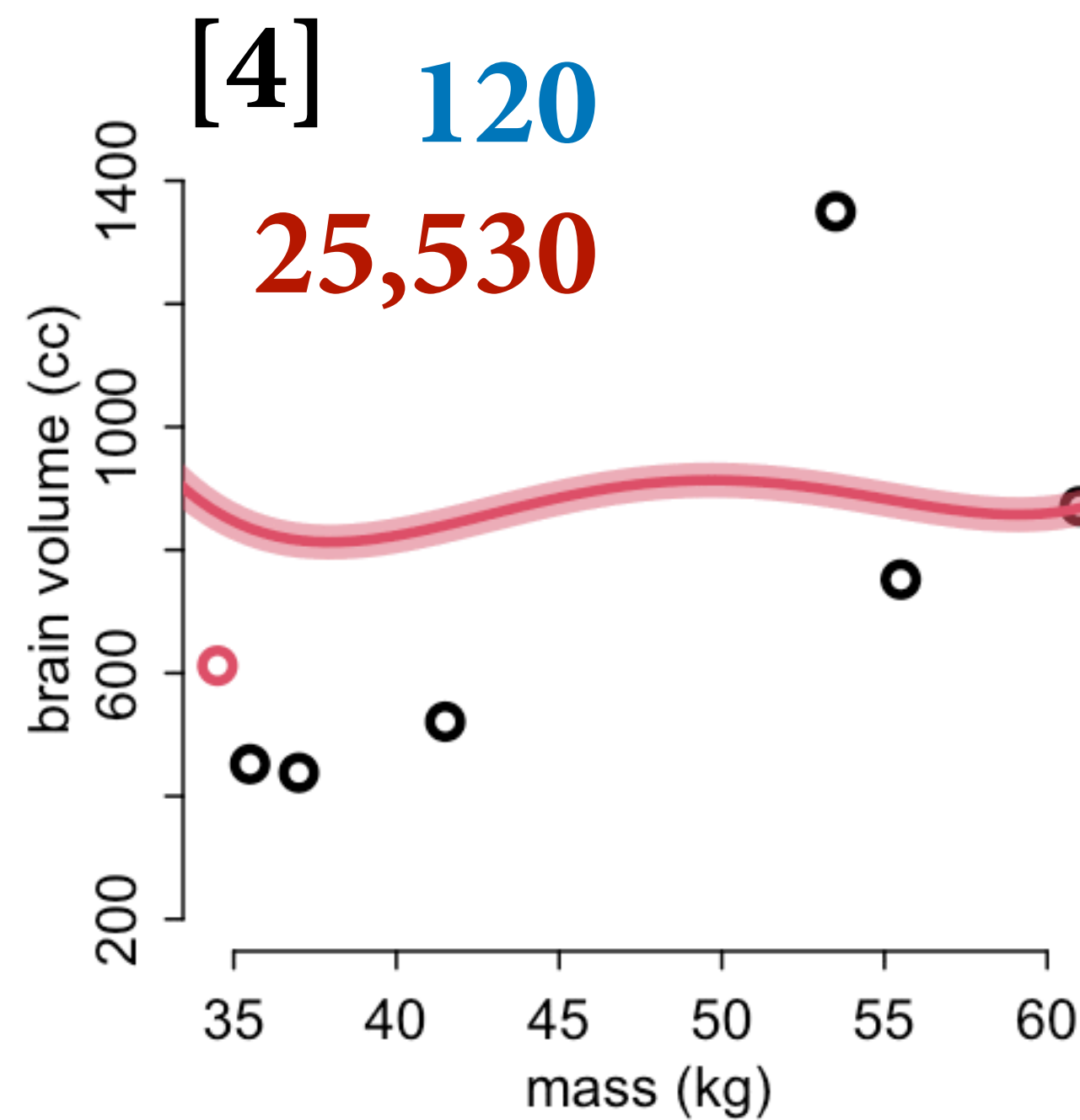
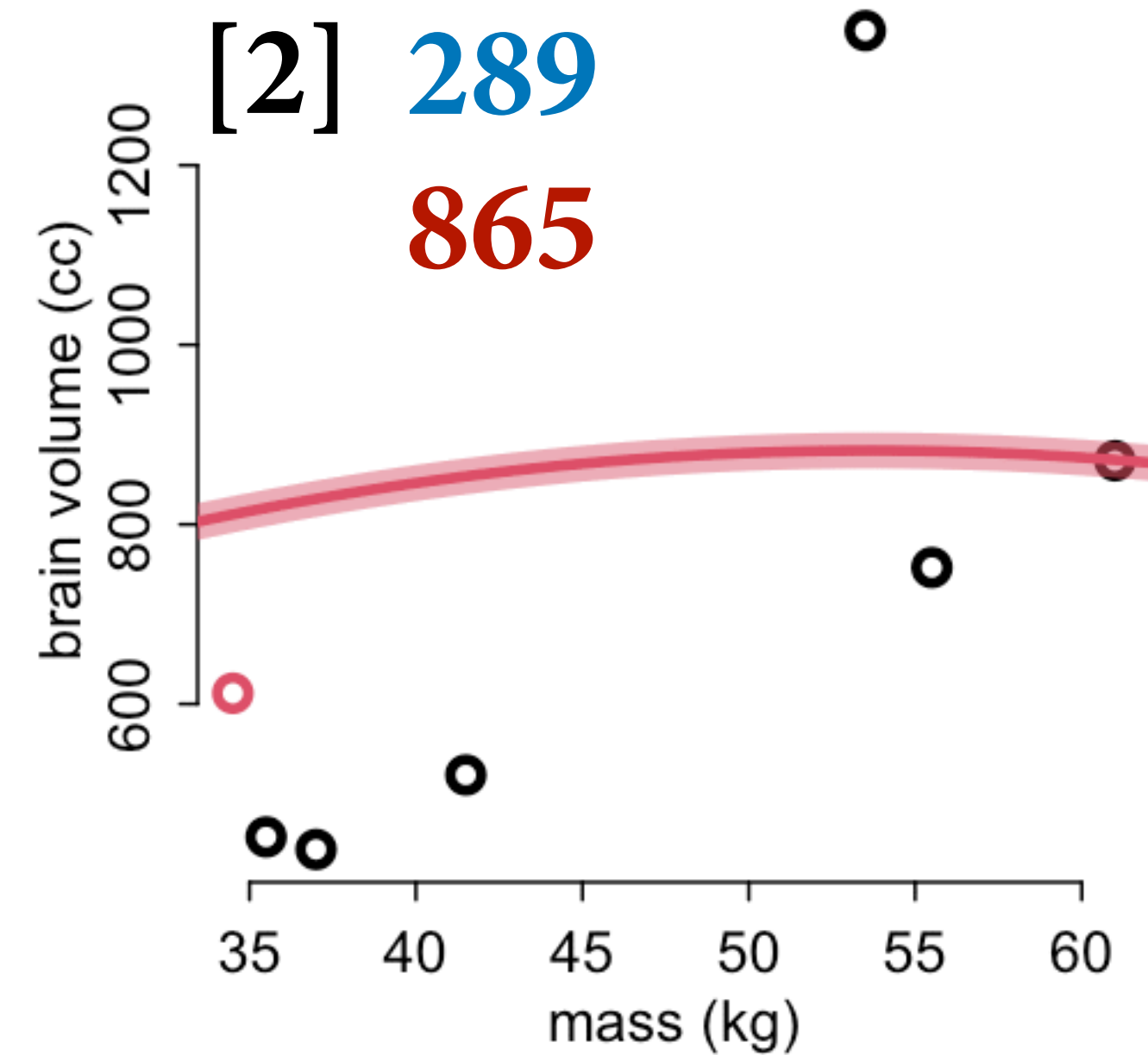
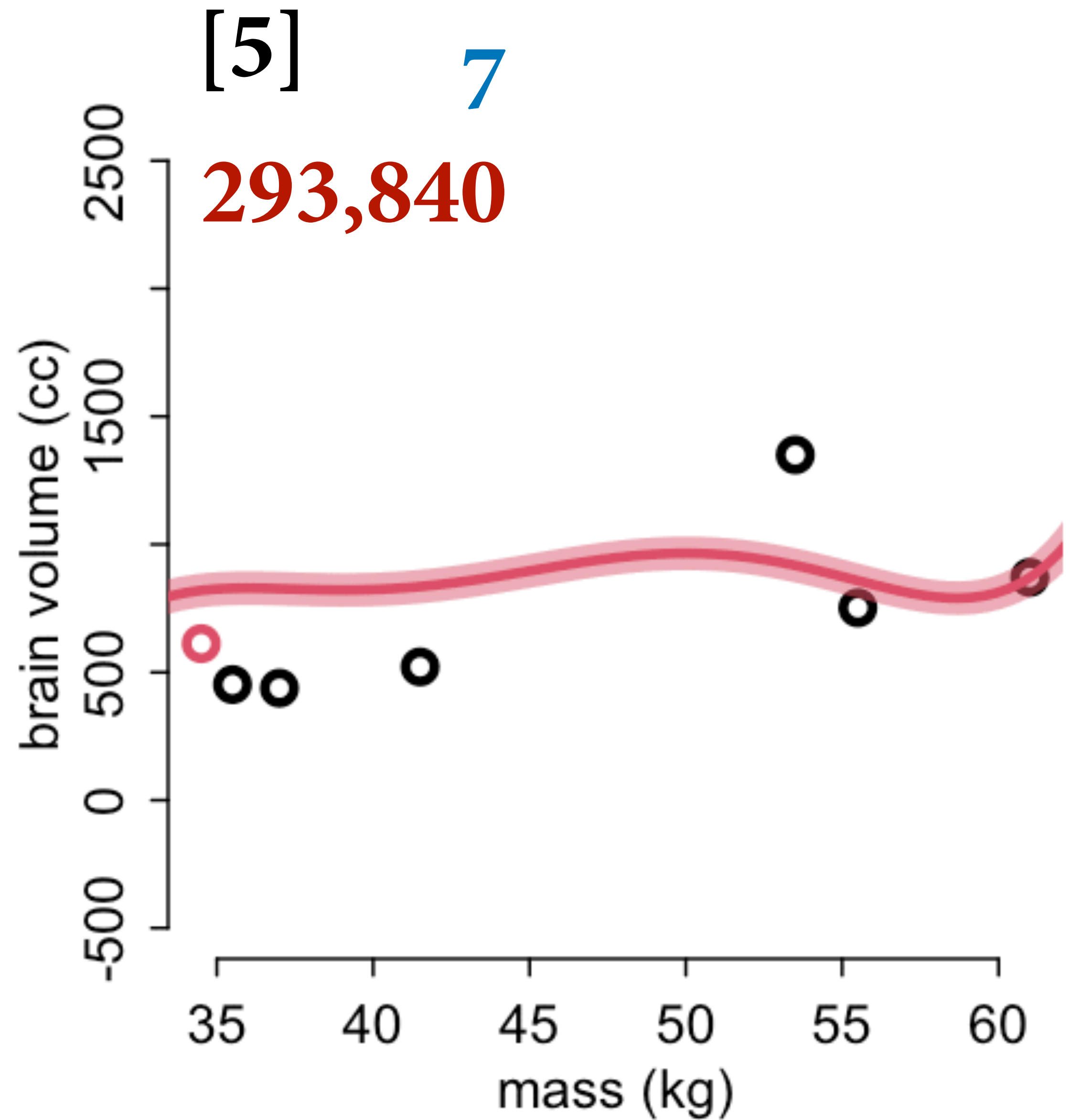
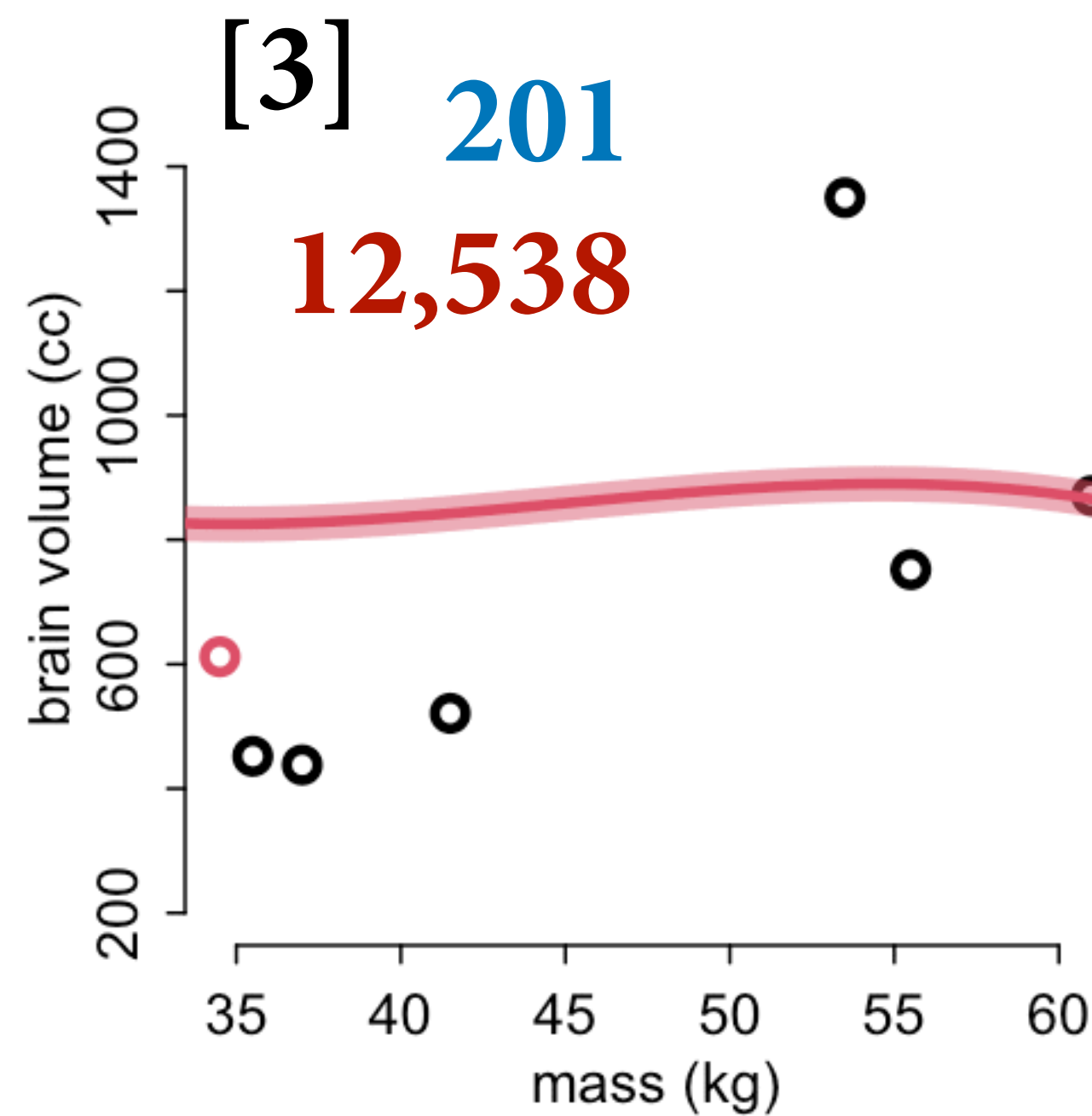
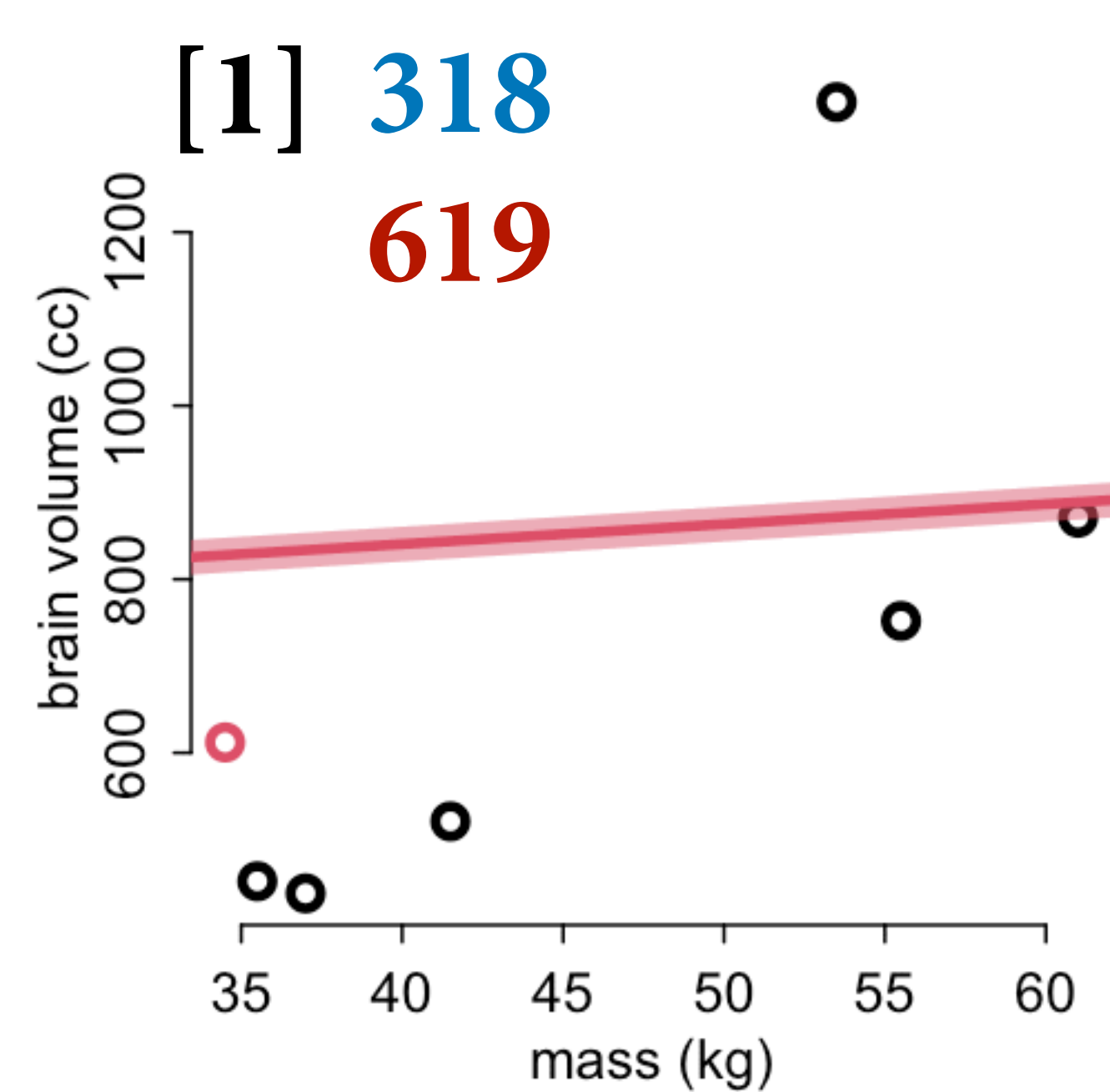


[2] In: 289

Out: 865





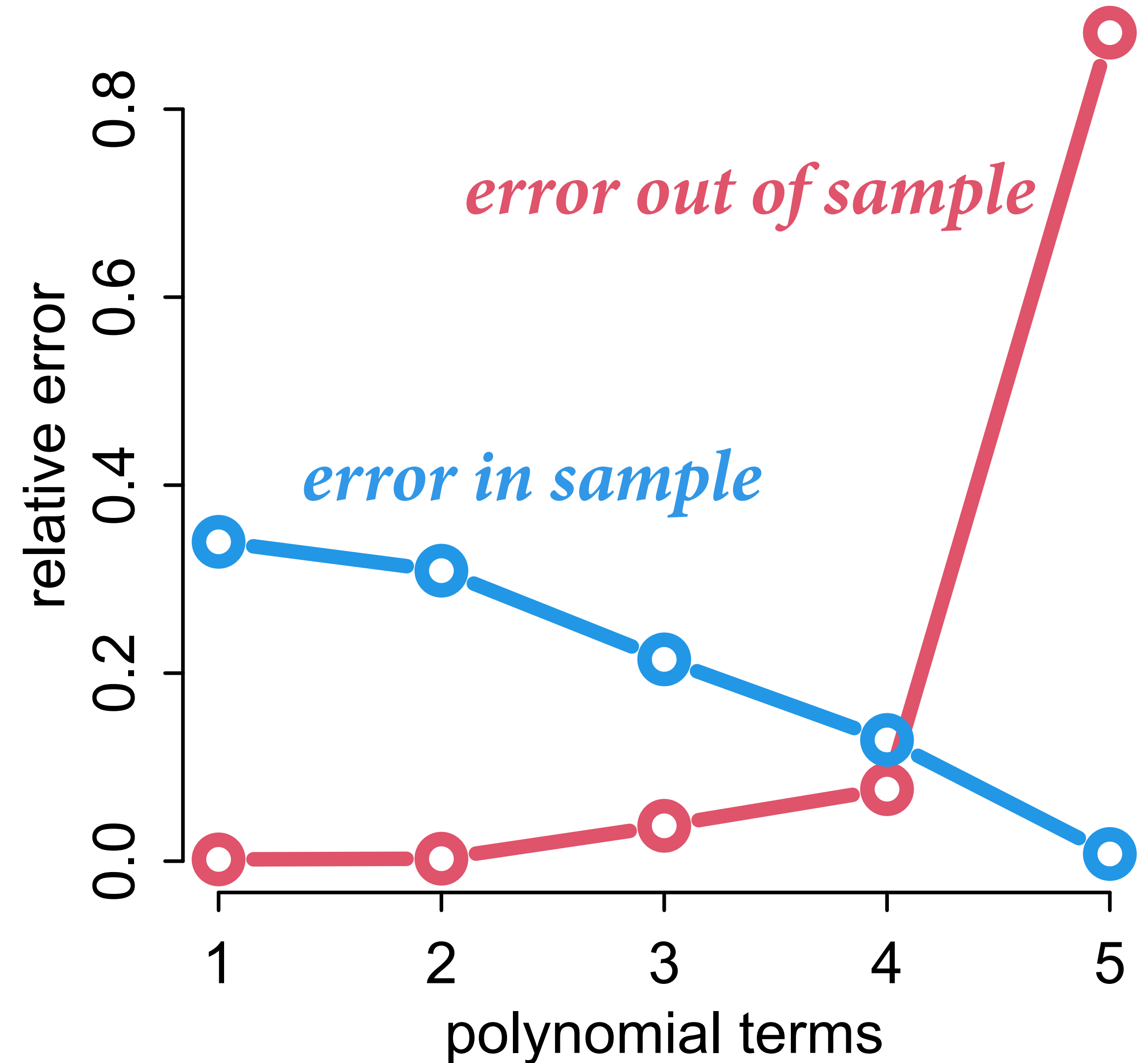


Cross-validation

For **simple** models, more parameters improves fit to sample

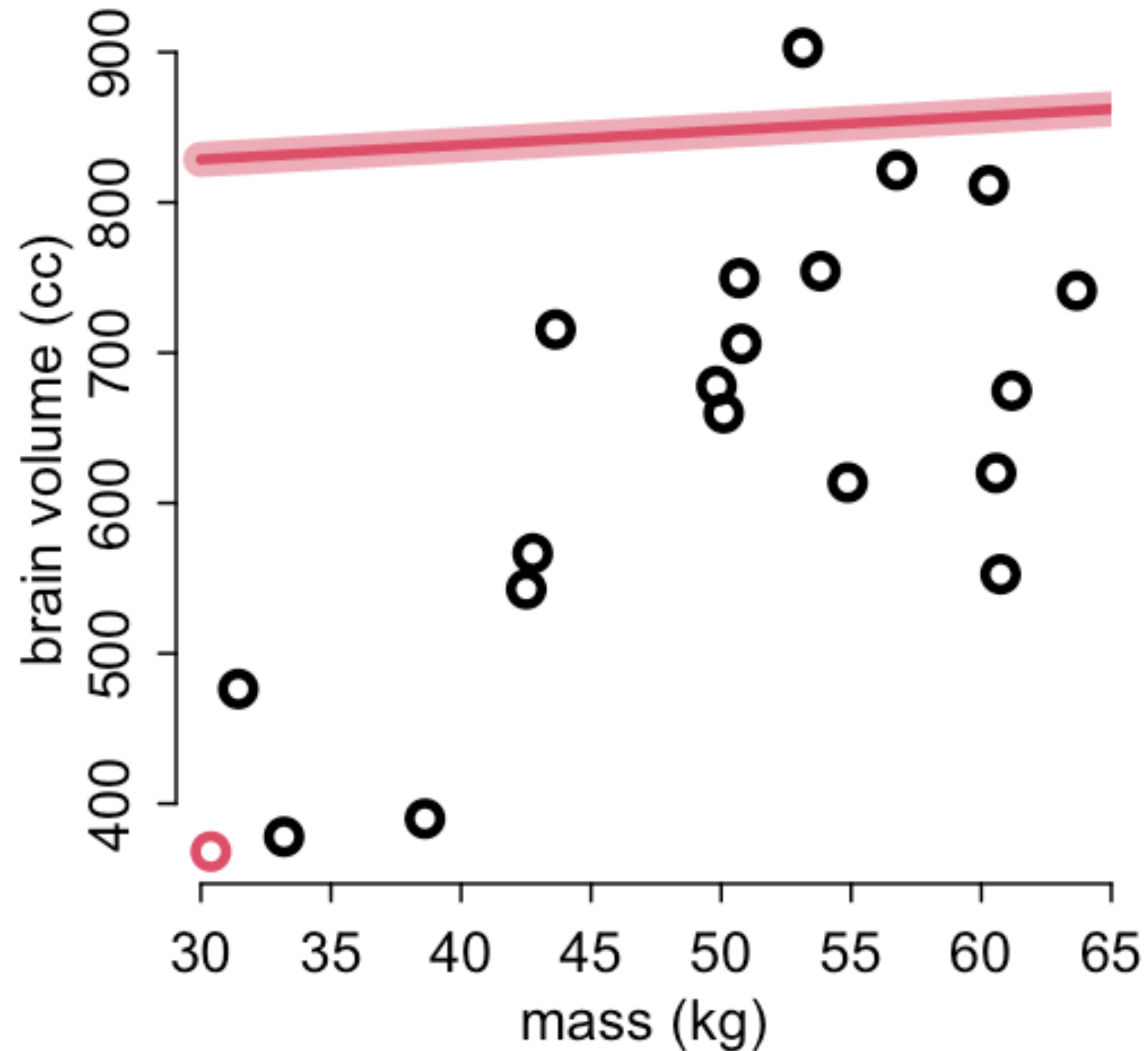
But may reduce accuracy of predictions out of sample

Most accurate model trades off flexibility with **overfitting**

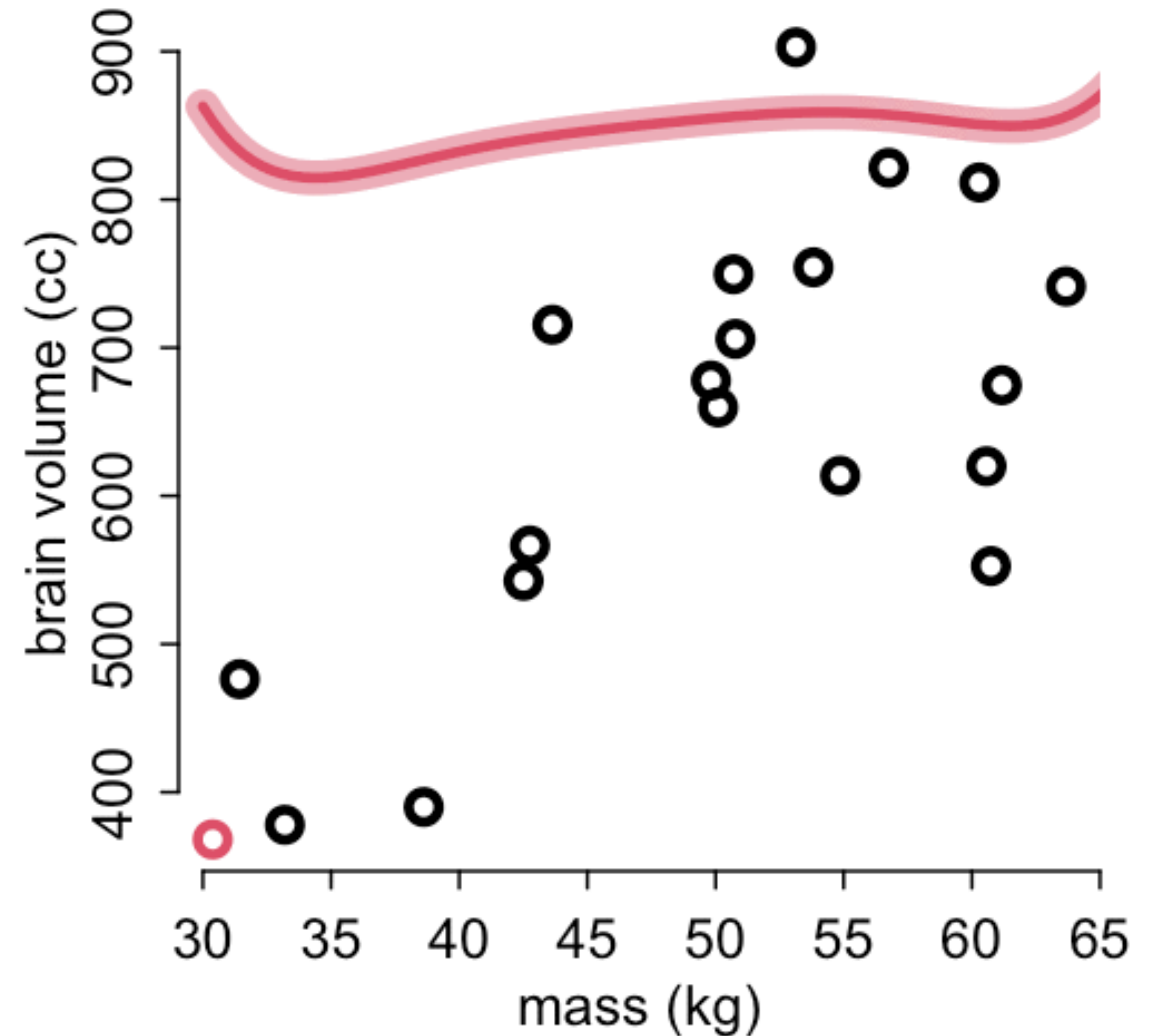


Flexibility vs overfitting tradeoffs depend on the amount of data

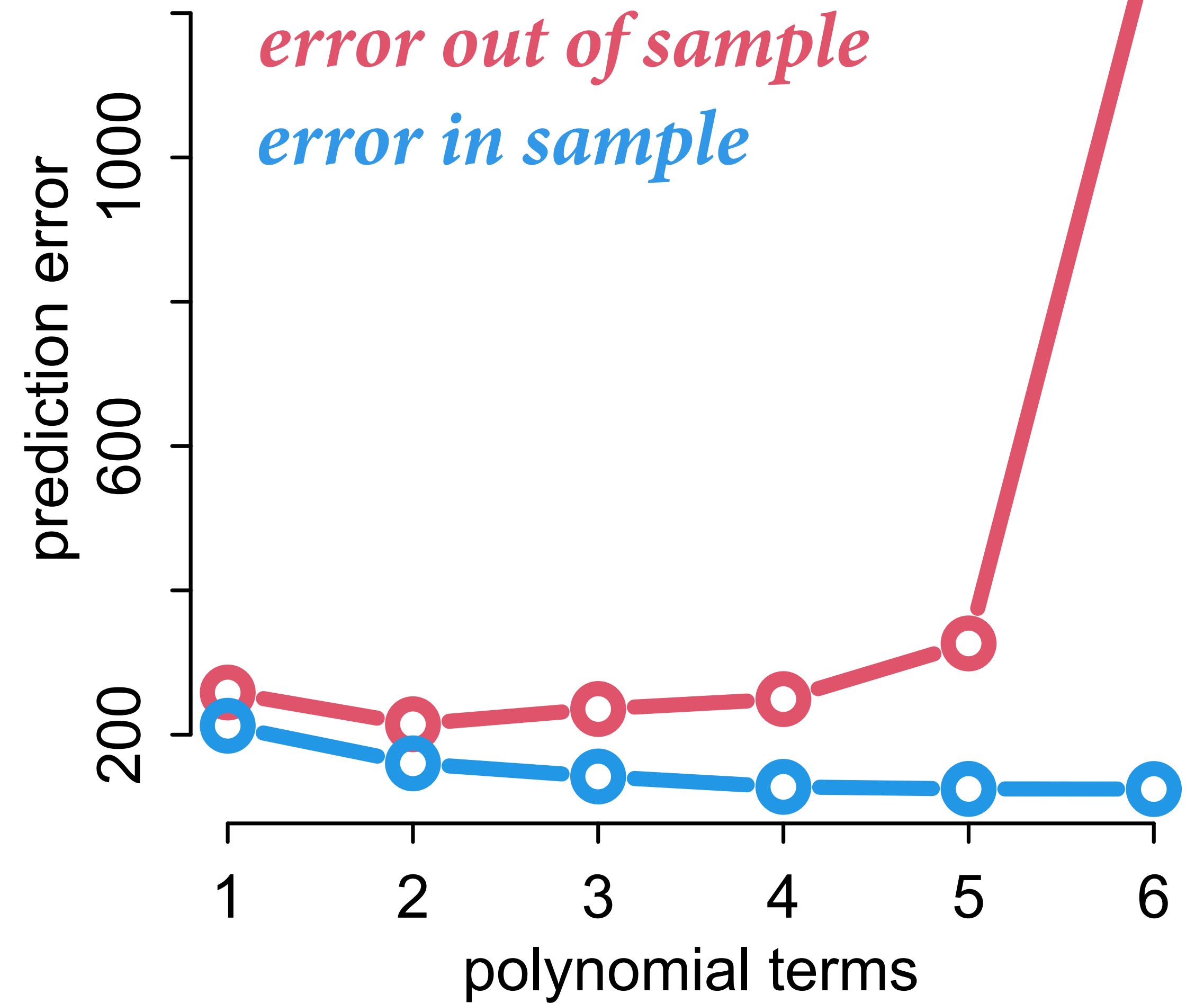
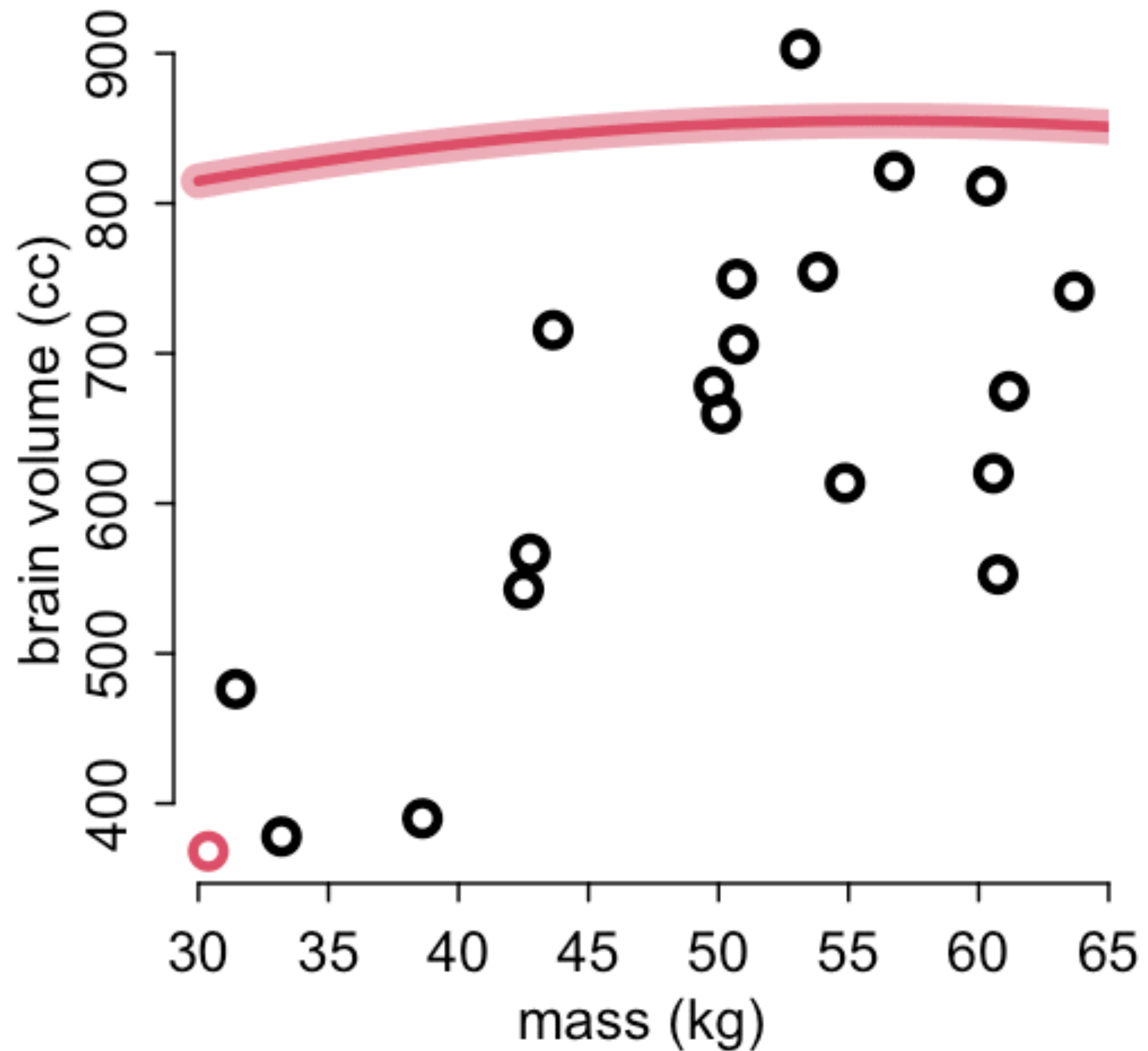
1st degree polynomial



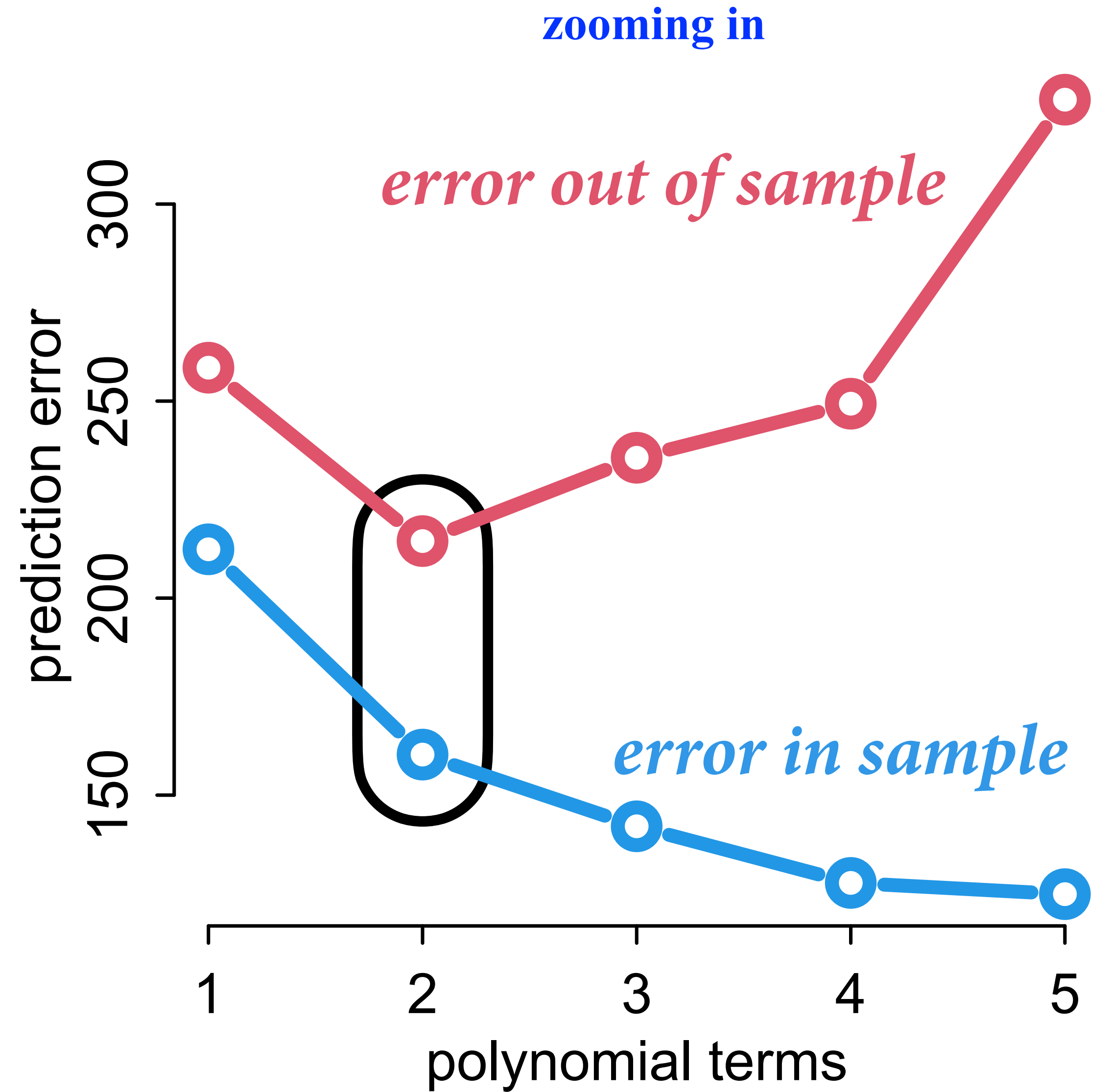
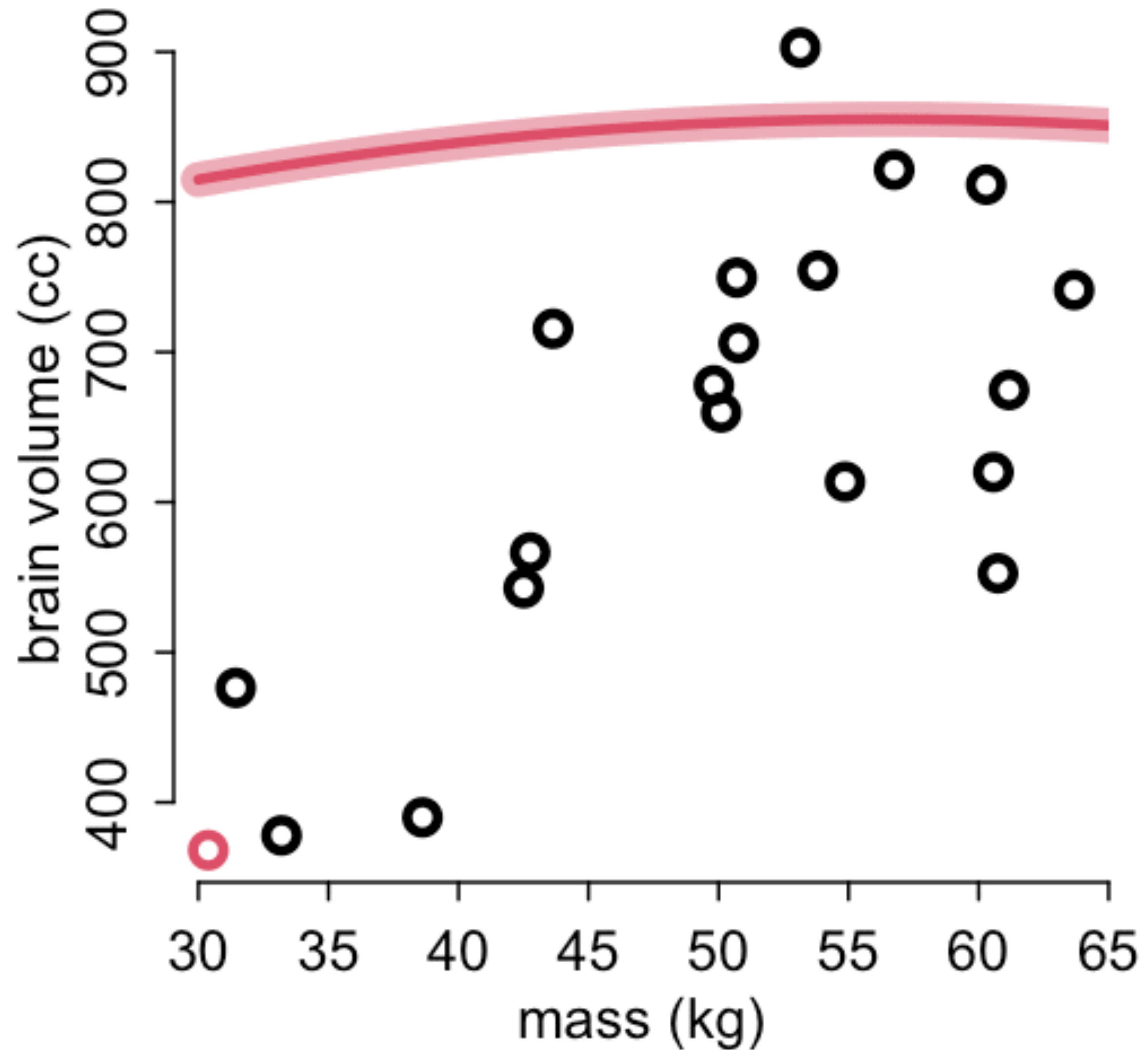
6th degree polynomial



2nd degree polynomial



2nd degree polynomial



Curves from lines

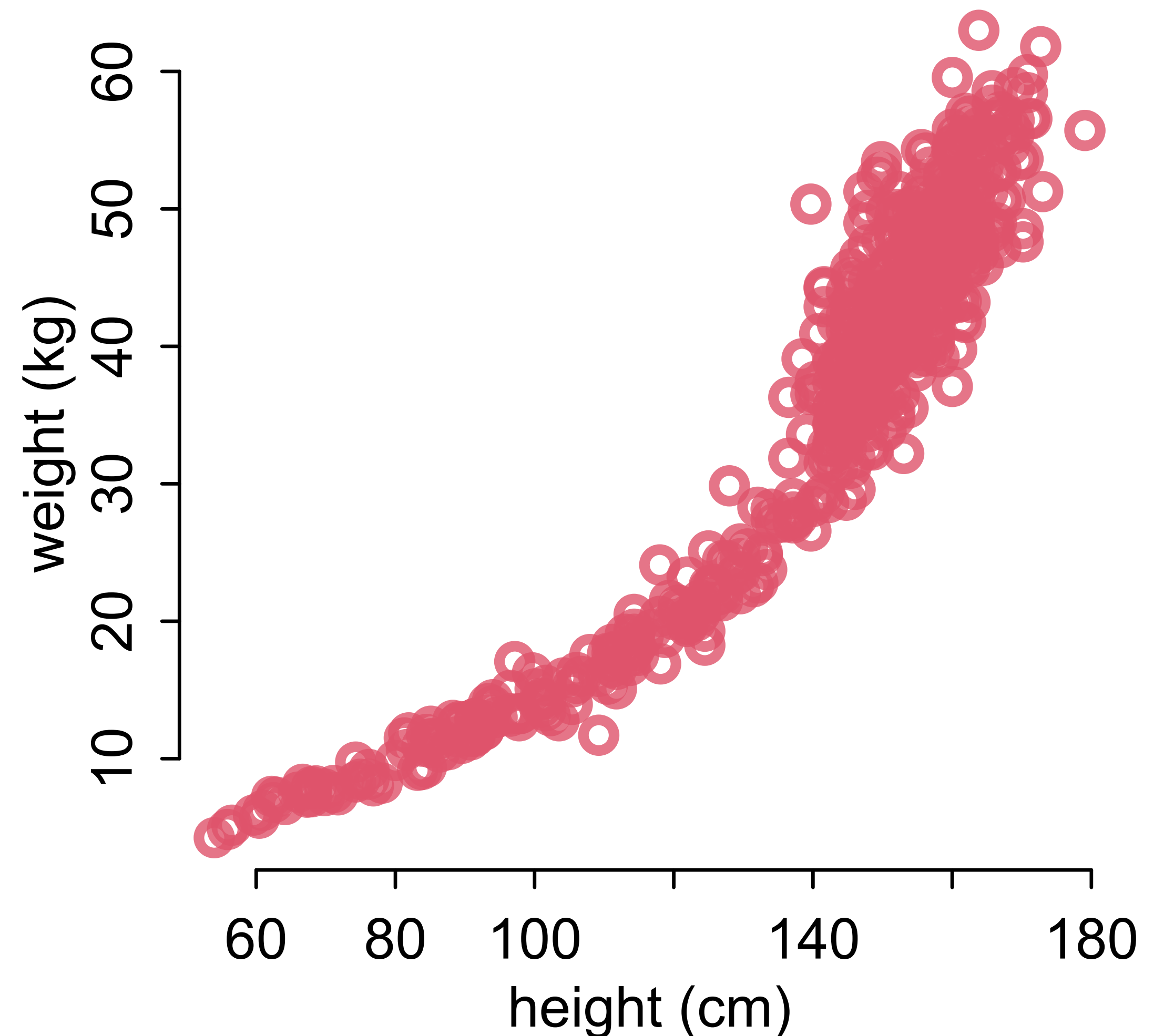
```
library(rethinking)  
data(Howell1)
```

Linear models can easily fit curves

Two popular strategies

(1) polynomials — awful

(2) splines and generalized additive models — less awful



Polynomial linear models

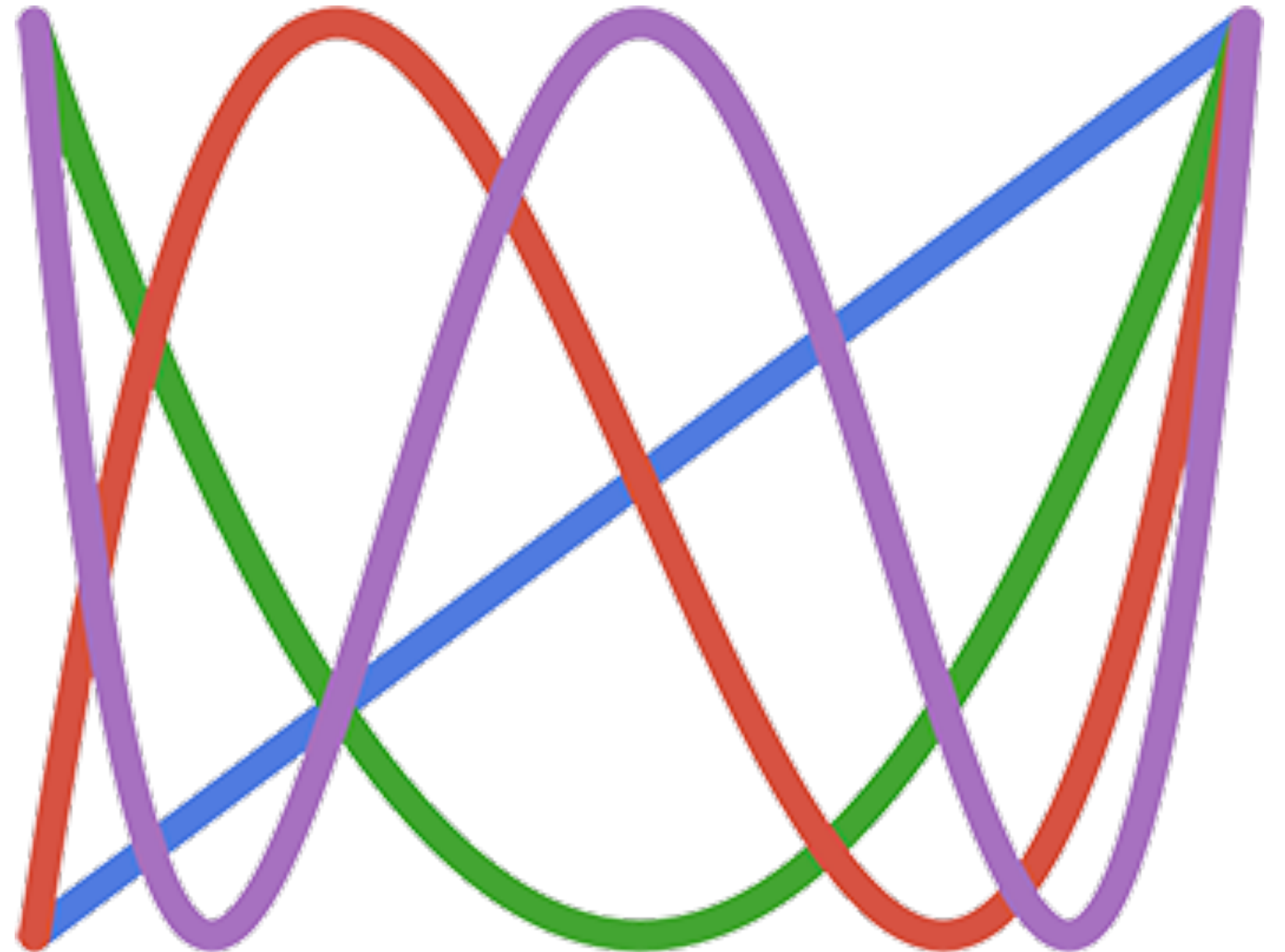
Strategy: polynomial functions

$$\mu_i = \alpha + \beta_1 x_i + \beta_2 x_i^2$$

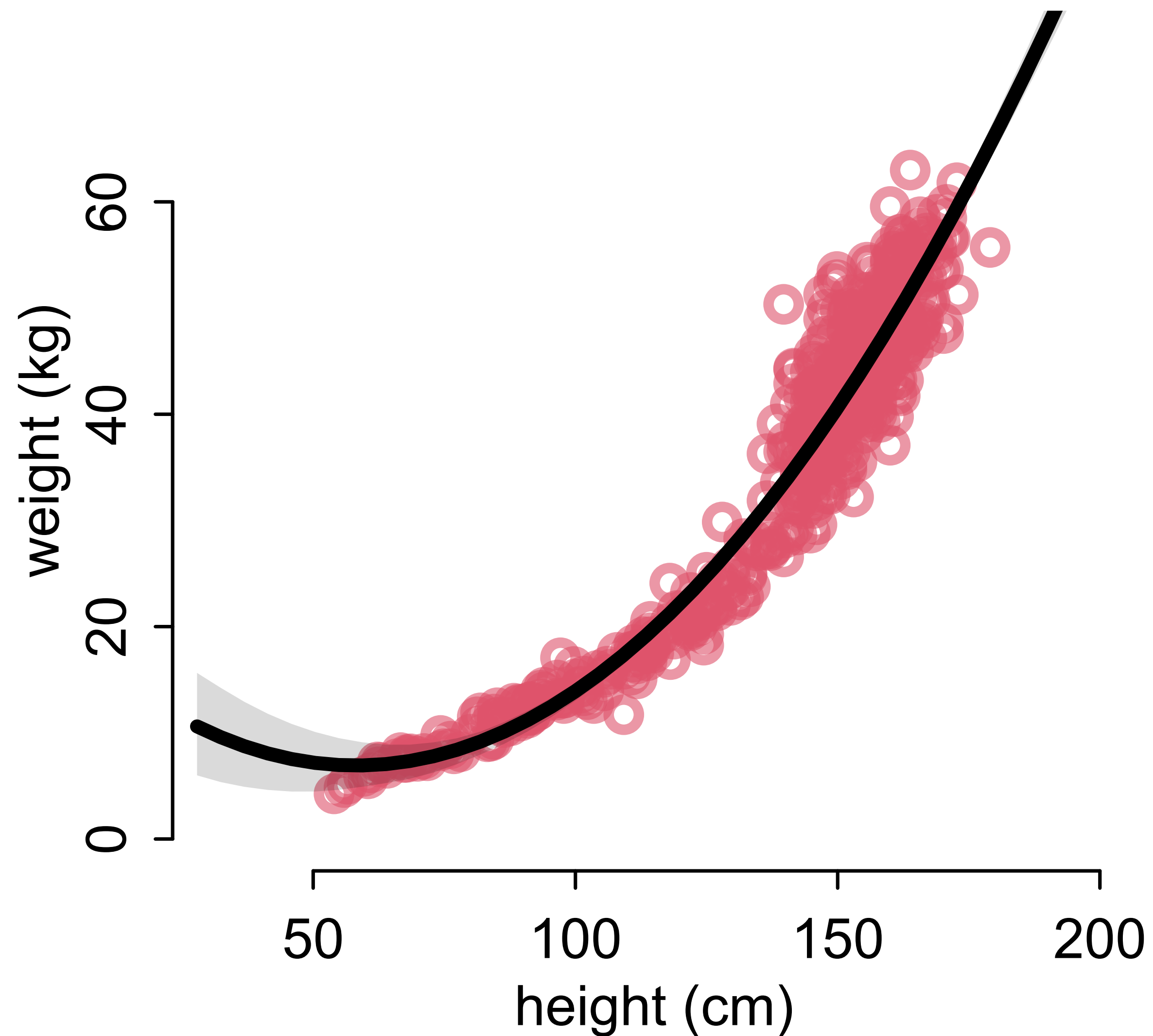
Problems: strange symmetries,
explosive uncertainty at edges

No *local* smoothing; only *global*

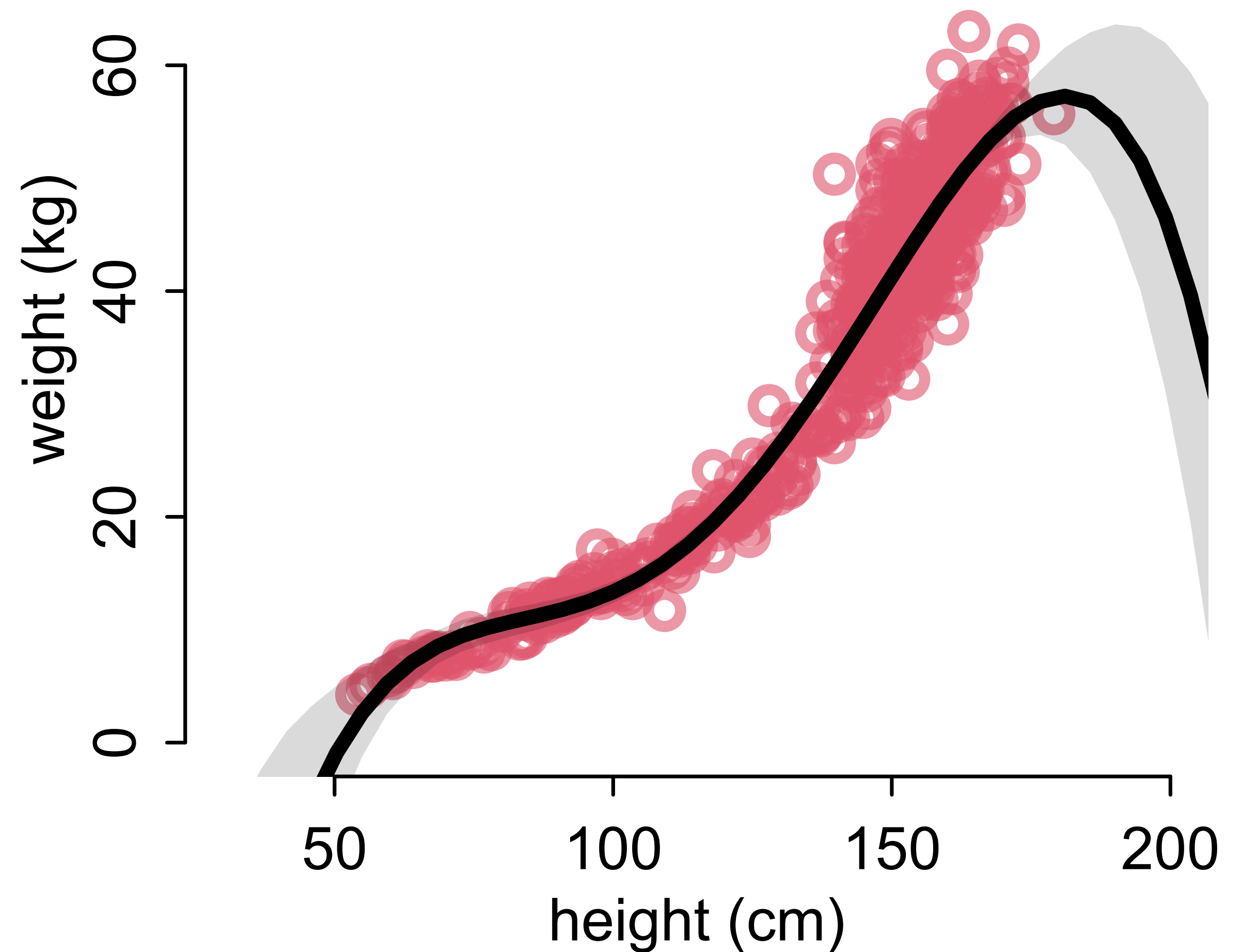
DO NOT USE



$$\mu_i = \alpha + \beta_1 H_i + \beta_2 H_i^2$$



$$\begin{aligned} \mu_i = & \alpha + \beta_1 H_i + \beta_2 H_i^2 \\ & + b_3 H^3 + b_4 H^4 \end{aligned}$$

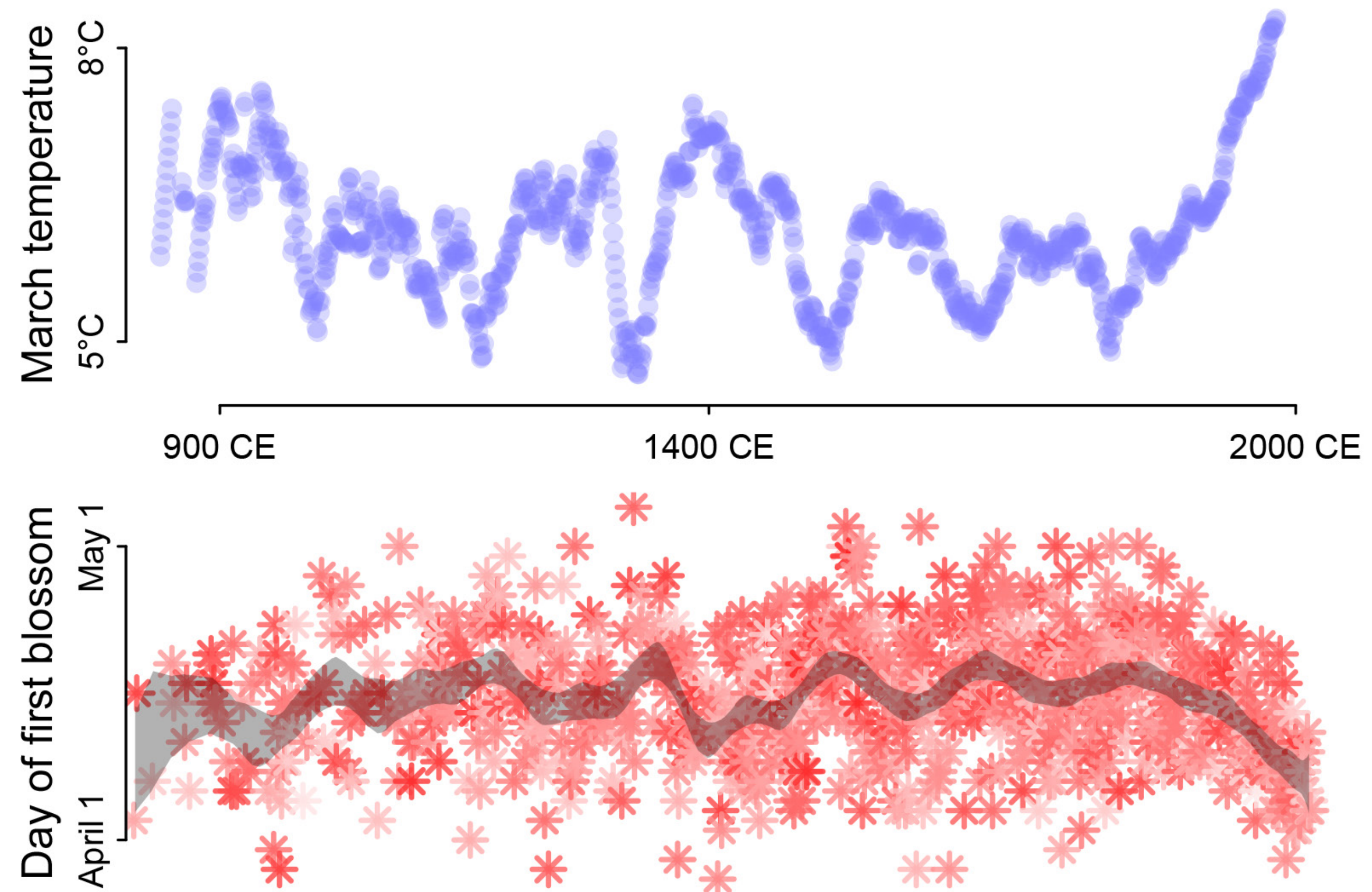


Splines

Basis-Splines: Wiggly function built from many local functions

Basis function: A local function

Local functions make splines better than polynomials, but equally geocentric



Regularization

Overfitting depends upon the priors

Skeptical priors have tighter variance, reduce flexibility

Regularization: Function finds regular features of process

Good priors are often tighter than you think!



1 skeptischer Hamster zu verkaufen

20 €

📍 25899 Niebüll >

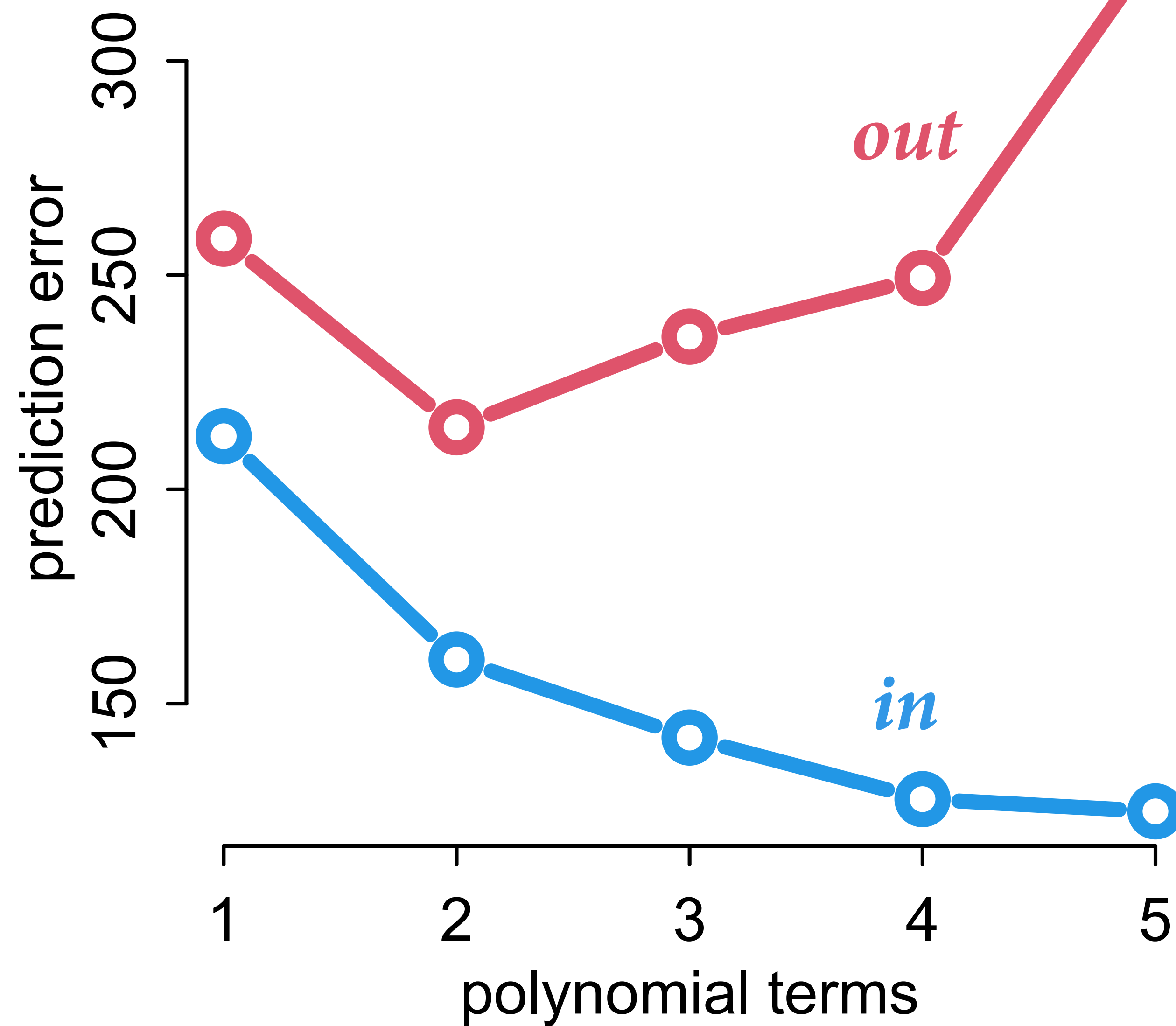
Art

Hamster

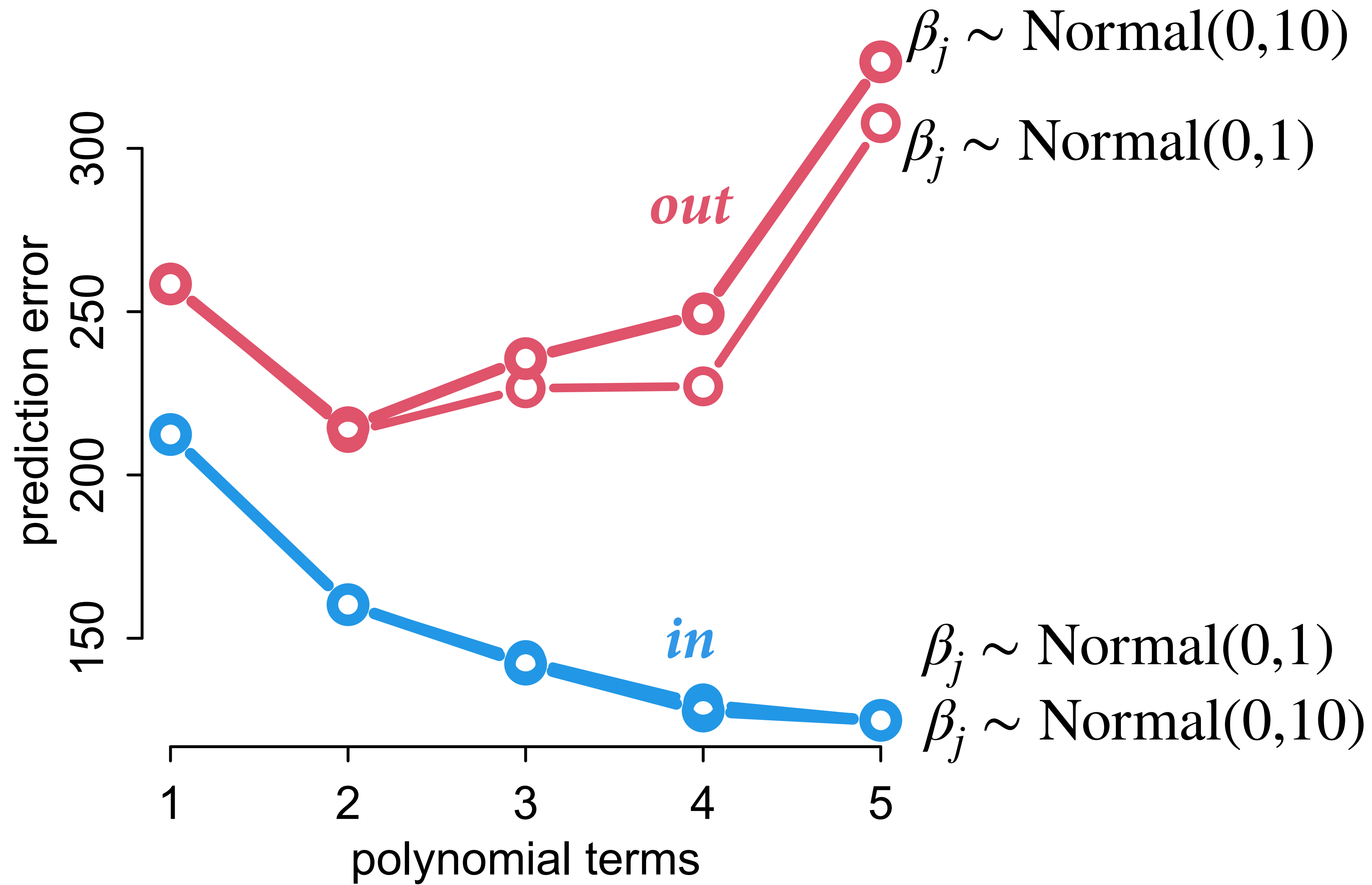
Er guckt einen skeptisch an, als würde man nichts richtig machen.

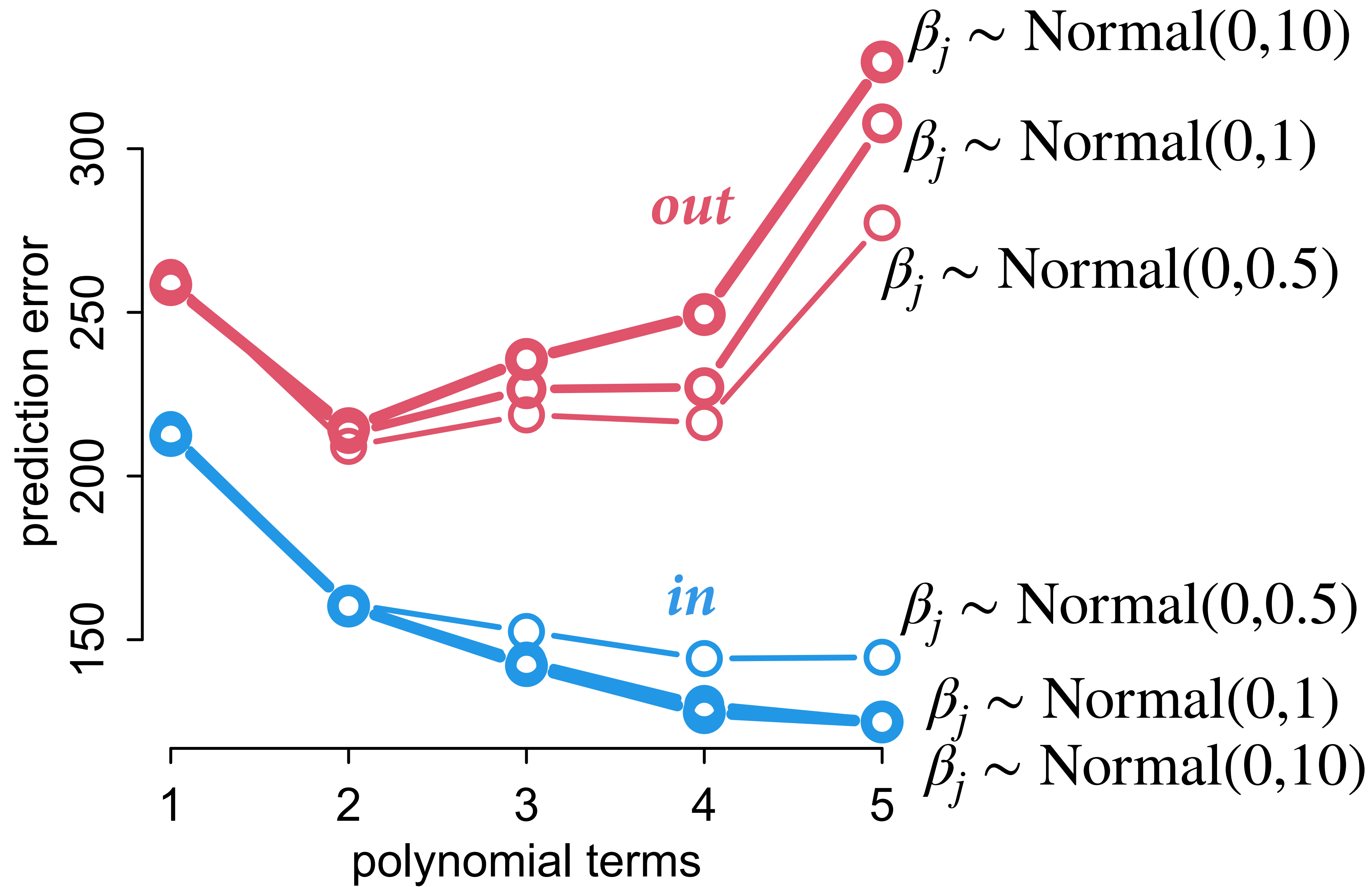
Es macht mich wahnsinnig, ich kann diesen vorwurfsvollen Blick nicht länger ertragen.

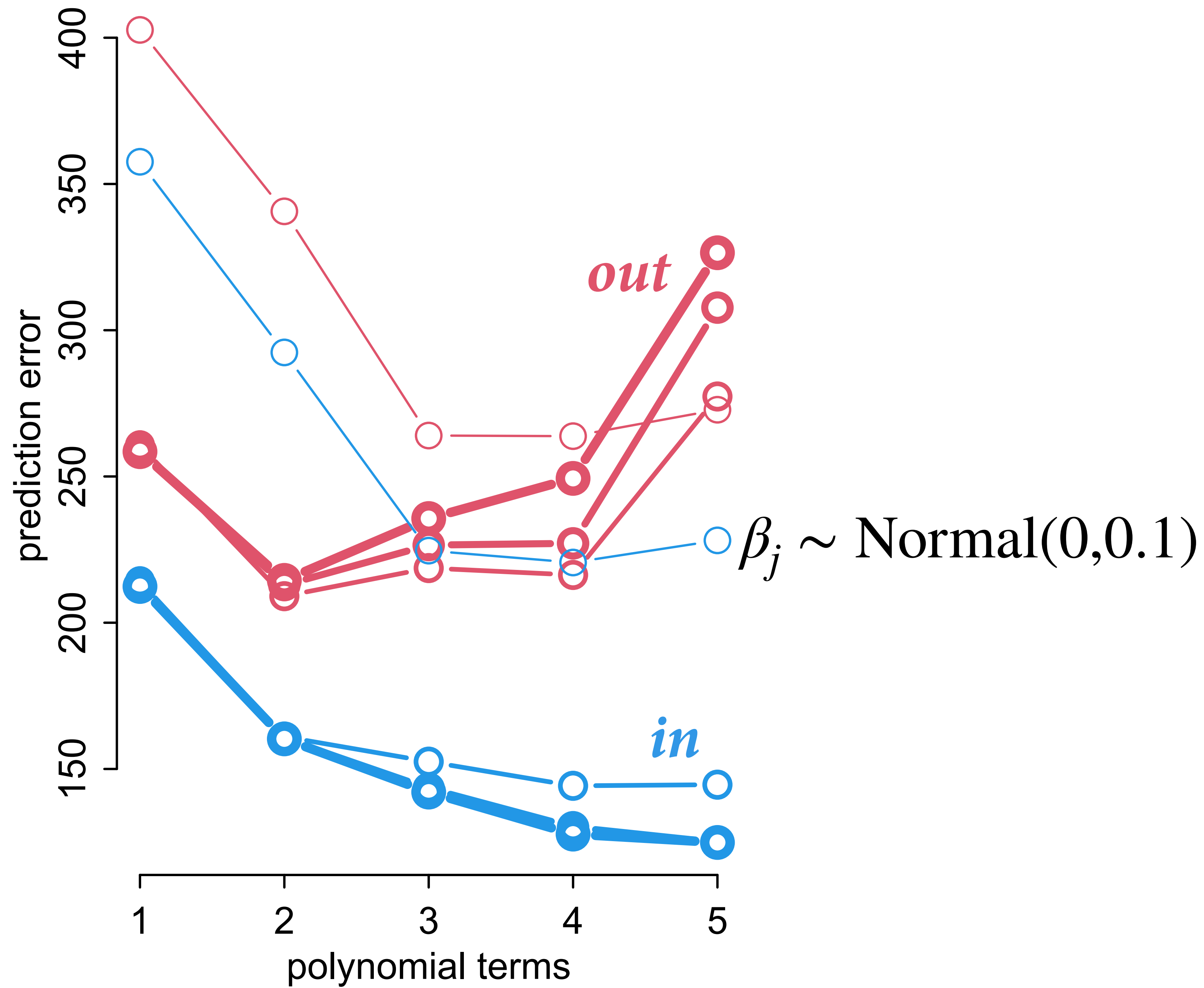
Sein Name ist Olaf.



$$\mu_i = \alpha + \sum_{j=1}^m \beta_j x_i^j$$
$$\beta_j \sim \text{Normal}(0, 10)$$







Regularizing priors

How to choose width of prior?

For **causal inference**, use science

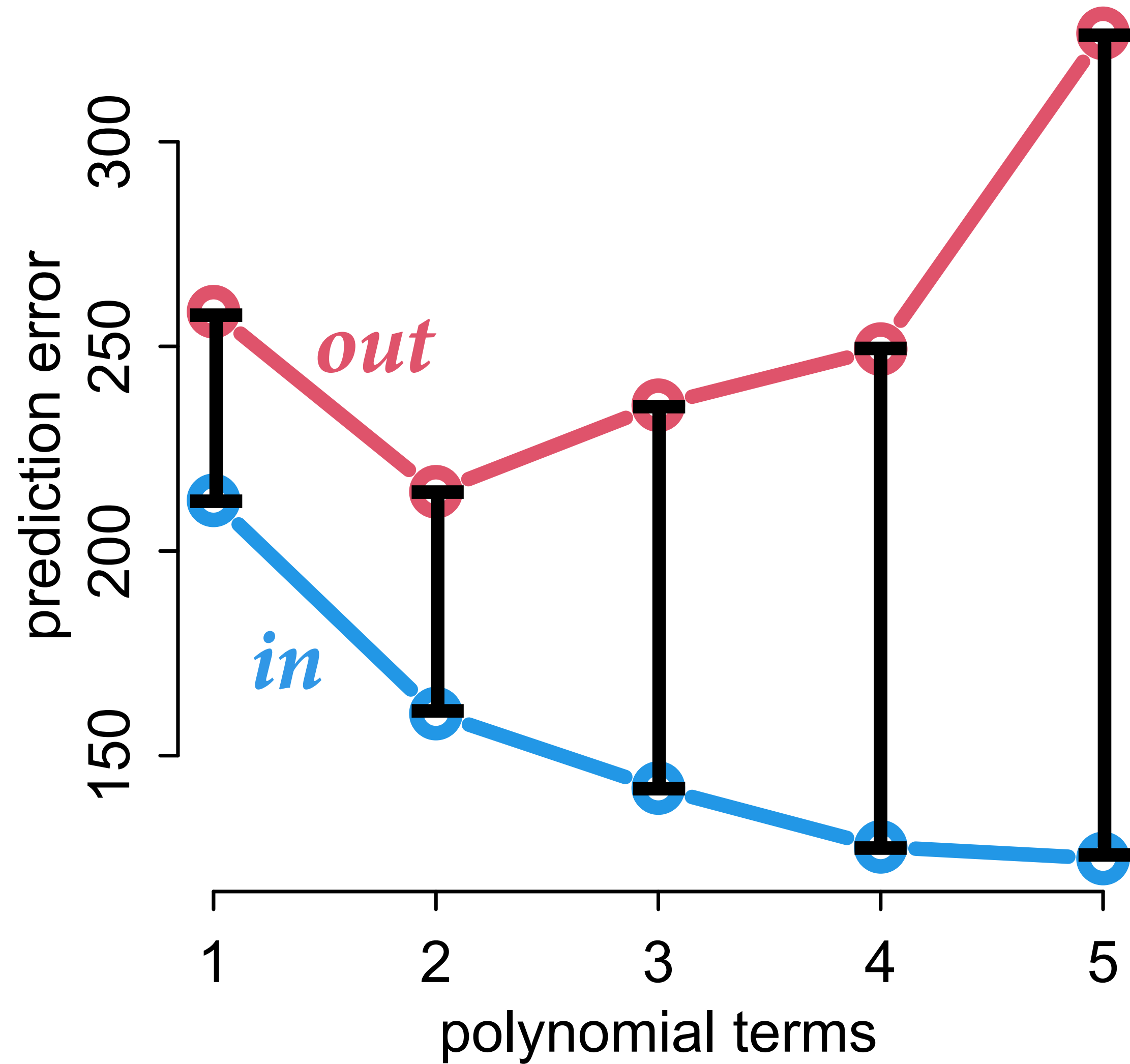
For **pure prediction**, can tune the prior using cross-validation

Many tasks are a mix of inference and prediction

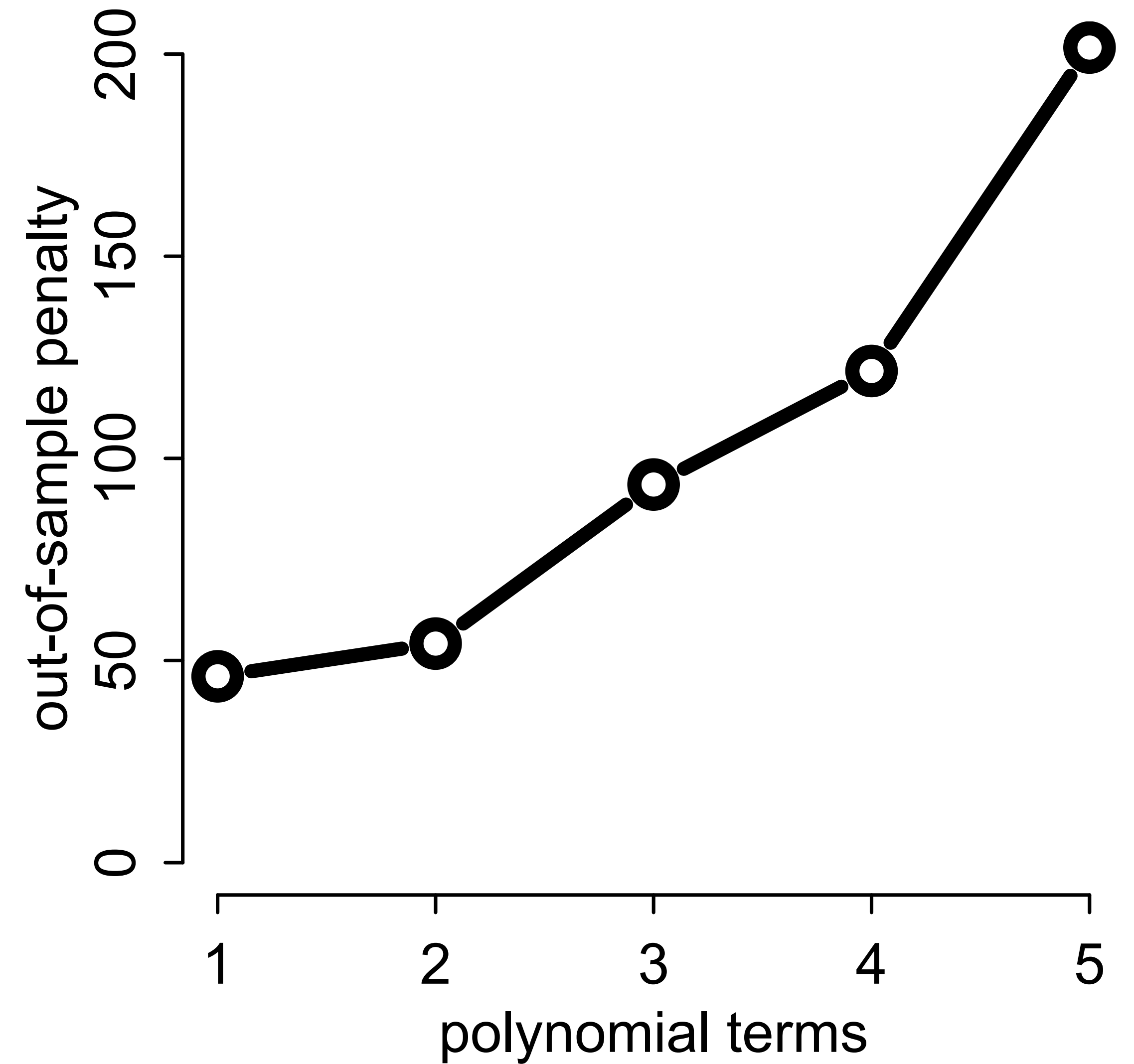
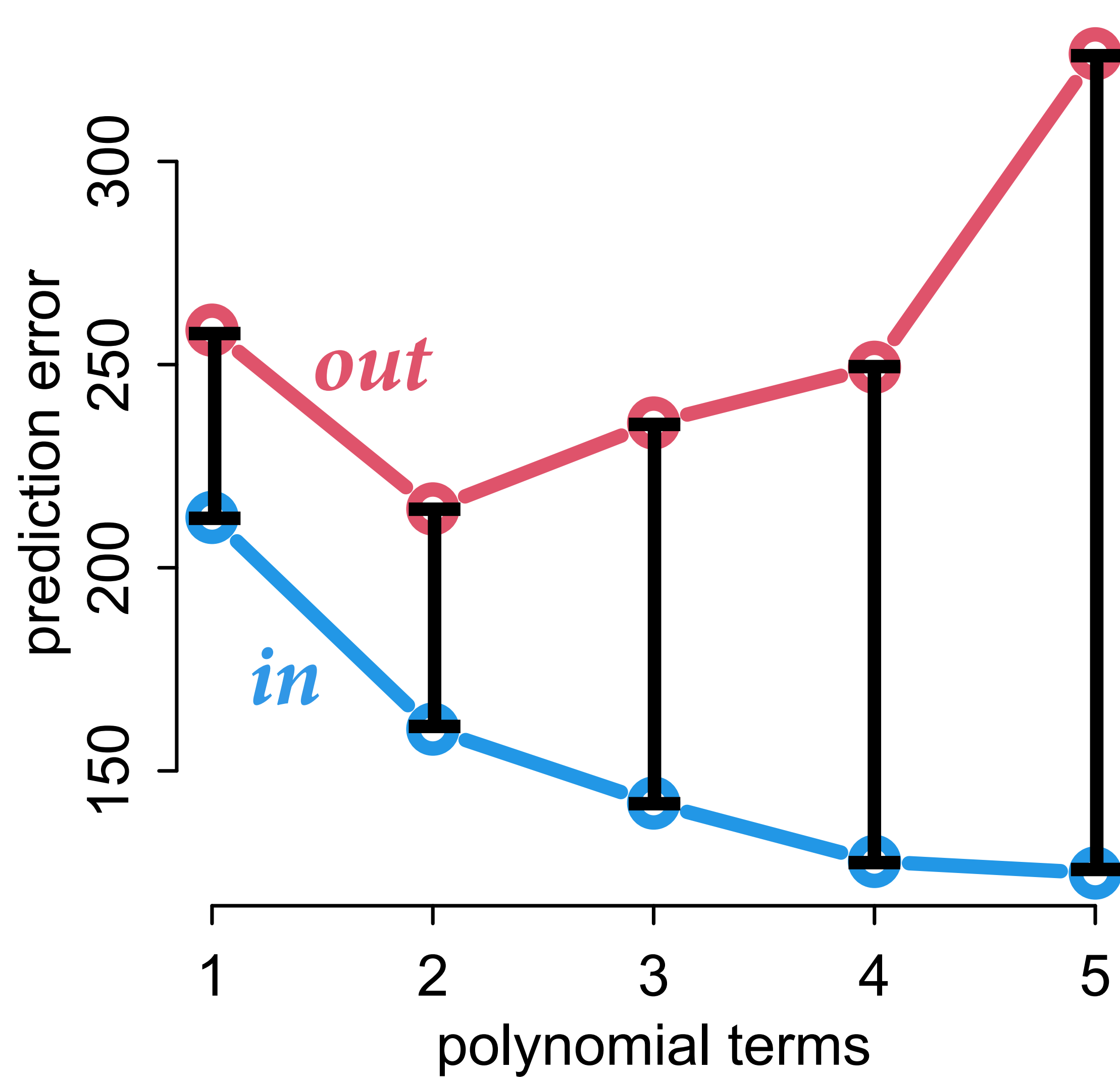
No need to be perfect, just better



Prediction penalty



Prediction penalty



Penalty prediction

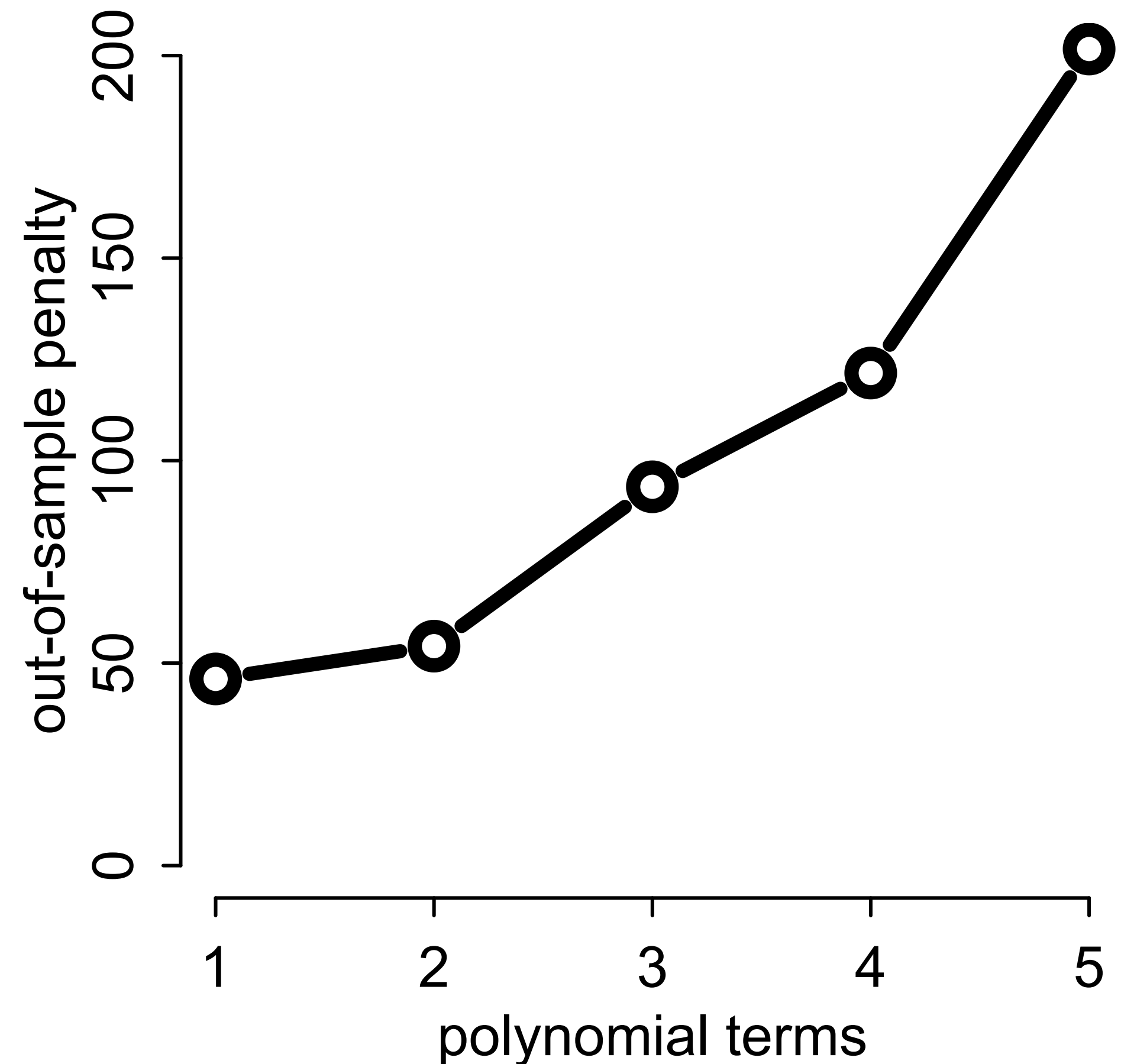
For N points, cross-validation requires fitting N models

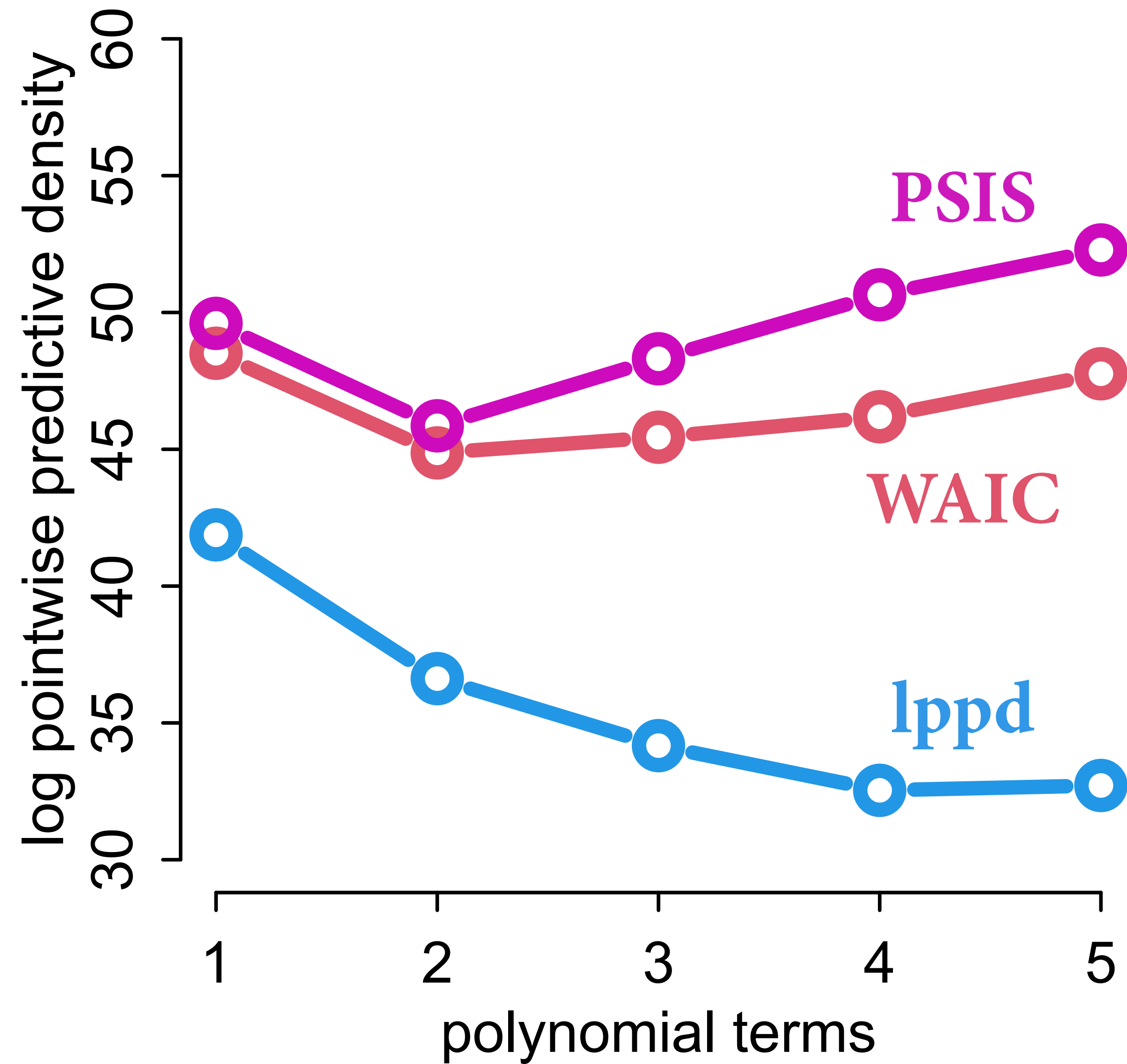
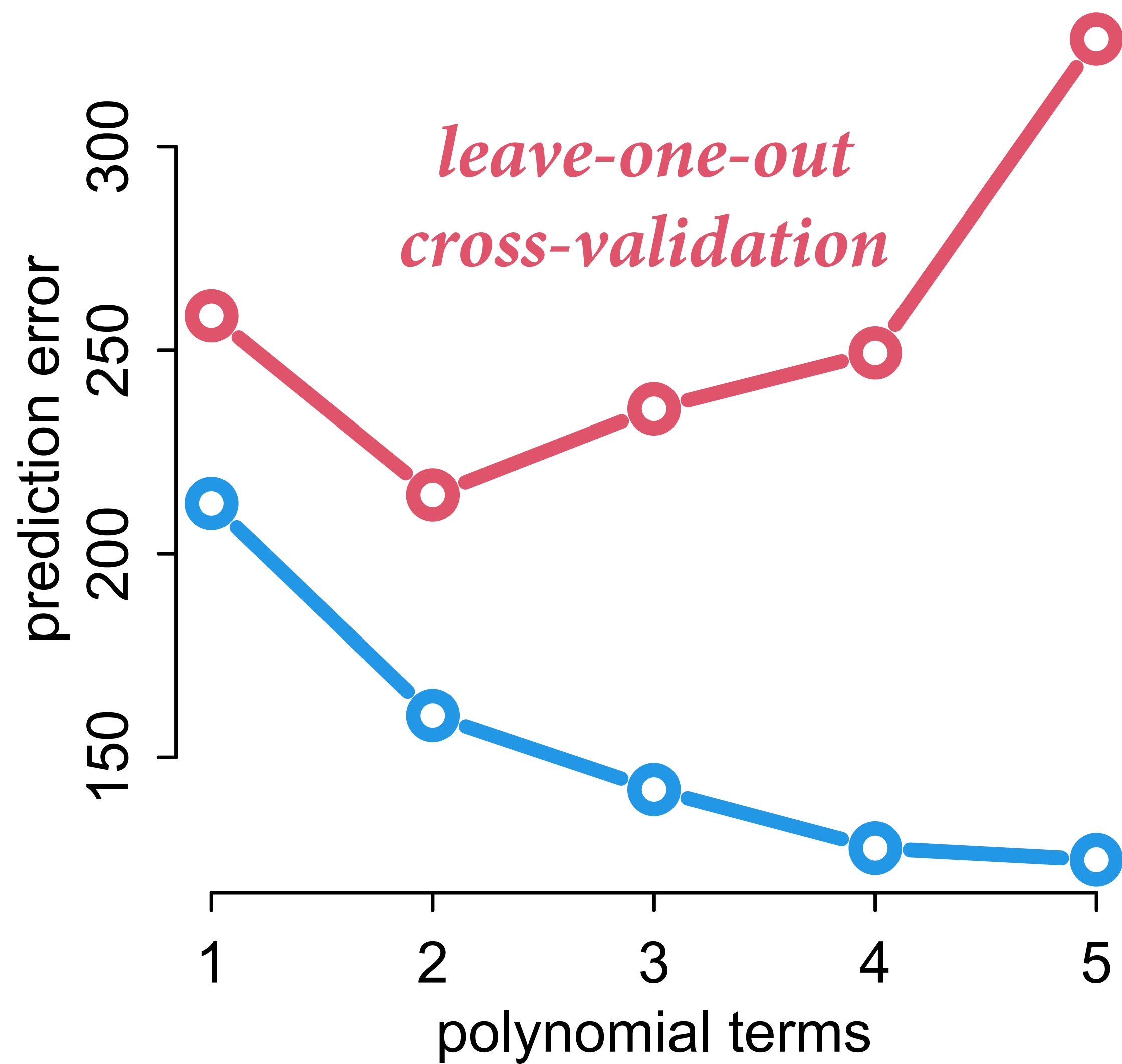
What if you could estimate the penalty from a single model fit?

Good news! You can:

Importance sampling (PSIS)

Information criteria (WAIC)





WAIC, PSIS, CV measure overfitting

Regularization manages overfitting

None directly address causal inference

Important for understanding statistical inference



Overfit



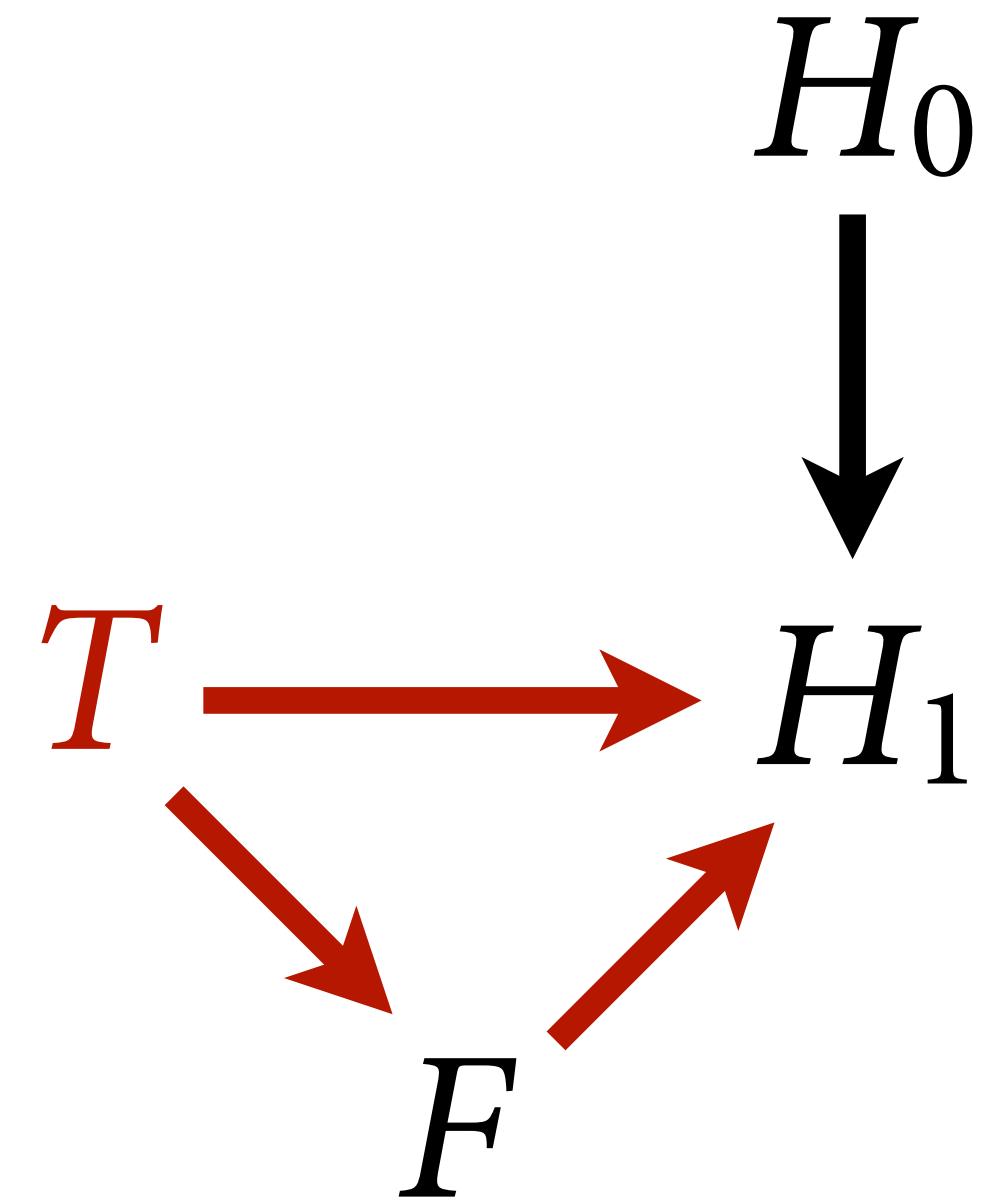
Underfit

Model Mis-selection

Do not use predictive criteria (WAIC, PSIS, CV) to choose a causal estimate

Predictive criteria actually prefer confounds & colliders

Example: Plant growth experiment

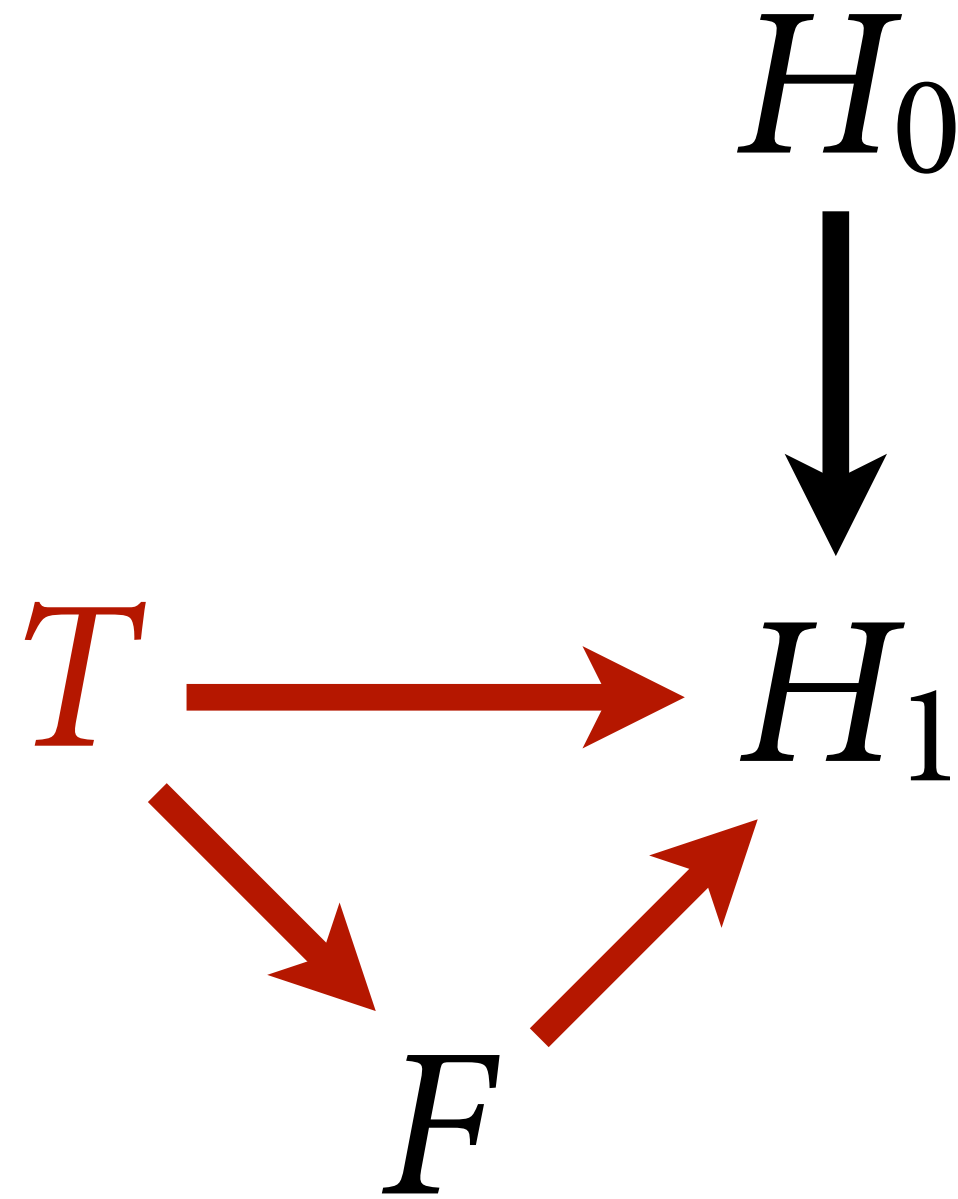


$$H_1 \sim \text{Normal}(\mu_i, \sigma)$$

$$\mu_i = H_0 \times p_i$$

$$p_i = \alpha + \beta_T T_i + \beta_F F_i$$

*Wrong adjustment set
for total causal effect of
treatment (blocks
mediating path)*



$$H_1 \sim \text{Normal}(\mu_i, \sigma)$$

$$\mu_i = H_0 \times p_i$$

$$p_i = \alpha + \beta_T T_i$$

*Correct adjustment set for
total causal effect of
treatment*

$$H_1 \sim \text{Normal}(\mu_i, \sigma)$$

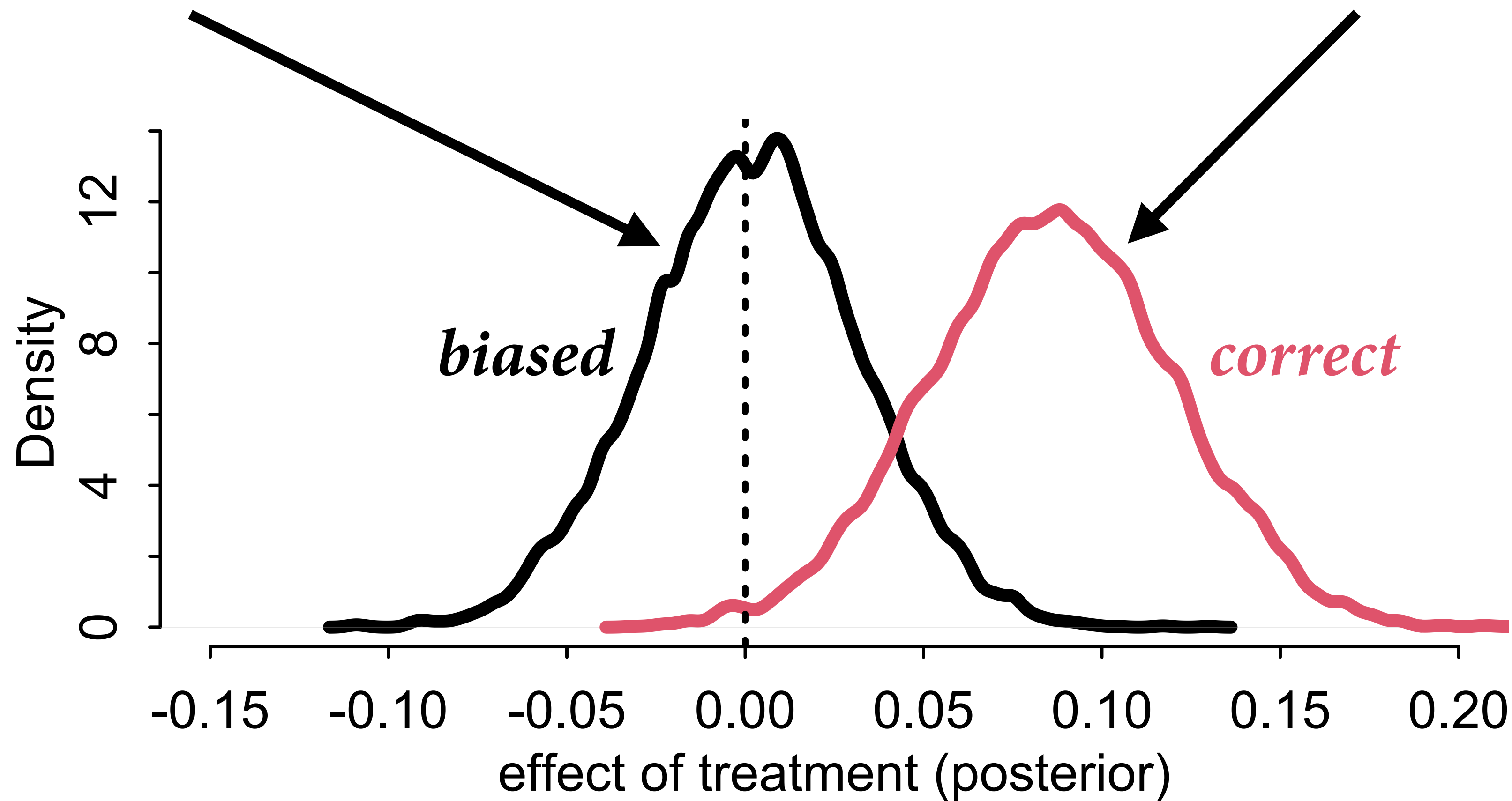
$$\mu_i = H_0 \times p_i$$

$$p_i = \alpha + \beta_T T_i + \beta_F F_i$$

$$H_1 \sim \text{Normal}(\mu_i, \sigma)$$

$$\mu_i = H_0 \times p_i$$

$$p_i = \alpha + \beta_T T_i$$



$$H_1 \sim \text{Normal}(\mu_i, \sigma)$$

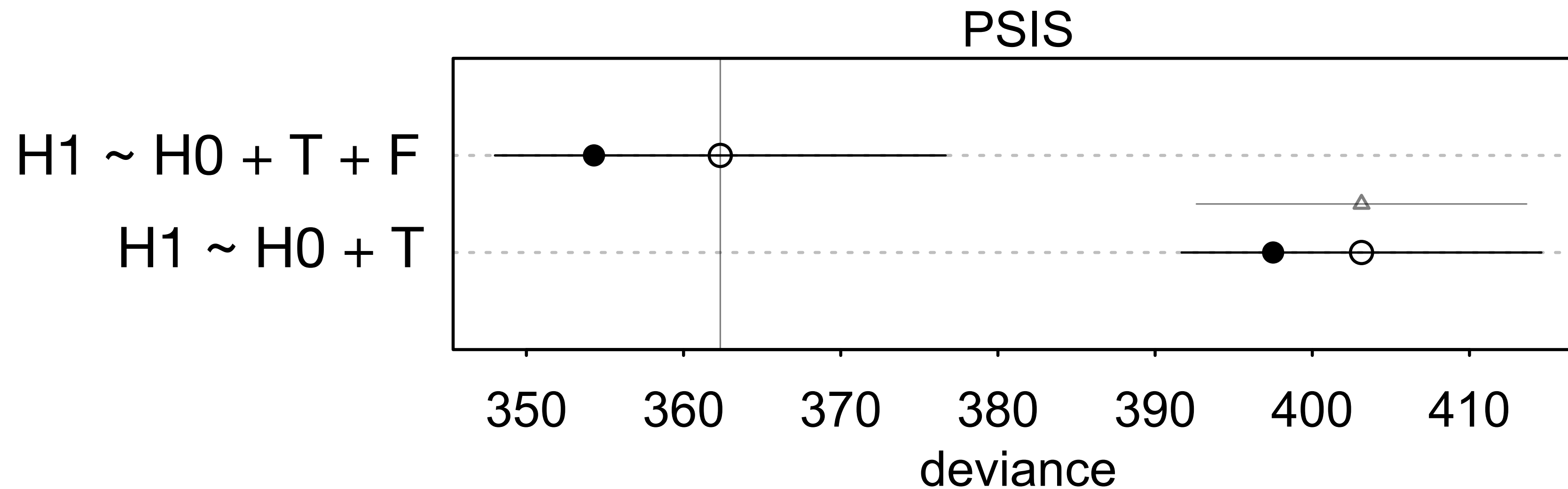
$$\mu_i = H_0 \times p_i$$

$$p_i = \alpha + \beta_T T_i + \beta_F F_i$$

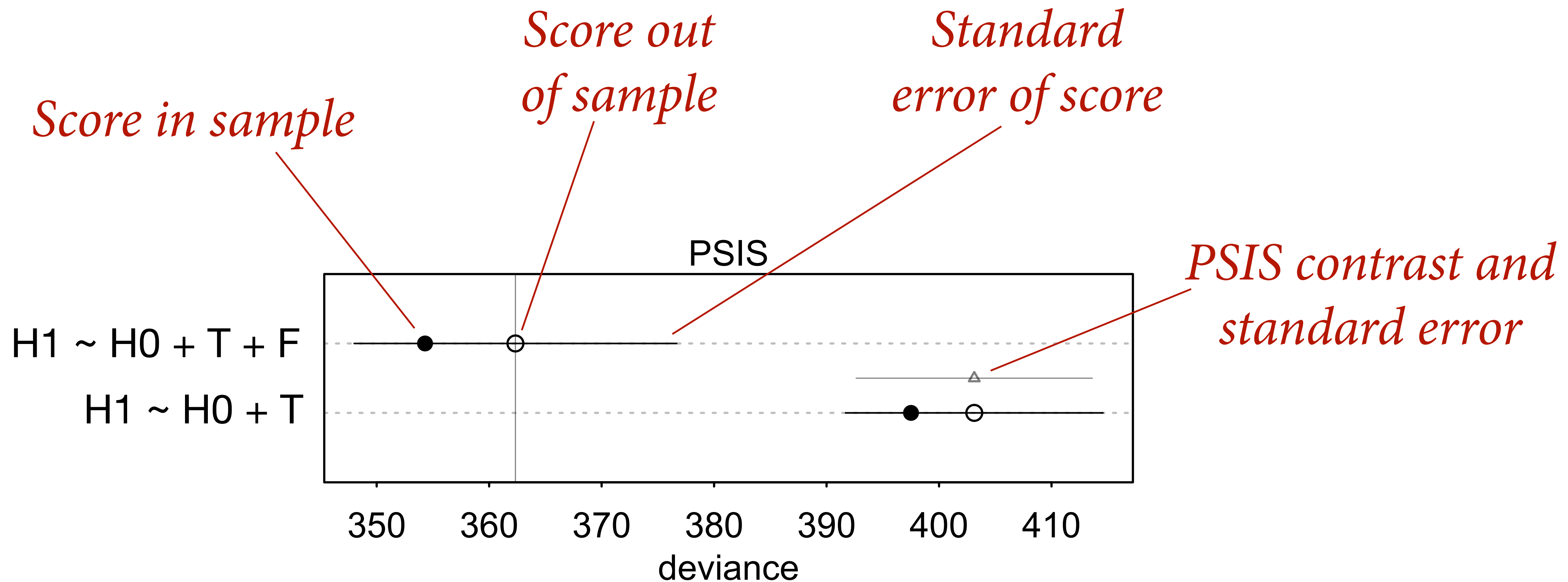
$$H_1 \sim \text{Normal}(\mu_i, \sigma)$$

$$\mu_i = H_0 \times p_i$$

$$p_i = \alpha + \beta_T T_i$$

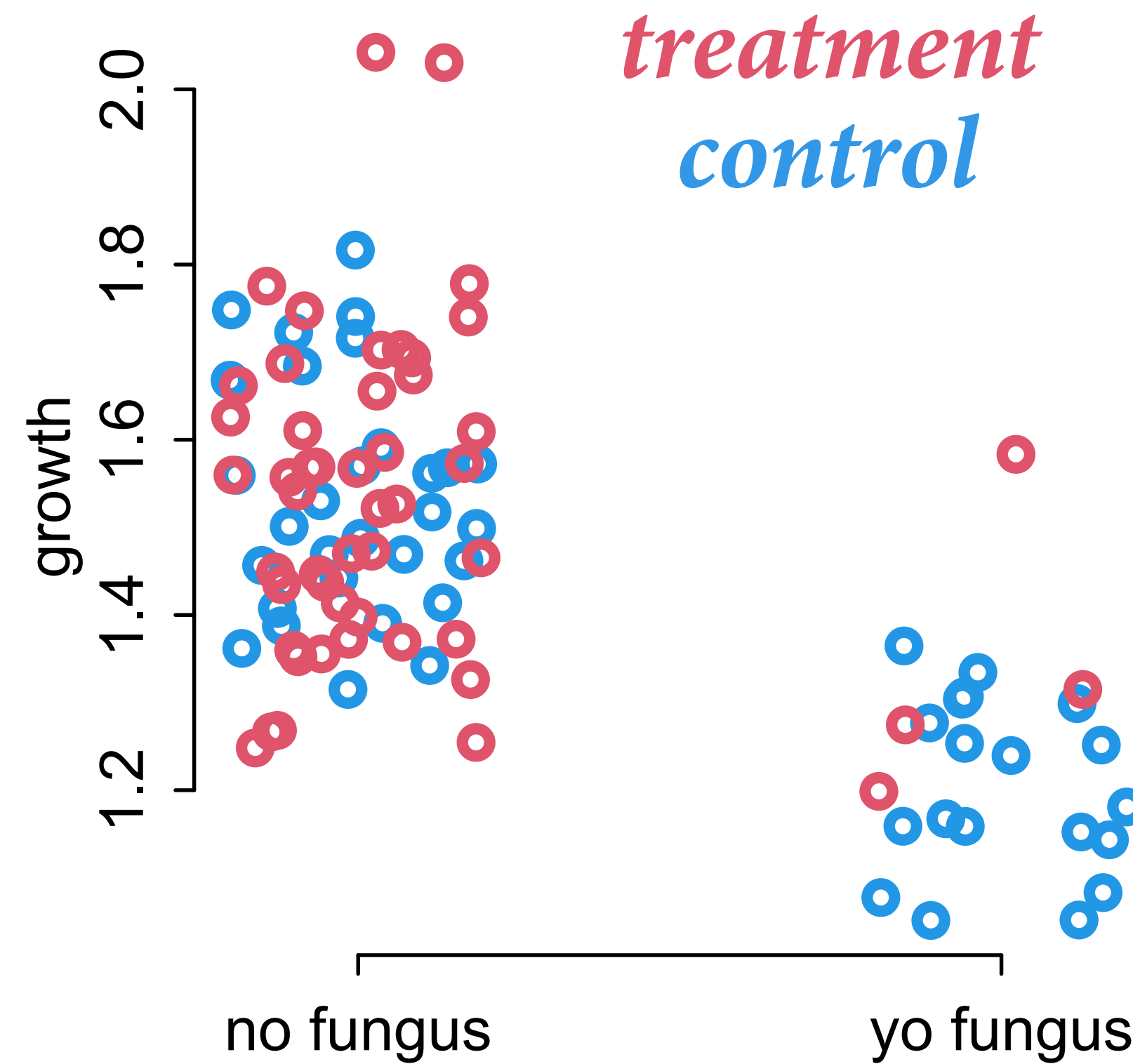
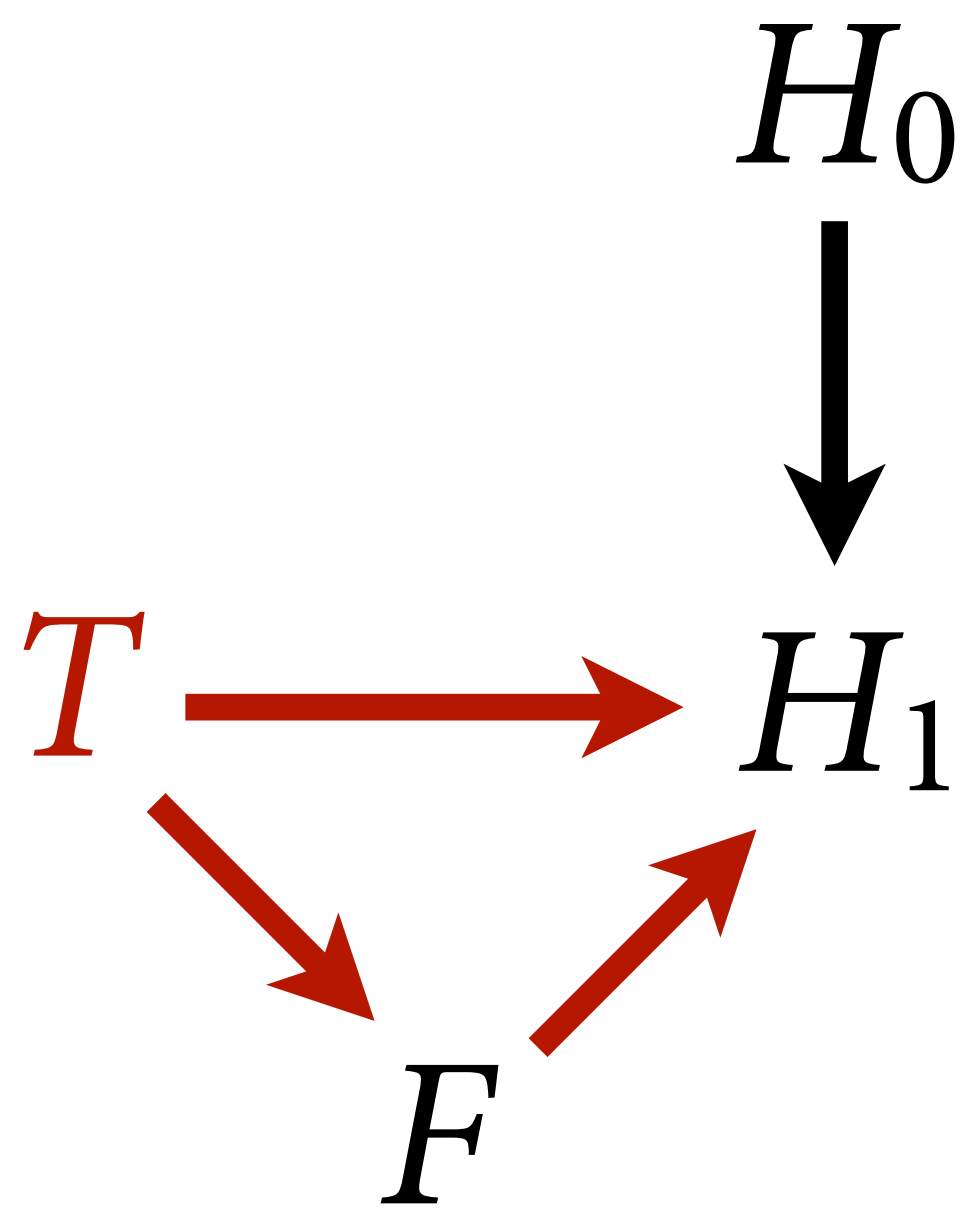


Wrong model wins at prediction

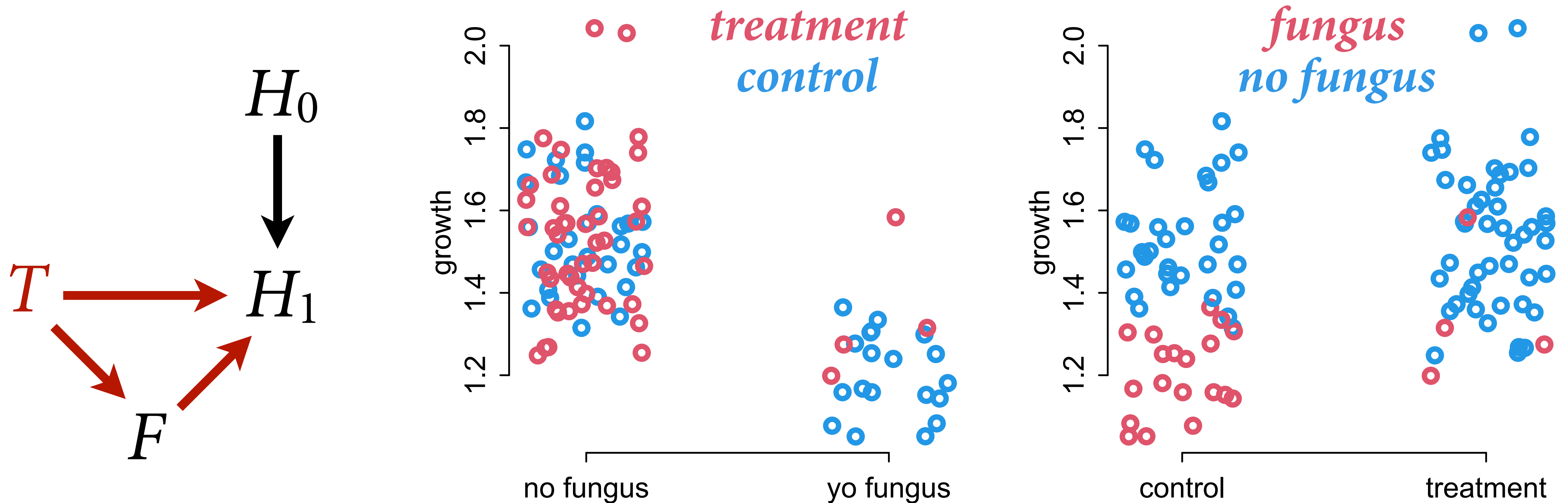


Wrong model wins at prediction

Why does the wrong model win at prediction?



Why does the wrong model win at prediction?



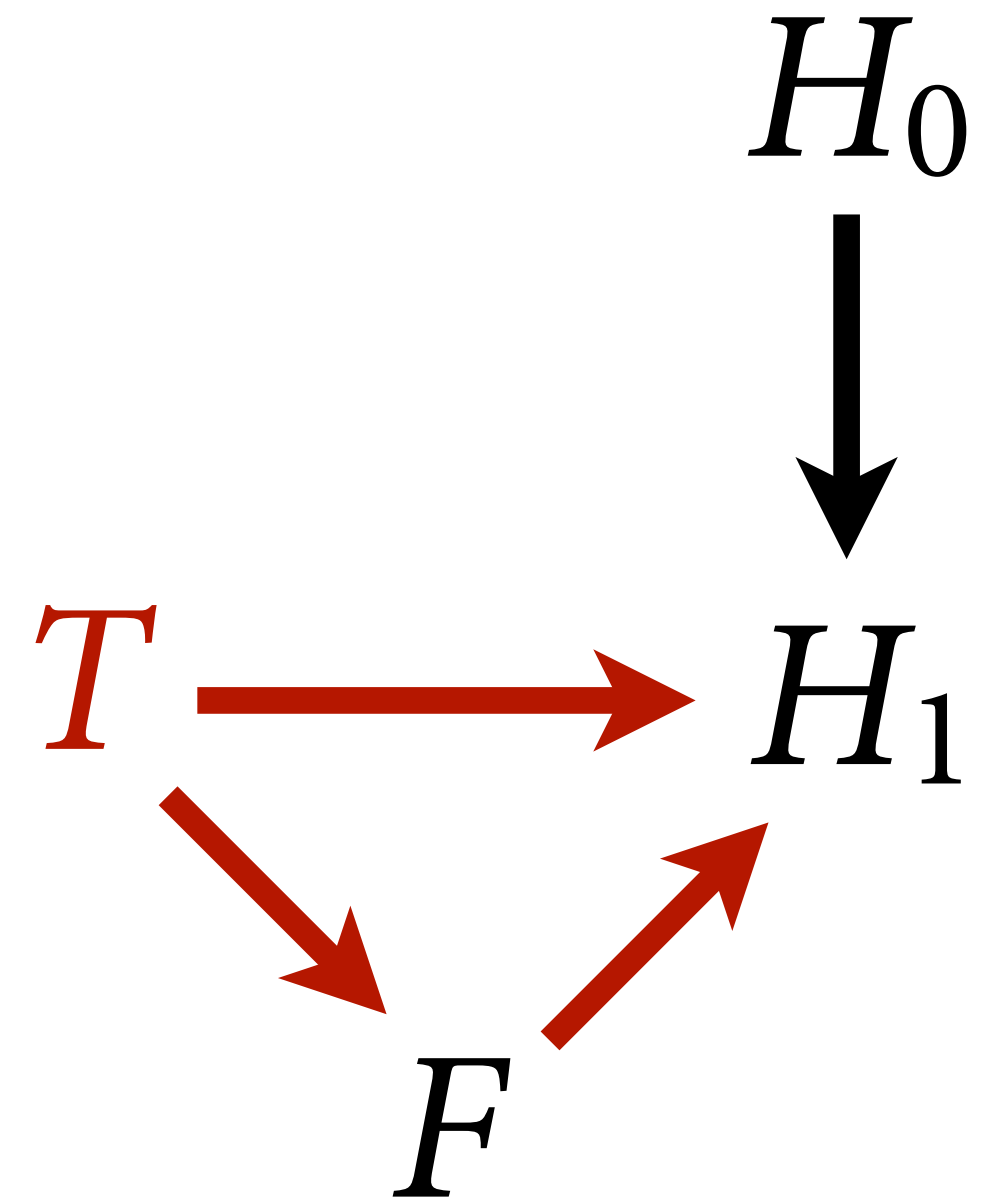
Fungus is in fact a better predictor than treatment

Model Mis-selection

Do not use predictive criteria (WAIC, PSIS, CV) to choose a causal estimate

However, many analyses are mixes of inferential and predictive chores

Still need help finding good functional descriptions while avoiding overfitting



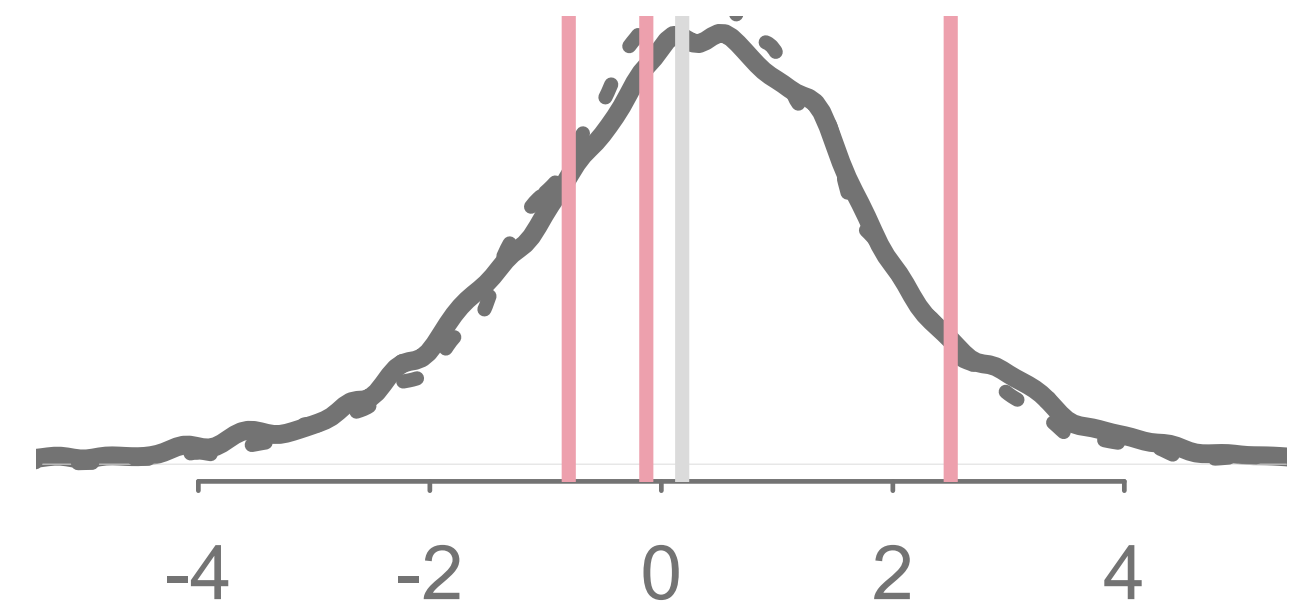
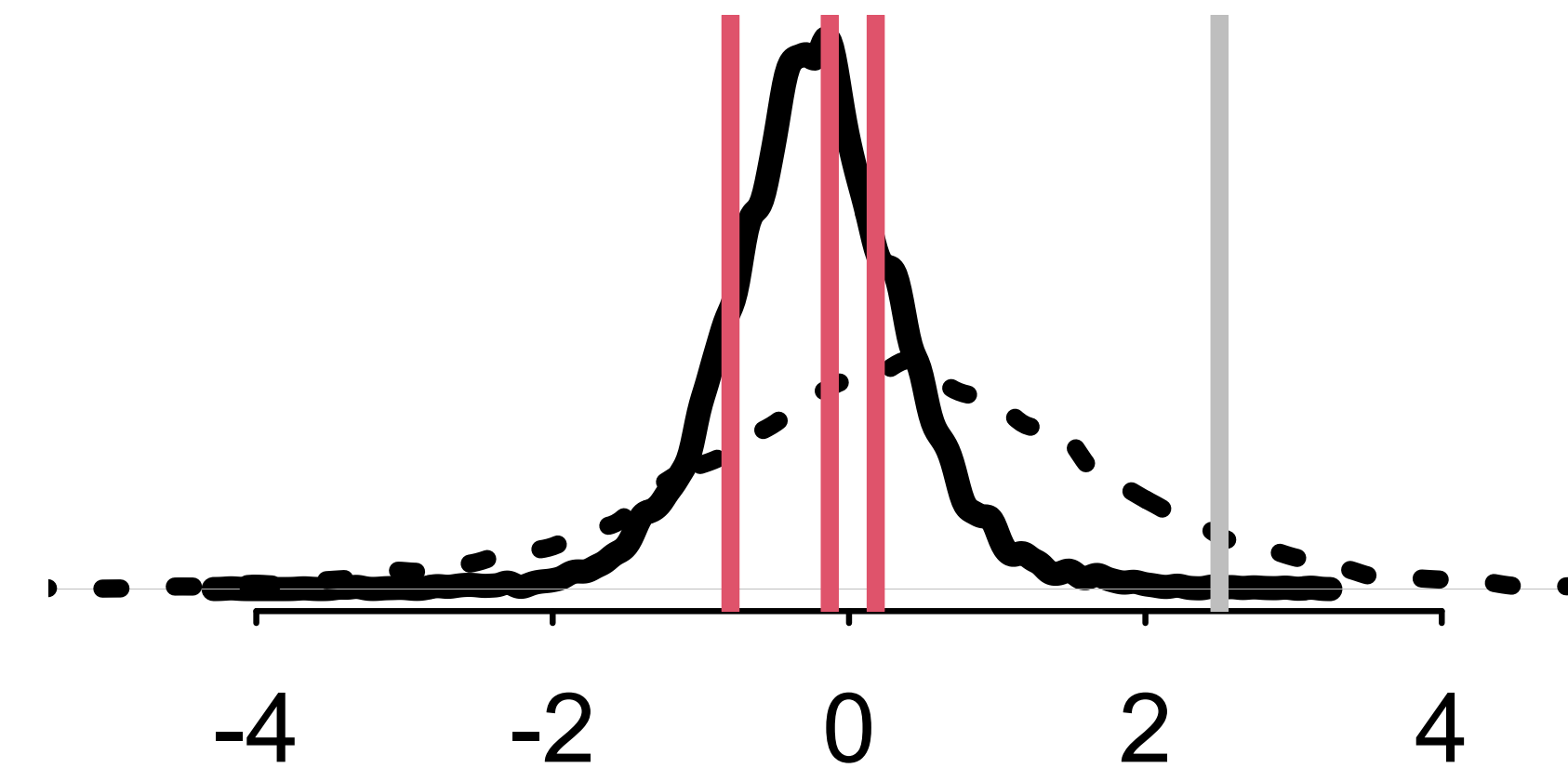
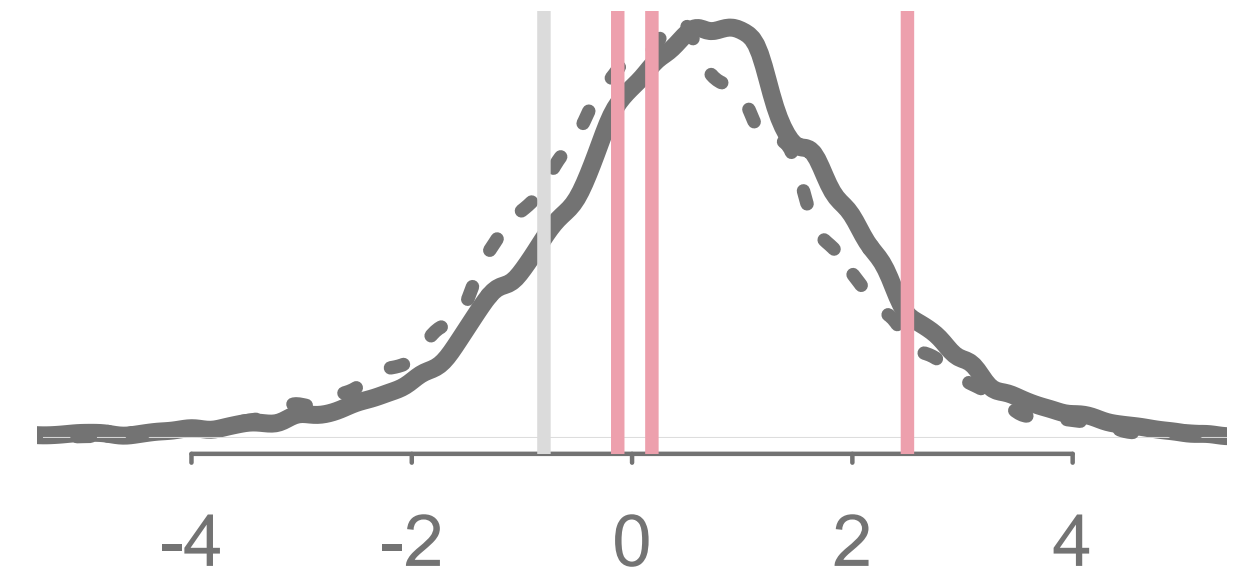
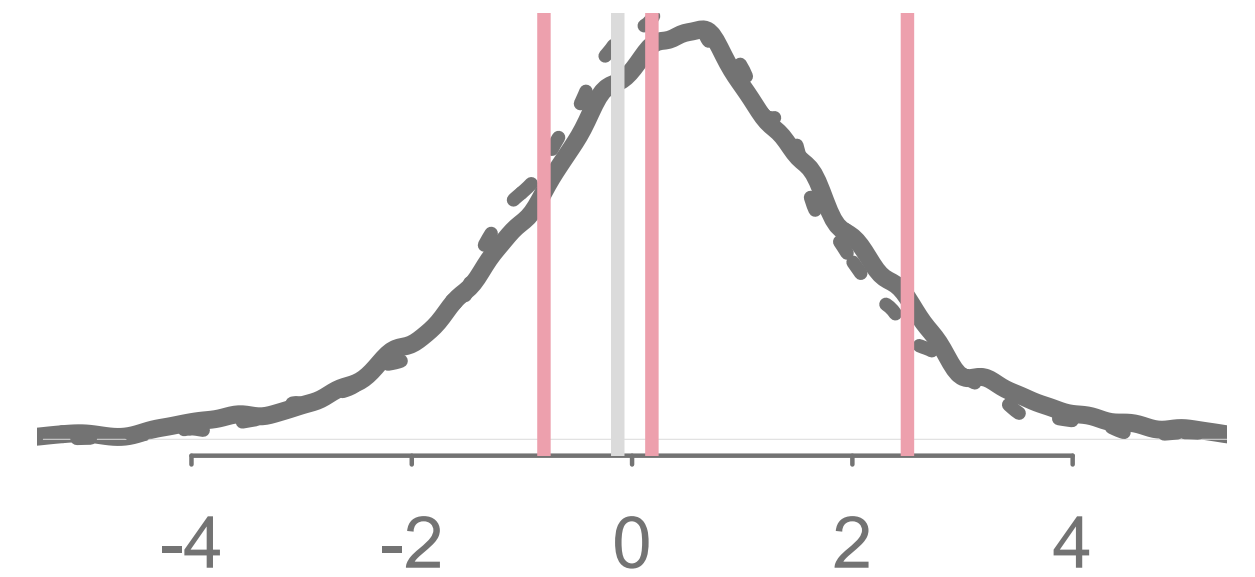
Outliers & Robust Regression

Some points are more influential than others

“Outliers”: Observations in the tails of predictive distribution

Outliers indicate predictions are possibly overconfident, unreliable

The model doesn't expect enough variation



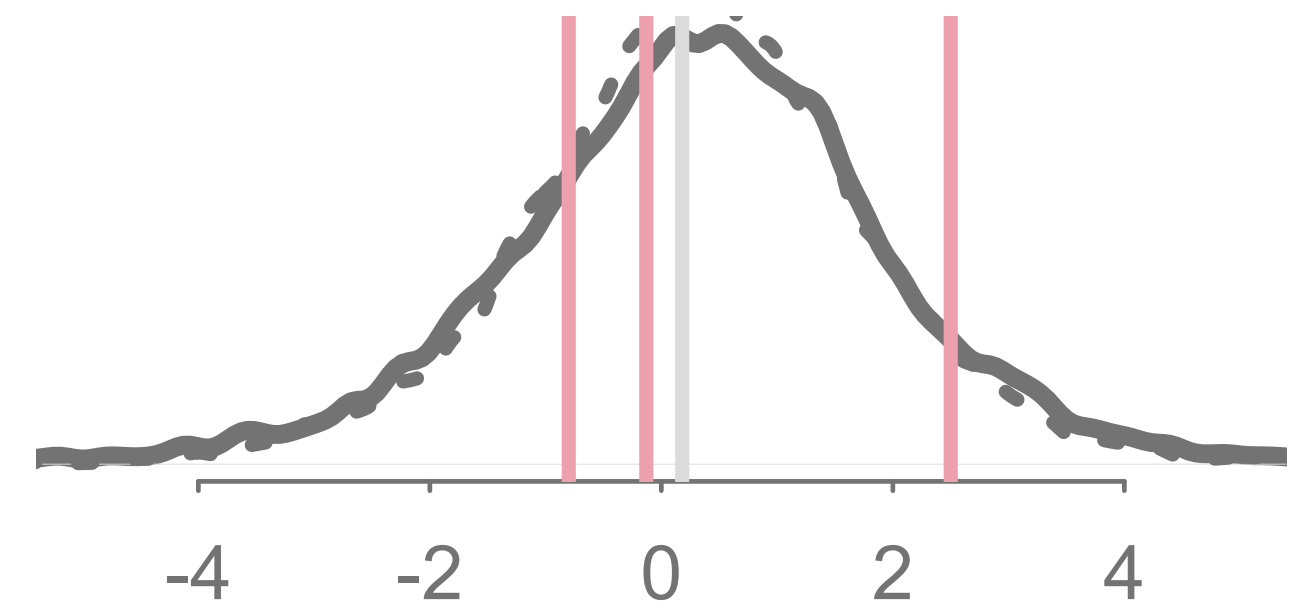
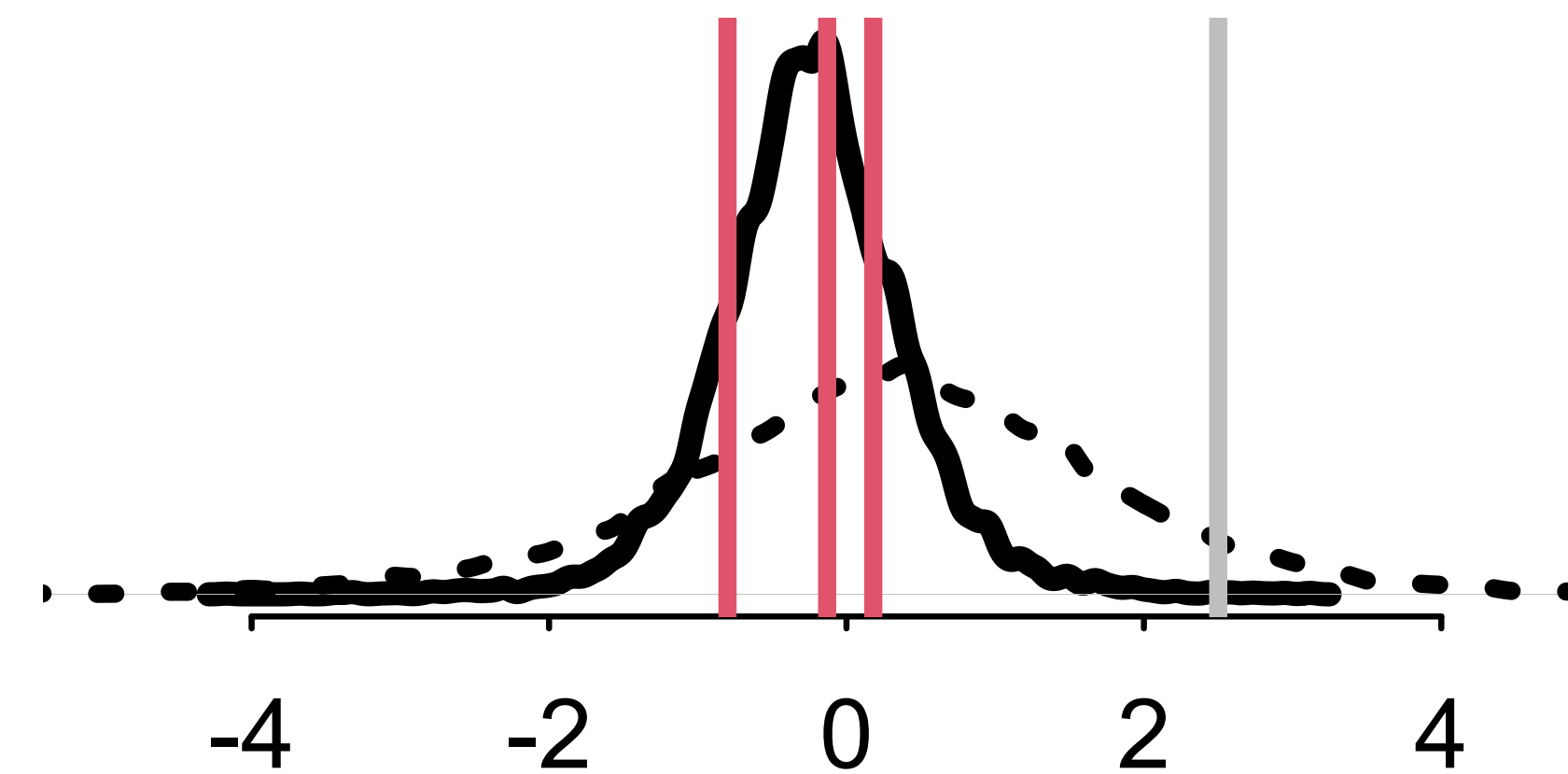
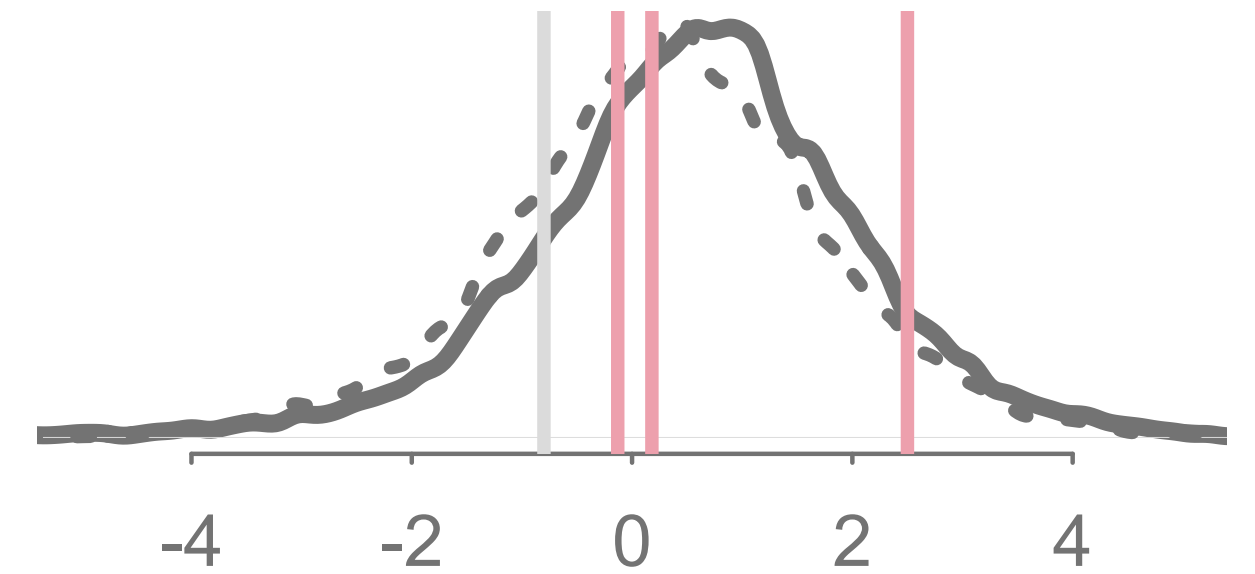
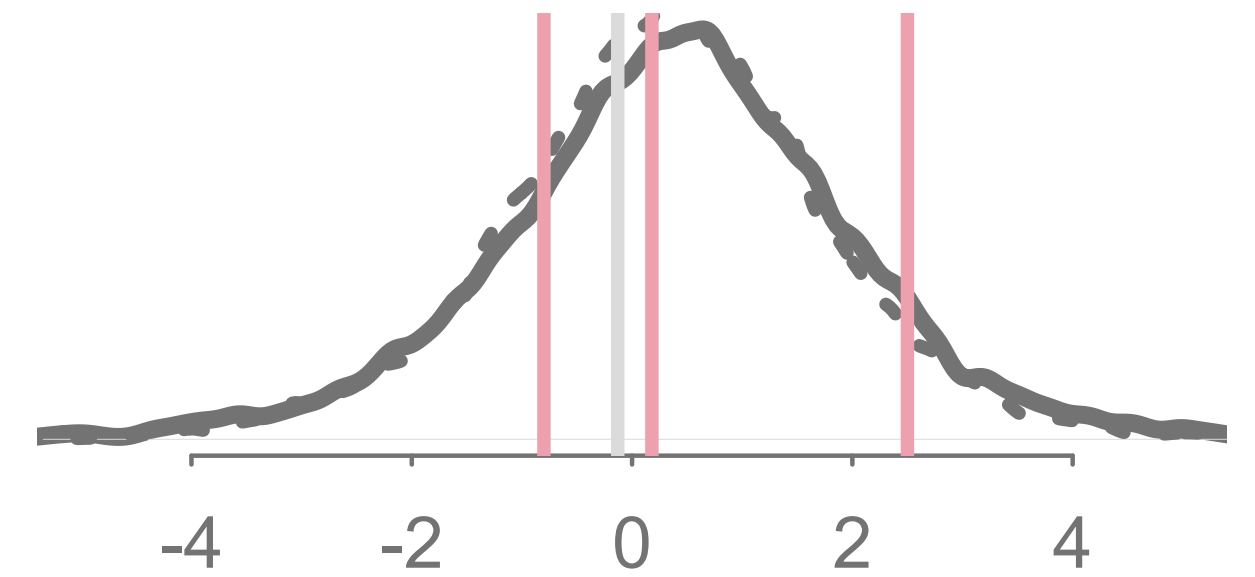
Outliers & Robust Regression

Dropping outliers is bad: Just ignores the problem; predictions are still bad!

It's the model that's wrong, not the data

First, quantify influence of each point

Second, use a mixture model (robust regression)



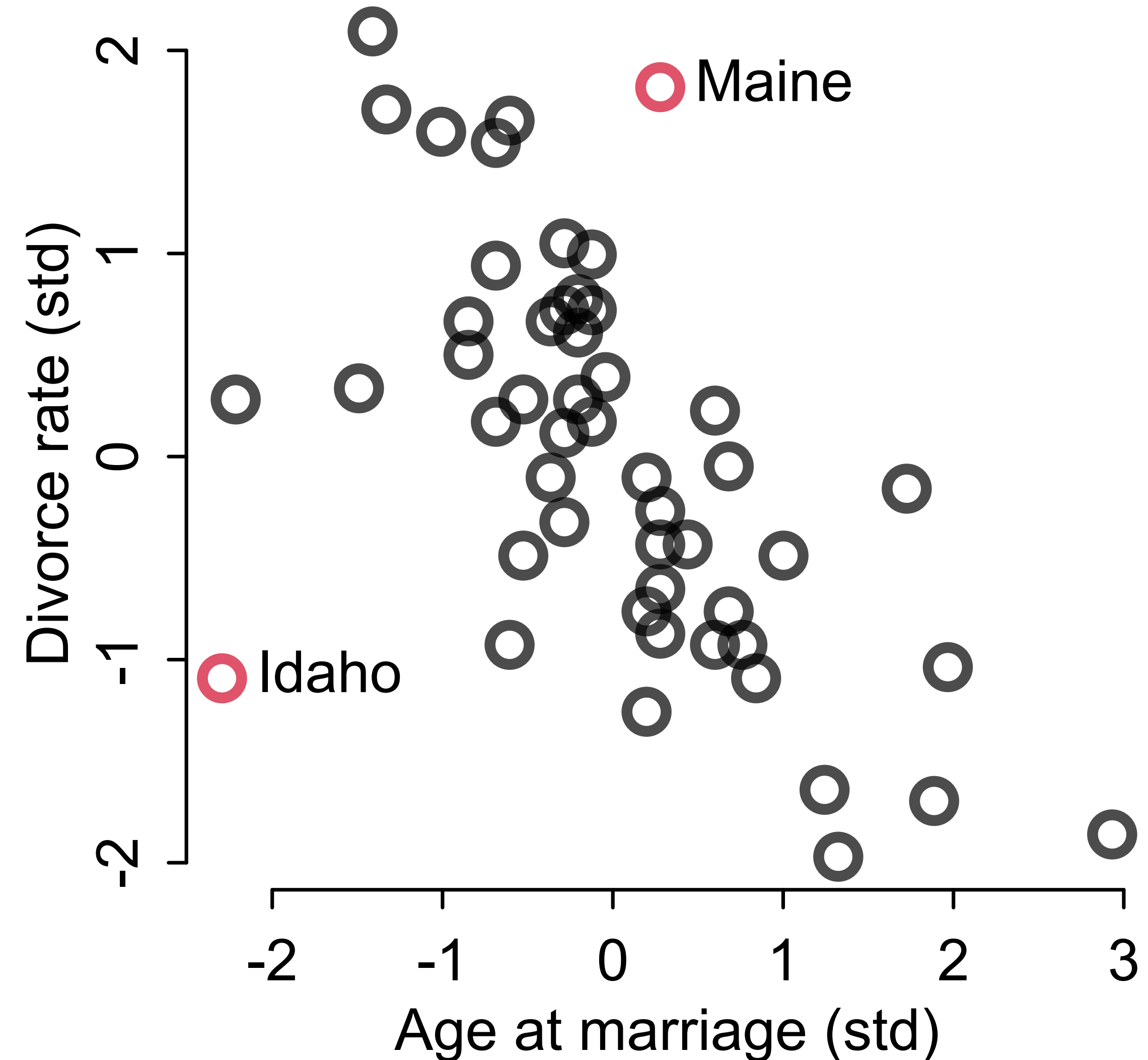
Outliers & Robust Regression

Divorce rate example

Maine and **Idaho** both unusual

Maine: high divorce for trend

Idaho: low divorce for trend

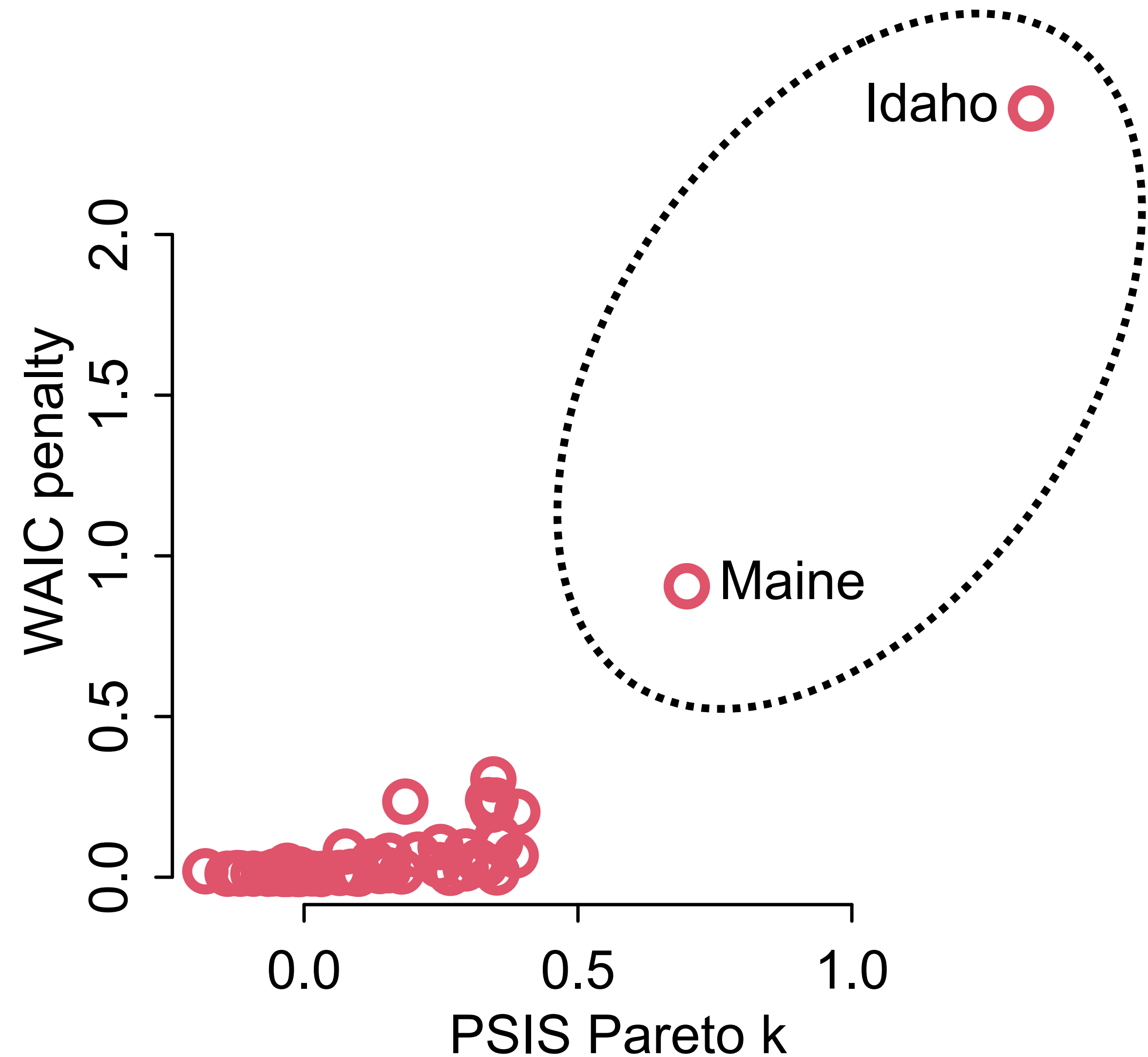


Outliers & Robust Regression

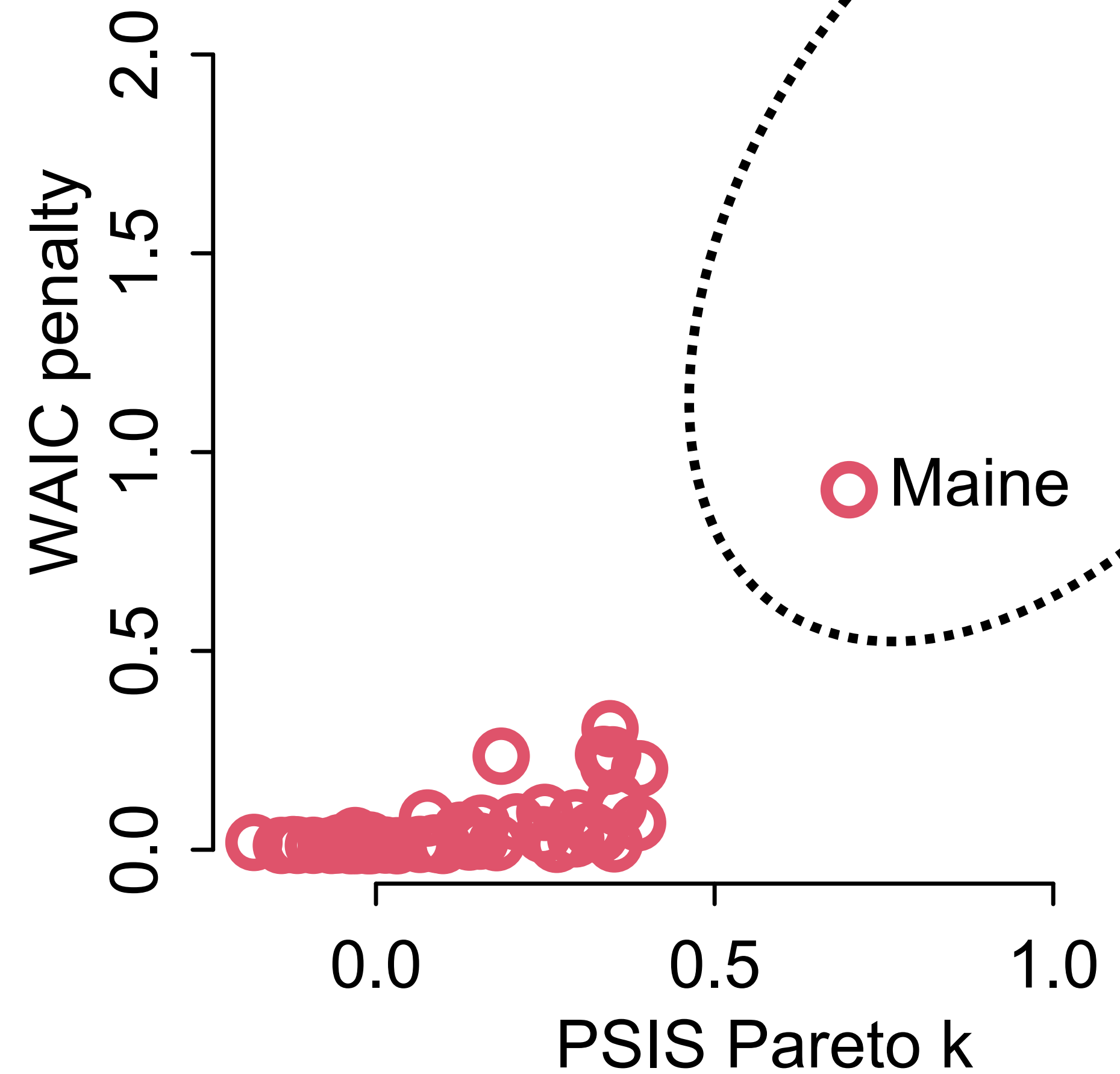
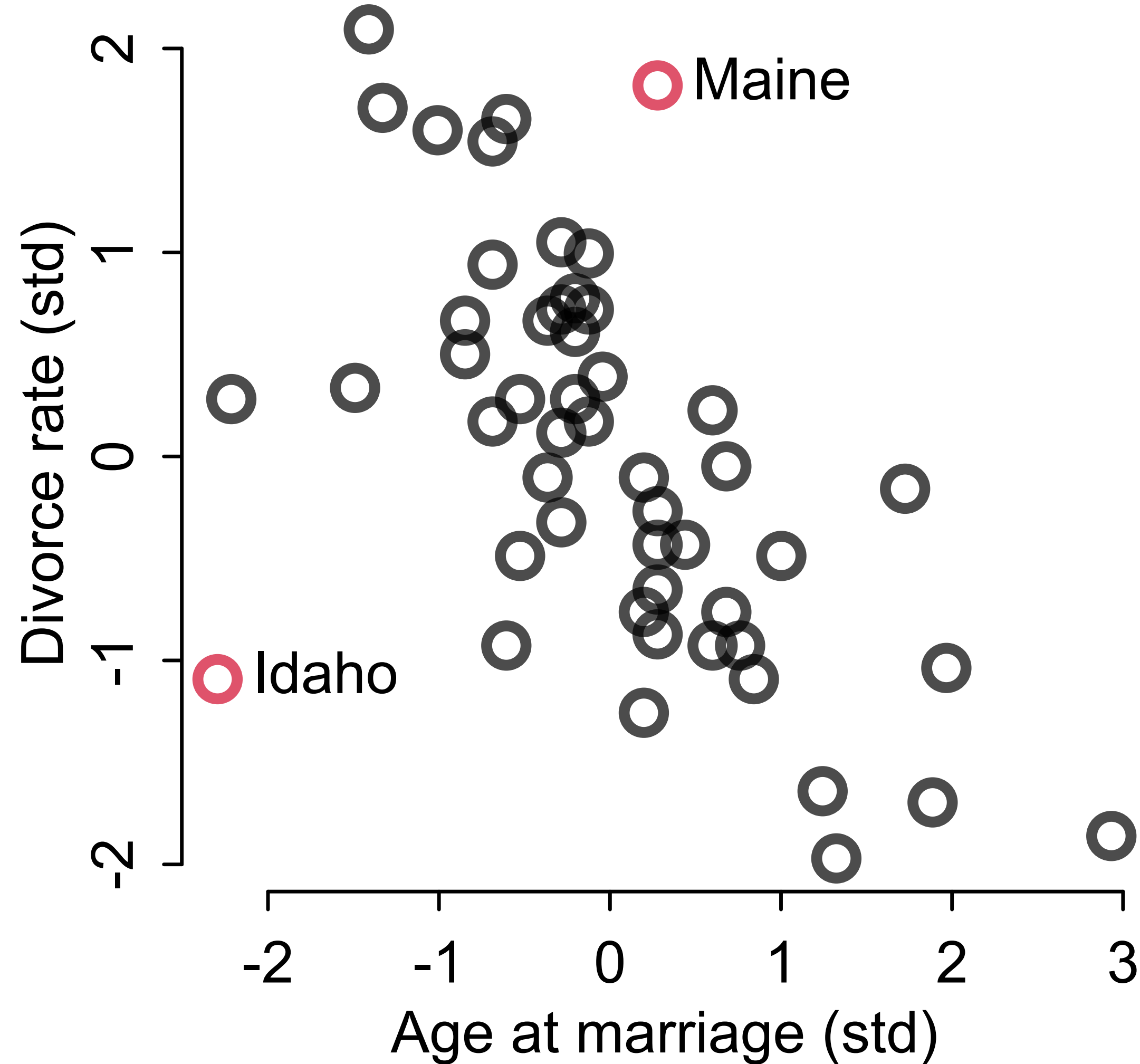
Quantify influence:

PSIS k statistic

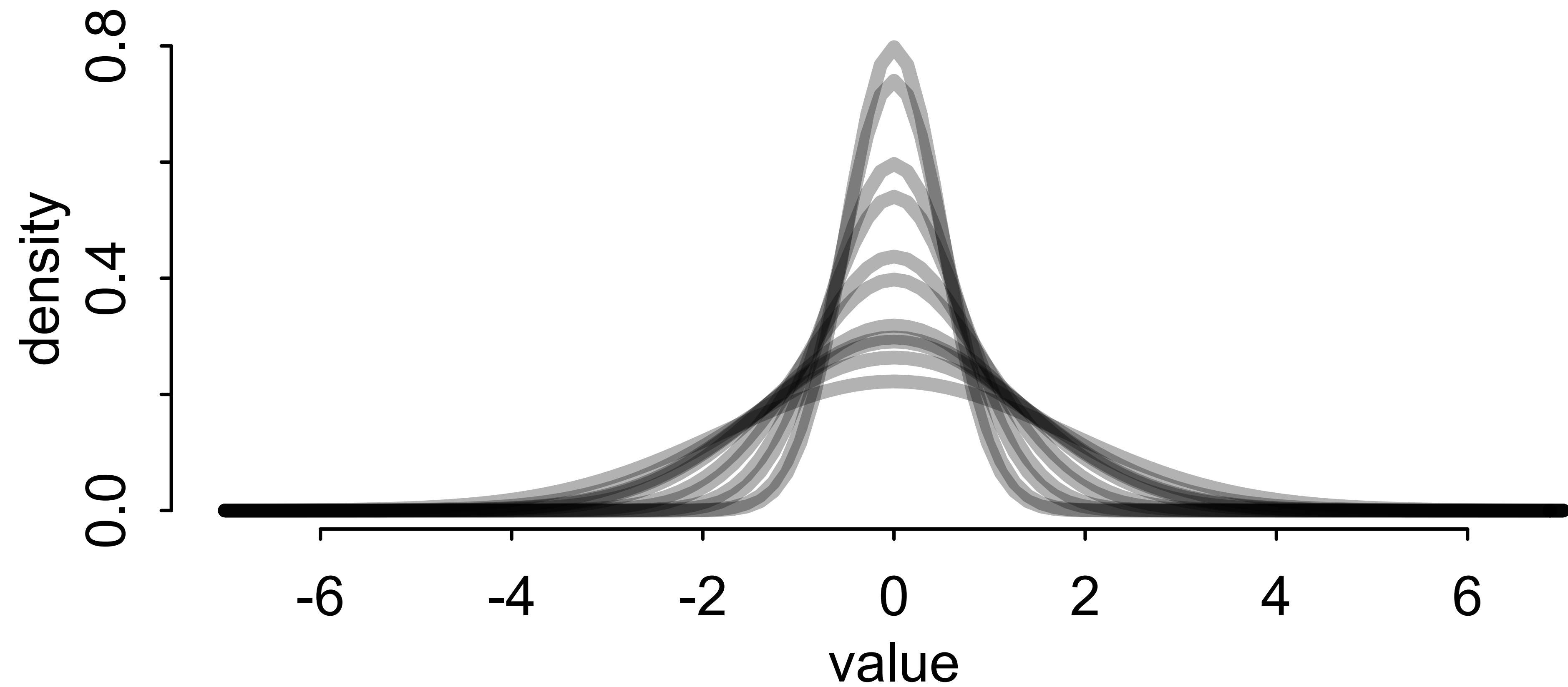
WAIC penalty term (“effective number of parameters”)



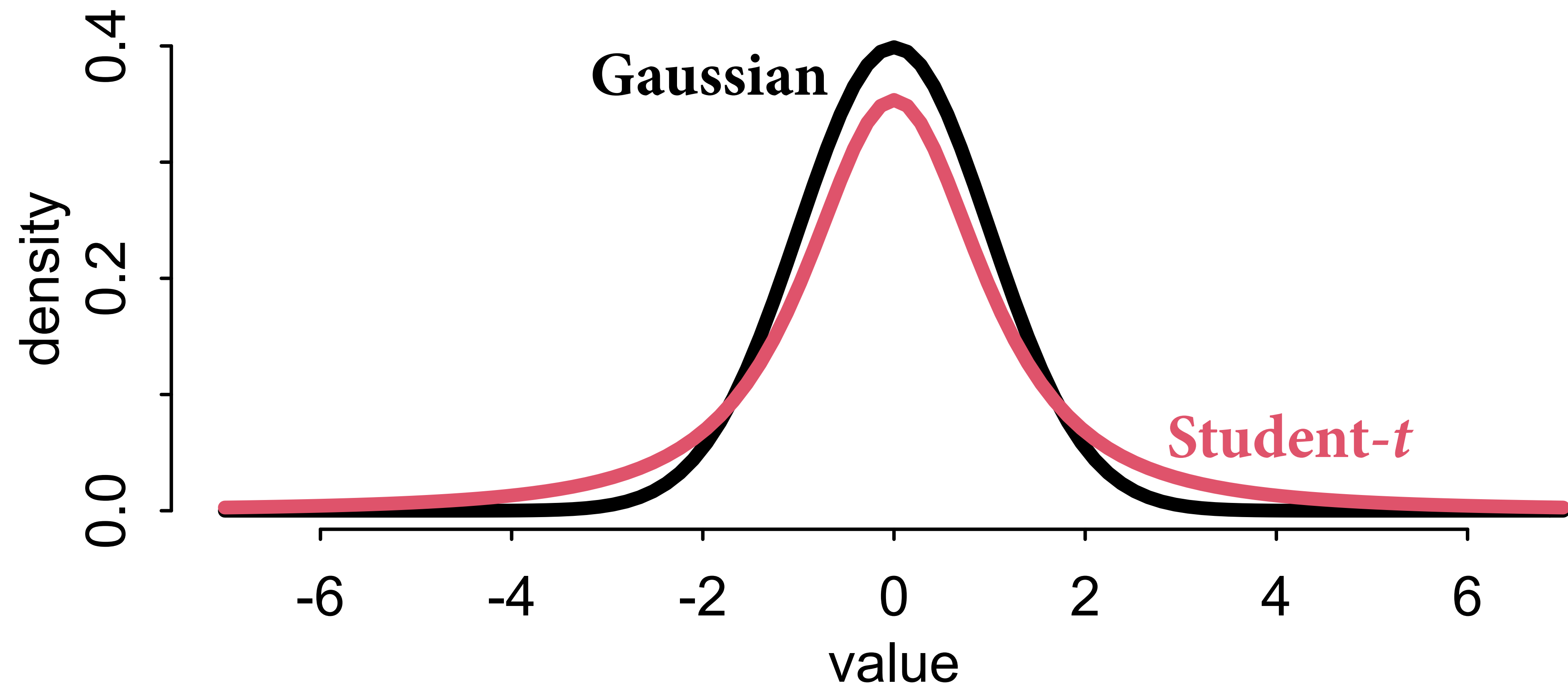
Outliers & Robust Regression



Mixing Gaussians



Mixing Gaussians




```
m5.3 <- quap(
  alist(
    D ~ dnorm( mu , sigma ) ,
    mu <- a + bM*M + bA*A ,
    a ~ dnorm( 0 , 0.2 ) ,
    bM ~ dnorm( 0 , 0.5 ) ,
    bA ~ dnorm( 0 , 0.5 ) ,
    sigma ~ dexp( 1 )
  ) , data = dat )

m5.3t <- quap(
  alist(
    D ~ dstudent( 2 , mu , sigma ) ,
    mu <- a + bM*M + bA*A ,
    a ~ dnorm( 0 , 0.2 ) ,
    bM ~ dnorm( 0 , 0.5 ) ,
    bA ~ dnorm( 0 , 0.5 ) ,
    sigma ~ dexp( 1 )
  ) , data = dat )
```

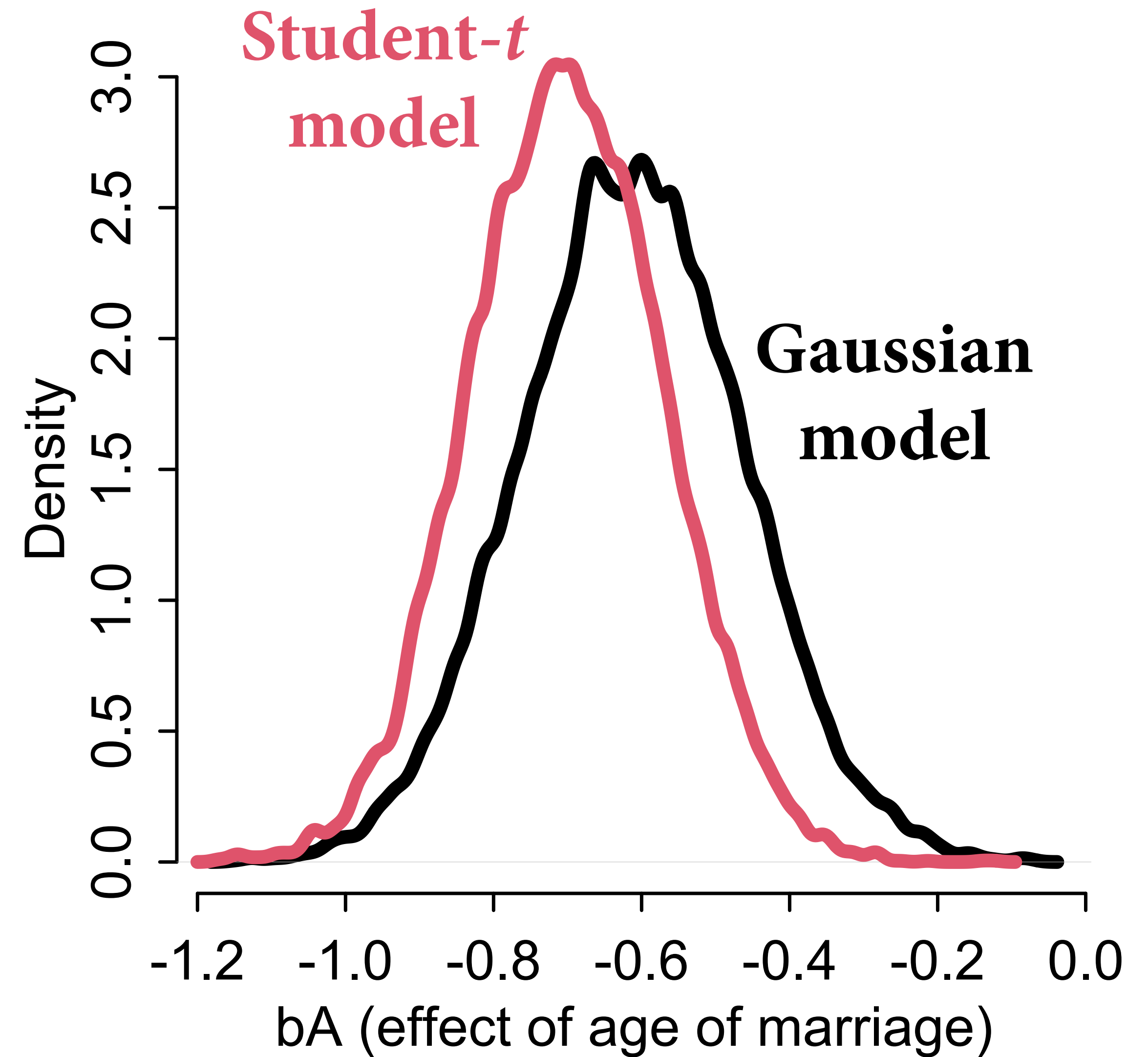


```

m5.3 <- quap(
  alist(
    D ~ dnorm( mu , sigma ) ,
    mu <- a + bM*M + bA*A ,
    a ~ dnorm( 0 , 0.2 ) ,
    bM ~ dnorm( 0 , 0.5 ) ,
    bA ~ dnorm( 0 , 0.5 ) ,
    sigma ~ dexp( 1 )
  ) , data = dat )

m5.3t <- quap(
  alist(
    D ~ dstudent( 2 , mu , sigma ) ,
    mu <- a + bM*M + bA*A ,
    a ~ dnorm( 0 , 0.2 ) ,
    bM ~ dnorm( 0 , 0.5 ) ,
    bA ~ dnorm( 0 , 0.5 ) ,
    sigma ~ dexp( 1 )
  ) , data = dat )

```



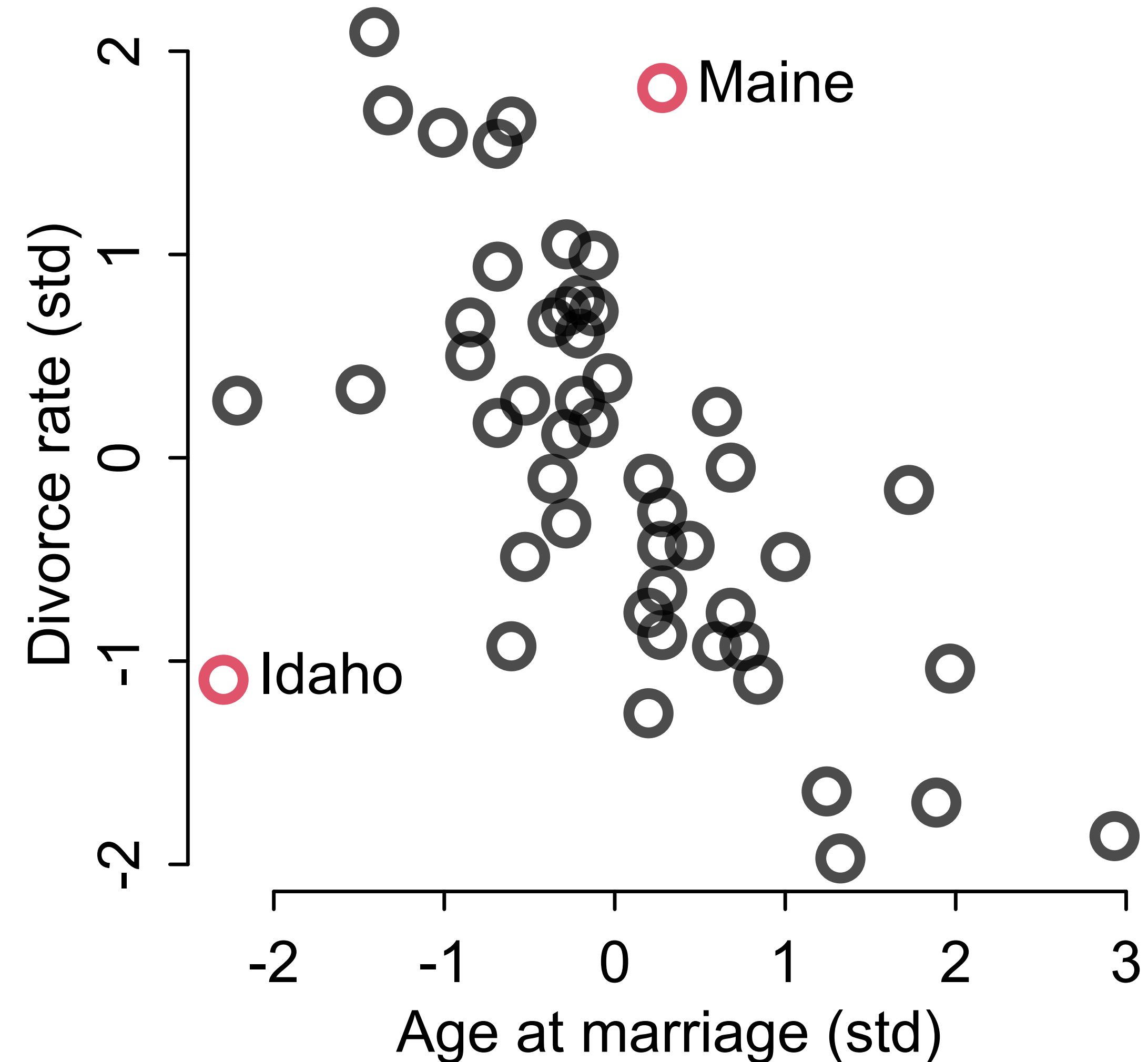
Robust Regressions

Unobserved heterogeneity =>
mixture of Gaussians

Thick tails means model is less
surprised by extreme values

Usually impossible to estimate
distribution of extreme values

Student- t regression as default?



Prediction Summary

What is the next observation from the same process? (prediction)

Possible to make very good predictions without knowing causes

Optimizing prediction does not reliably reveal causes

Powerful tools (PSIS, regularization) for measuring and managing accuracy

